FACULTY OF COMPUTING
UTM Johor Bahru

UNIVERSITI TEKNOLOGI MALAYSIA

SECJ2203 - SOFTWARE ENGINEERING

Semester 1, 2024/2025

Section 01

Group: **A3 - MoodyMate**

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Figma link:

https://www.figma.com/design/bIggyW74G6NmbhCkOPUT1s/Untitled?node-id=0-1&t=ZO2lRJCw5cosIb4k-1

Revision Page

a. Overview

MoodMate is an integrated mental health support application designed to enhance the emotional well-being of children and their parents while ensuring oversight and management by administrators. The system fosters a proactive approach to mental health through structured tracking, therapy resources, and engaging activities.

b. Target Audience

The MoodMate system is designed for three key audiences: parents, children, and administrators.

c. Project Team Members

Member Name	Role	Task	Status
Ainurrulnajwa binti Sahlan	Group Member	1. Introduction 2. specific requirements 2.1 persona 2.2system features 2.4user story details 4.3 Logical view	completed
Ravinesh	Group Leader	2.3 launch phase 4.4 Process View 4.5 Deployment View	completed
Bernice	Group Member	2.5 Performance and Other Requirements 4.1 Use Case View 4.2 Implementation View	completed
Karen	Group Member	2.6 Design Constraints 3.0 Architectural rationale 3.1 architectural style and rationale 4 architectural views	completed

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Ainurrulnajwa (Team Member)	Completed Chapter 1 and 2. Intan work out on chapter 1 and chapter 2 (2.1 , 2.2 , 2.4) . All members did their part .	23/12/2024
1.0	Ainurrulnajwa (Team Member)	Completed Chapter 1 and 2. Intan work out on chapter 1 and chapter 2 (2.1 , 2.2 , 2.4) . All members did their part .	23/12/2024
1.0	Ainurrulnajwa (Team Member)	compile all task given from team member	28/12/2024

Table of Contents

1. Introduction

Children's mental health is a crucial aspect of their overall well-being, shaping how they navigate relationships, learn, and grow into adulthood. In recent years, there has been a growing recognition of the mental health challenges faced by children, from anxiety and depression to difficulties coping with academic and social pressures. Despite its importance, kids' mental health often goes unnoticed, leaving many children without the support they need.

This report aims to shed light on the state of children's mental health, examining the key factors contributing to mental health challenges and exploring effective strategies to address them. By highlighting the role of families, schools, and communities in fostering positive mental health, we hope to inspire actionable steps toward a supportive environment where every child can thrive emotionally and psychologically.

Through this exploration, we seek to empower parents, educators, and policymakers to recognize the importance of mental well-being in shaping a brighter future for *children and society alike*

1.1 Purpose

The purpose of this Software Development (SD) is to outline the development and intended use of a mobile application designed to monitor and support children's mental health. This application seeks to empower parents, educators, and healthcare professionals by providing them with real-time data and insights into children's emotional and psychological well-being.

The primary goals of the application are:

- To enable early identification of potential mental health concerns through behavior and mood tracking.
- To offer tailored recommendations and resources for promoting mental well-being.
- To bridge the gap between mental health observation and professional intervention by offering actionable insights.

This application is intended for use by caregivers, teachers, and child psychologists who play a key role in fostering children's mental health. It is designed to complement professional support systems by providing an easy-to-use platform for continuous mental health monitoring.

1.2 Scope

The proposed mobile application, **Moodmate**, is a digital tool aimed at observing and supporting children's mental health.

Key Features:

- **Behavioral and Mood Tracking:** Enables users to log daily behavioral patterns, emotions, and activities for children, providing a comprehensive view of their mental health over time.
- **Alerts and Notifications:** Uses predefined thresholds to alert caregivers of significant changes or concerns in mental health indicators, encouraging early intervention.
- **Resource Library:** Offers access to articles, videos, and practical exercises aimed at improving mental resilience and understanding mental health.
- **Secure Data Sharing:** Allows authorized users (e.g., parents and psychologists) to share data**Interactive Tools:** Includes mood calendars, self-reflection prompts for children, and securely for collaborative decision-making.
- gamified mental health check-ins.

Application Benefits:

1. **Proactive Mental Health Monitoring:** Ensures early detection of potential concerns, reducing long-term risks.
2. **User-Focused Design:** Designed for ease of use by non-experts, with clear visuals and minimal technical requirements.
3. **Customizable Features:** Adaptable to i*ndividual children's needs, enabling personalized care.*

1.3 Definitions, Acronyms and Abbreviations

This subsection provides definitions of key terms, acronyms, and abbreviations used in the SD.

- **Behavioral Tracking:** Monitoring and recording patterns in actions, reactions, and interactions to identify potential mental health trends.
- **Mental Health Indicator:** Specific factors (e.g., mood, sleep patterns, social interactions) used to assess emotional well-being.
- **CBT (Cognitive Behavioral Therapy):** A common therapy approach that helps individuals manage thoughts and behaviors.

- **UI/UX (User Interface/User Experience):** Refers to the app's design and how users interact with its features.
- Intervention Alert: Notifications generated by the application when critical mental health thresholds are crossed, prompting users to take action.
- Data Privacy: Ensuring that all user data is securely stored and accessible only to authorized individuals.

1.4 References

PMC Home, https://pmc.ncbi.nlm.nih.gov/. Accessed 23 December 2024.

Exploring how informed mental health app selection may impact user engagement and satisfaction, https://journals.plos.org/digitalhealth/article?id=10.1371%2Fjournal.pdig.0000219&utm_source=chatgpt.com#abstract1.

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1.5 Overview

Overview of the MoodMate System

The MoodMate system is a comprehensive mental health support application designed to promote the emotional well-being of children and their families. The system focuses on fostering a collaborative approach between parents and children while maintaining robust administrative oversight to ensure secure and efficient operations.

Parents begin their journey by registering on the platform, providing essential information about themselves and their children. This process includes completing a mental health assessment to identify any potential parental stressors that may indirectly affect their children's mental health. Parents then establish secure access for their children through a PIN system, enabling monitored app usage.

MoodMate offers diverse features, including tools for tracking emotions, educational modules for parents, interactive therapy games for children, and shared progress monitoring. The application is designed to be engaging for children while equipping parents with actionable insights and resources to support their child's mental health journey.

Administrative users ensure the system operates smoothly by managing accounts, monitoring usage, and safeguarding user data. Overall, MoodMate bridges the gap between technology and emotional well-being, empowering families to nurture resilience, awareness, and support in mental health.

2. Specific Requirements

This section outlines the software requirements for the mobile application, ensuring it meets the needs of the intended users and operates effectively. The requirements describe the inputs, outputs, and functions performed by the system in response to various inputs.

Functional Requirements:

1. **Data Input**:
    - Users (parents, educators, or healthcare professionals) can input daily observations of a child's mood, behavior, and activities through structured forms or quick prompts.
    - Children can interact with the app to provide self-reflections using simple emoji-based mood trackers or age-appropriate questions.
2. **Data Analysis**:
    - The system processes the data to generate insights, such as identifying patterns, trends, or deviations in mental health indicators.
    - The app flags significant behavioral changes and sends alerts to the caregiver or user.
3. **Output**:
    - The application provides visual dashboards summarizing the child's mental health status over a given time (e.g., weekly or monthly).
    - It generates personalized recommendations, including activities or resources, based on the child's logged data.
4. **Security and Privacy**:
    - All user data is encrypted to ensure confidentiality.
    - Access controls allow data sharing only with authorized users, such as parents or designated psychologists.
5. **Integration with Other Systems**:
    - The app can sync with wearable devices to monitor physical activities and sleep patterns, complementing mental health observations.

2.1 Persona

This section describes the intended users of the system and their characteristics, including their technical expertise, knowledge, and physical abilities. These personas provide a clear understanding of whom the system is designed for and ensure the application meets the specific needs of its intended users.

2.1.1 Persona 1 (Parent User)

User Need

- A parent needs a way to easily track and monitor their child's mental health patterns to proactively identify issues and take necessary actions.

User Stories

- As a parent, I want to sign up and fill in specific information so that the system tailors its features to my needs.
- As a parent, I want to complete a mental health assessment so that I can identify any potential issues that may affect my child's mental health.
- As a parent, I want to create a secure PIN for my child's account so that I can monitor their app usage.
- As a parent, I want to receive notifications every time my child uses the app so that I can stay informed.
- As a parent, I want access to features like mood tracking, parenting modules, online therapy videos, an alarm, daily goals, and profile supervision to support my child's mental health effectively.

2.1.2 Persona 2 (Child User)

User Need

A child needs a supportive and engaging way to express their emotions, track their mental health, and engage in interactive therapy games, with parental oversight.

User Stories

- As a child, I want to enter my PIN to securely access the app under my parent's supervision.
- As a child, I want to log my feelings daily using a mood tracker so that I can better understand and monitor my emotions.
- As a child, I want to play interactive therapy games so that I can learn coping skills in a fun and engaging way.
- As a child, I want to see a scoreboard after completing a game so that I can track my progress and feel a sense of accomplishment.

2.2 System Features

The MoodMate Application is a software system that operates on mobile devices, including both Android and iOS operating systems. The system provides a means for kids to better manage their emotional health and at the same time simplifies the process of seeking emotional support from parents or adults .The system also implemented a real-time cloud-based database for better user experiences.

The system features are illustrated in Figure 2.2 below. The detailed description of each module and functions is tabulated in Table 2.2.

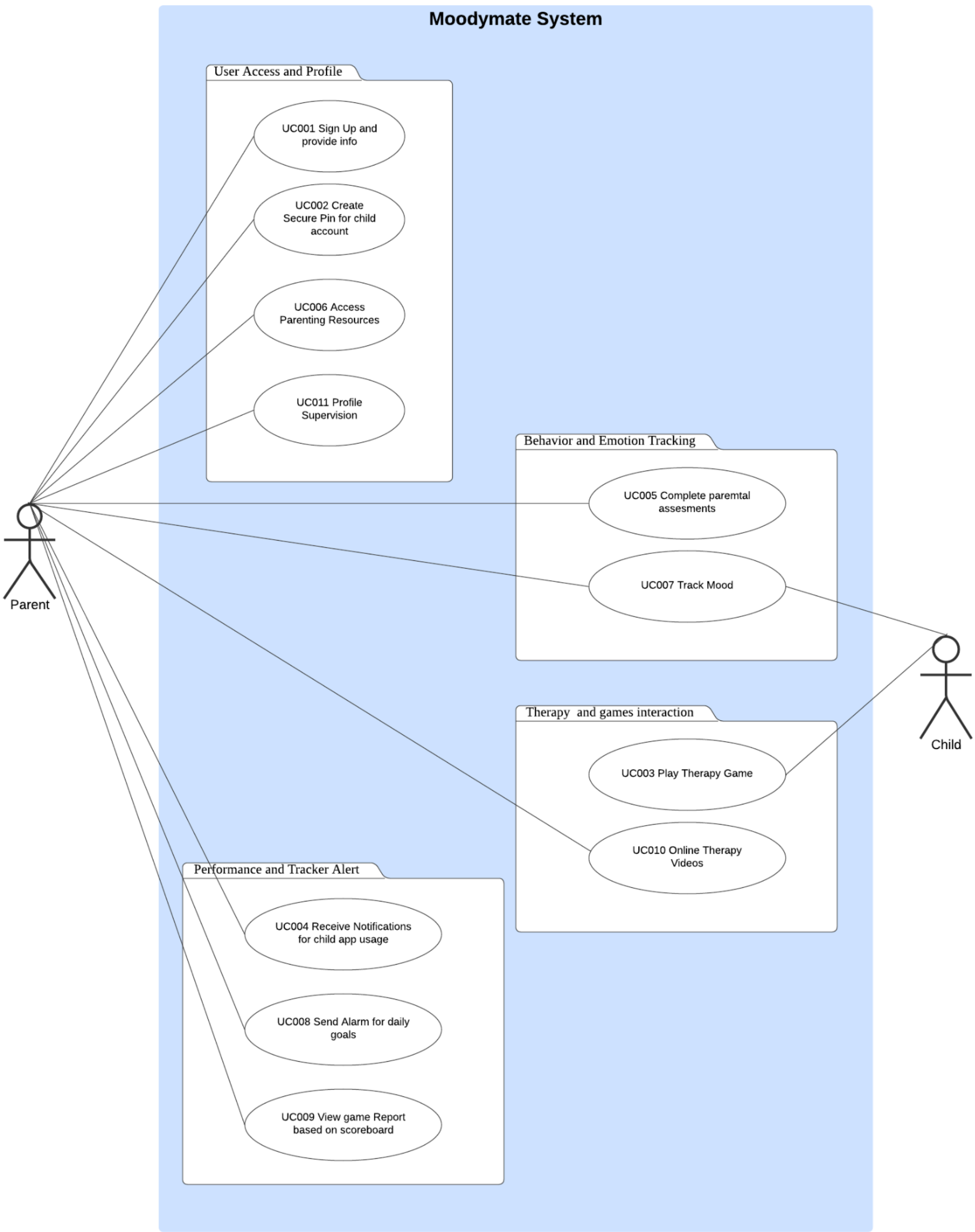


Figure 2.2: Use Case Diagram for moodmate System

Module	*Use Case*	*Function*	*Description*
User Access and Profile	***UC001***	**Sign Up and Provide Information**	Allows parents to create an account and provide specific information about themselves and their child to personalize the app's features.
	UC002	**Create Secure PIN for Child Account**	Allows parents to set up a secure PIN for their child’s app access, ensuring controlled and monitored usage.
	UC006	**Access parenting resources**	Offers educational resources for parents on supporting their child's mental health and building resilience.
	UC011	**Profile Supervision**	Enables parents to review their child’s app activity, including mood logs, activity logs. It provides real-time updates, generates activity reports. If no activity is logged, the system suggests ways to encourage usage.
Behavior and Emotion Tracking	***UC005***	**Complete Parental Assessment**	Enables parents to fill out a mental health assessment, helping identify any issues that may impact their child’s well-being.
	UC007	**Track Mood**	Provides a tool for parents and children to log moods and track emotional patterns over time.
Performance and Tracker Alert	***UC004***	**Receive Notifications for child app usage**	Sends notifications to parents whenever their child accesses the app, keeping them informed of usage.
	UC008	**Send Alarm for daily goals**	Lets parents set reminders or alarms for mental health activities of daily tasks.

	UC009	**View Game Report based on Scoreboard**	Displays children's game results, encouraging engagement and allowing parents to track their child's progress.
Therapy and Game Interaction	***UC003***	**Play Therapy Games**	Offers interactive games for children to learn coping skills and enhance emotional resilience in a fun way.
	UC010	**online therapy video**	Provides interesting therapy interaction

Table 2.2: Description of Functions for <MoodMate System>

The use case diagram illustrates the interactions between two main actors—Parent and Child—and the system's modules. Each module represents a key functional area, addressing specific user

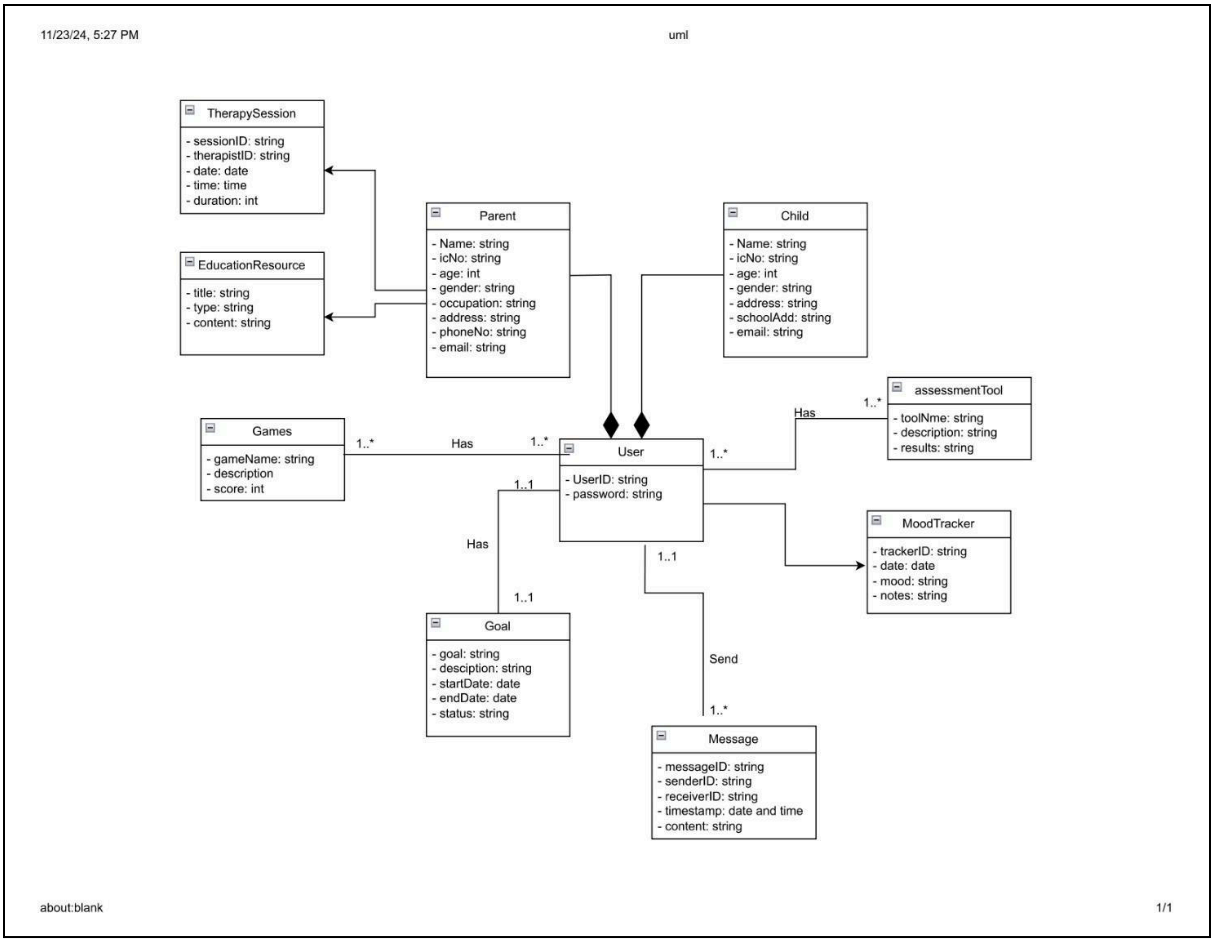


Figure 2.3: Domain Model for MoodMate Application

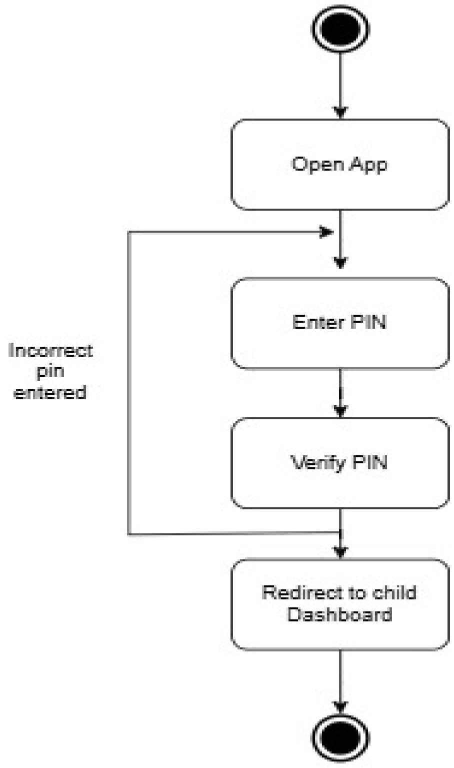


Figure 2.4 : State machine diagram for Enter PIN to Access App

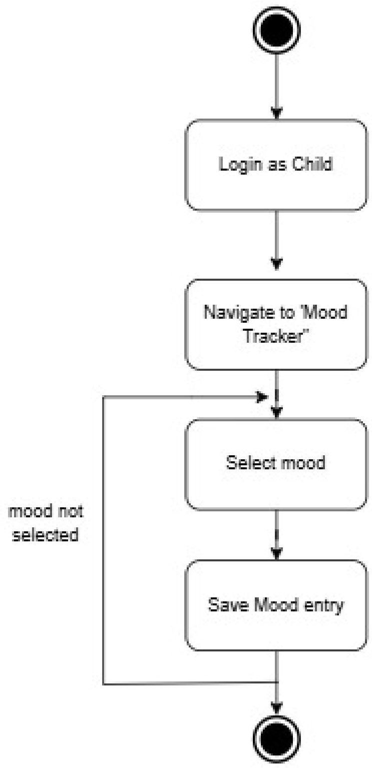


Figure 2.5: State Machine Diagram for Log mood

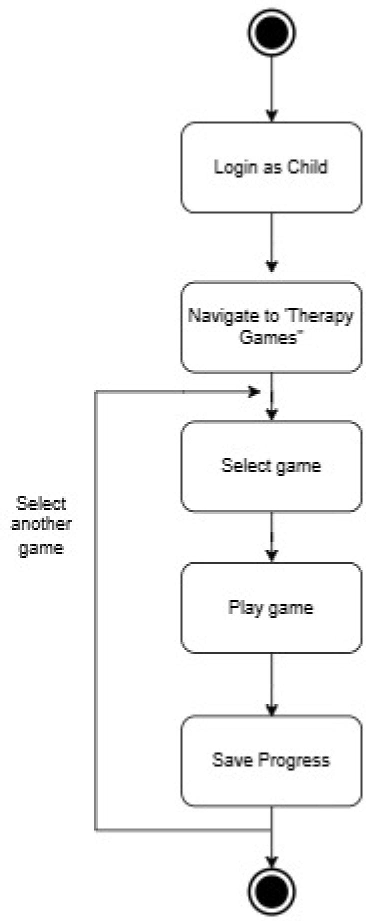


Figure 2.6: State Machine Diagram for Play therapy games

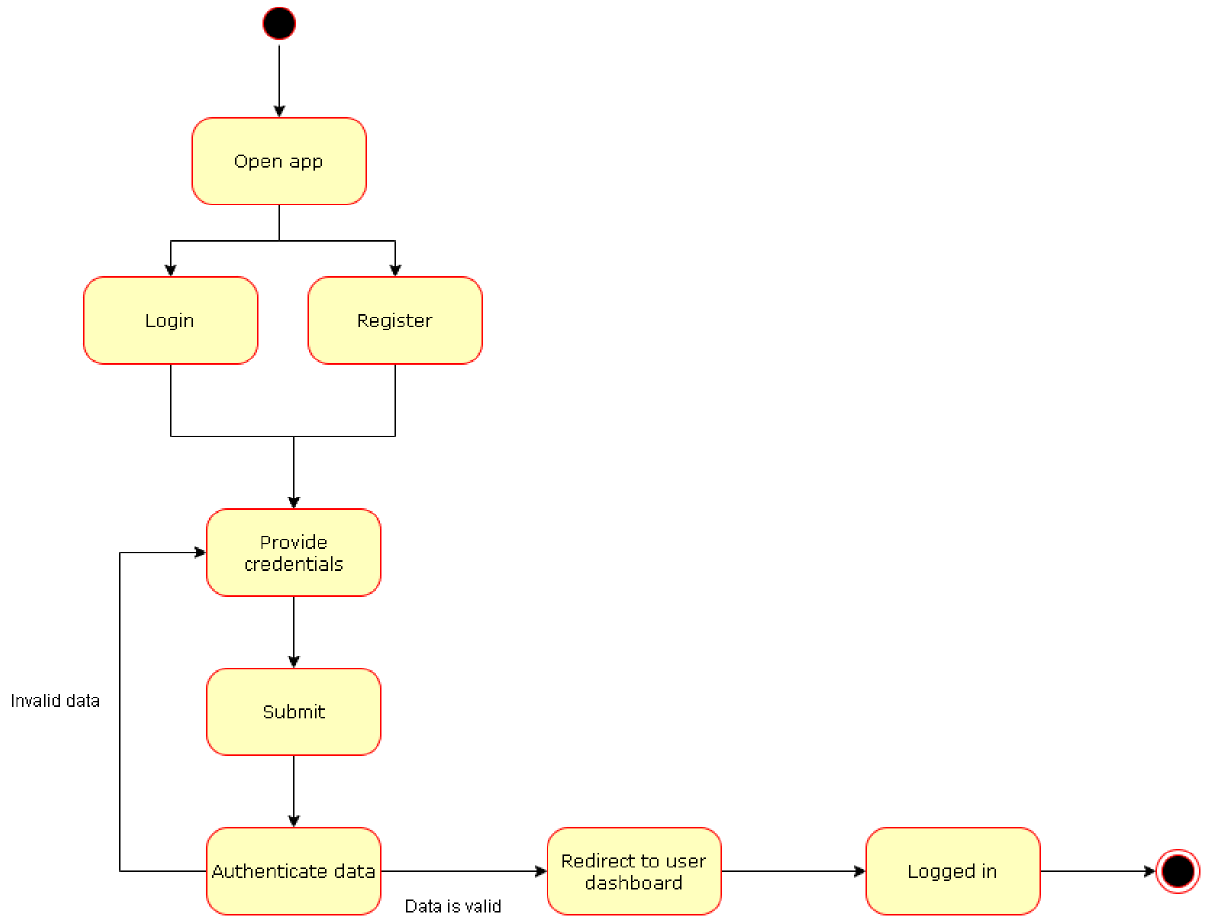


Figure 2.7: State Machine Diagram for Registration

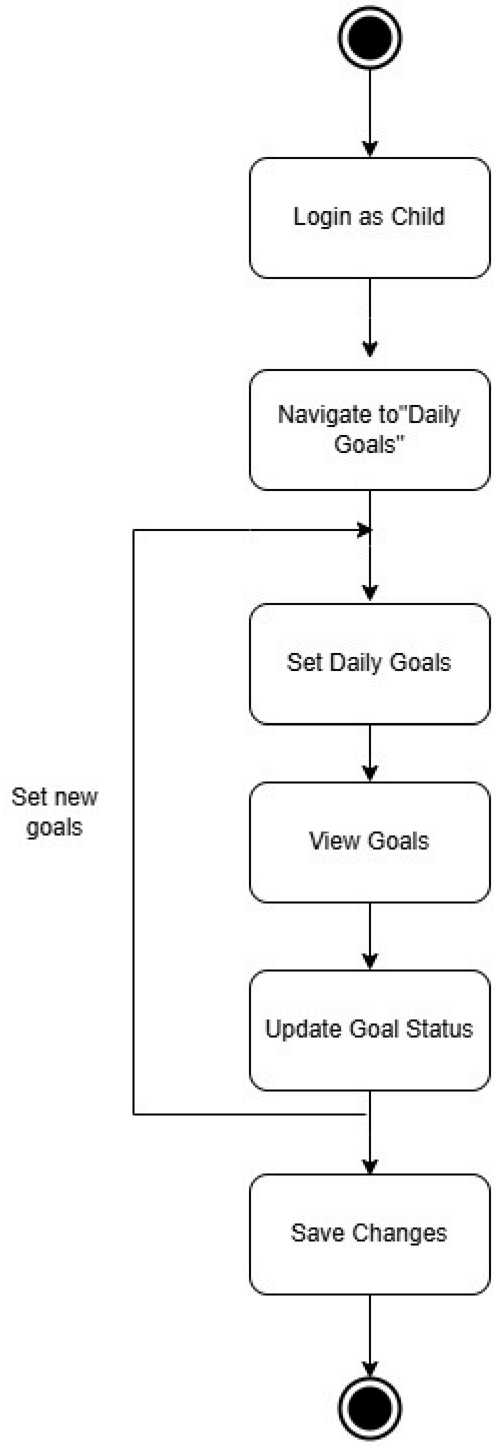


Figure 2.8State machine diagram for track daily goals

2.3 Launch Phase

Sprint	**Team members assigned**
Sprint 1 **Mood Tracking:** Daily mood logging, visual summaries, and weekly/monthly trend analysis. This feature helps users identify patterns in their emotional well-being over time. **User story:** Users can log their daily moods and view trends over weekly or monthly periods to better understand their emotional patterns. This information can also be shared with trusted adults like parents or counselors for support.	Intan
Sprint 2 **Profile Parents:** Features for creating parent profiles, tracking child behaviour, and supervising activities. These tools provide parents with actionable insights while respecting privacy. **User Story**: Parents can create their own profiles to access helpful resources and supervision tools, enabling them to stay informed and actively support their child's mental health journey.	Bernice
Sprint 3 **Interactive learning module:** learning module for parents to help them observe and understand their child's mental health can empower them with skills and tools to support their child effectively. **User Story:** Parents gain access to an interactive learning module that explains mental health concepts and strategies, helping them better understand their child's emotional needs and effectively support them.	Ravi

Sprint 4 **Parental profile & supervision :** Allow parents to monitor their child's mental wellness, set up supportive structures, and gain insights into their child's behaviour in a responsible and non-invasive way. **User Story:** Parents have a platform to monitor their child's mental wellness and receive actionable insights into their behaviour, enabling them to provide constructive support and foster a healthy environment without being intrusive.	Karen
Sprint 5 **Online therapy session :** Access various video therapy, contact info therapist **User Story:** Users can schedule and attend video therapy sessions and access contact information for therapists, ensuring professional guidance is available whenever necessary.	Intan
Sprint 6 **Goal and habit Tracking :** Set child goals (e.g., handling tantrum, increasing mindfulness) and track progress with reminders. **User story:** Users can set personal goals (e.g., improving emotional regulation or practicing mindfulness) and receive progress reminders to stay motivated and effectively track personal development.	Bernice and Karen
Sprint 7 **Assessment tool :** Create cards with emotions, characters, and settings, and have kids draw cards to create short stories about how different characters handle emotions. **User Story:** Users can create stories using cards that represent different emotions, characters, and settings. This allows them to explore their feelings creatively and learn how to handle emotions through storytelling.	Ravi

2.4 User Story Details

Sign-Up and Provide Information

As a parent, I want to sign up and provide details about myself and my child so that the app can personalize features for our needs.

Details to include: child's name, age, and specific mental health concerns (if any).

Complete Parental Assessment

As a parent, I want to complete a mental health assessment so that I can identify potential stressors or issues affecting my ability to support my child.

Assessment feedback helps prioritize support strategies for both parent and child.

Create secure pin for child account

As a parent, I want to create a secure PIN for my child's account so that their app usage remains private and monitored.

The PIN ensures the parent controls when and how the child accesses the app.

Receive Notifications for child app usage

As a parent, I want to receive a notification whenever my child logs in to the app so that I can stay informed of their usage patterns.

Notifications could also include usage summaries.

Access Parenting Resources

As a parent, I want access to parenting resources to learn strategies for improving my child's emotional resilience and handling challenging situations.

Track Mood

As a parent, I want to track my child's mood entries to identify emotional trends over time and intervene if necessary.

Mood tracking integrates visual trends for easy analysis.

Send alarm for daily goals

As a parent, I want to set daily goals for my child to encourage positive habits and create alarms to remind them about tasks or routines.

Profile Supervision

As a parent, I want to supervise my child's app usage, including viewing their game scoreboards and mood logs, to understand their progress and emotional state.

Mood Tracking child

As a child, I want to log my feelings daily so that I can express my emotions and see patterns over time.

The system offers mood options with emojis or descriptors for easy selection.

Play Therapy Games

As a child, I want to play interactive therapy games to learn coping skills while having fun.

Games may include challenges like breathing exercises, puzzle-solving, or mindfulness tasks.

View Scoreboard

As a child, I want to see my game results on a scoreboard to track my progress and feel motivated to improve.

Progress is displayed visually with rewards or badges for encouragement.

2.4.1 UC001: User Story Parent Sign-Up and Provide Information

User Story description of UC001

User story: Parent Sign-Up and Provide Information
ID: UC001
User Story Description As a parent, I want to sign up and provide required information so that I can access tailored features for my child and myself.
Flow of events: 1. Parent navigates to the sign-up page. 2. Parents enter their personal information (name, email, and password) and child-related information (child's name and age). 3. Parents submit the form by clicking the "Sign Up" button. 4. System sends a confirmation email to the provided email address. 5. Parents verify their email by clicking the link in the confirmation email. 6. System confirms successful account creation and redirects the parent to the parent assessment.
Alternative flow : 1. If the parent does not verify the email within the specified time or the link expires, the system displays an error message. 2. Parents can request a new confirmation email.
Acceptance Criteria Precondition: Parent's account is successfully created and email verified. Postcondition: 1. Parent provides valid details. 2. Parent's email is active and capable of receiving confirmation emails. Other Conditions: The system sends a confirmation email.
Exception flow: 1. If the parent fails to complete any mandatory fields, the system displays an error message indicating the missing information. 2. Account creation is blocked until all required fields are provided.

Table 2.4.1: User Story Description for Parent Sign-Up and Provide Information

System Sequence Diagram of UC001

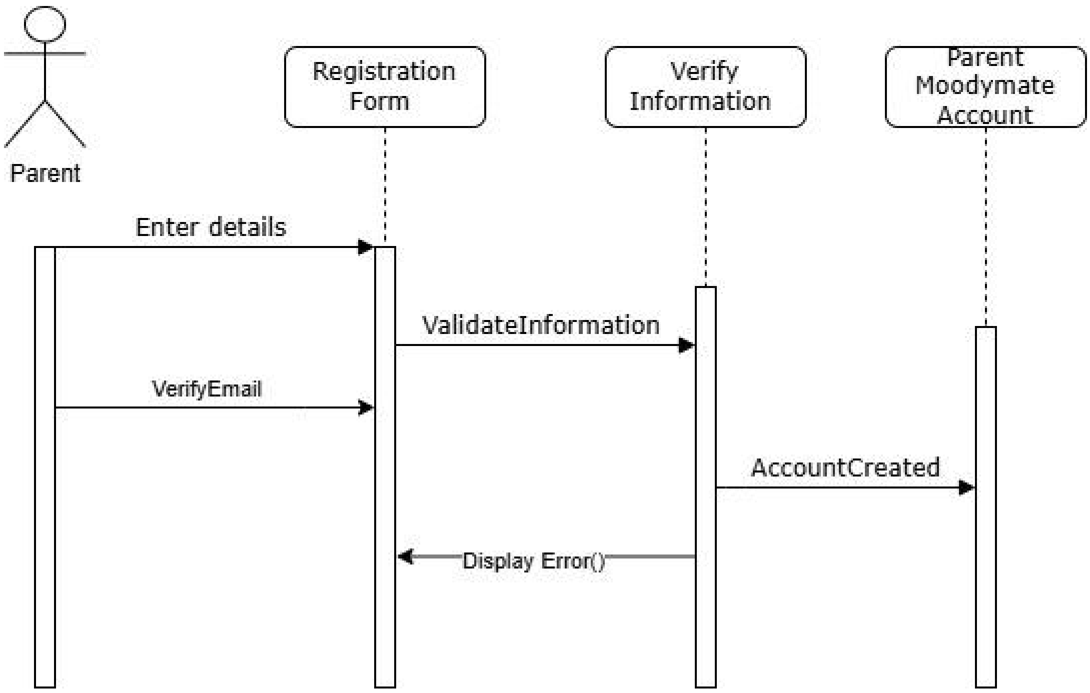


Figure 2.4.1: Sequence diagram for UC001

Activity Diagram of UC001

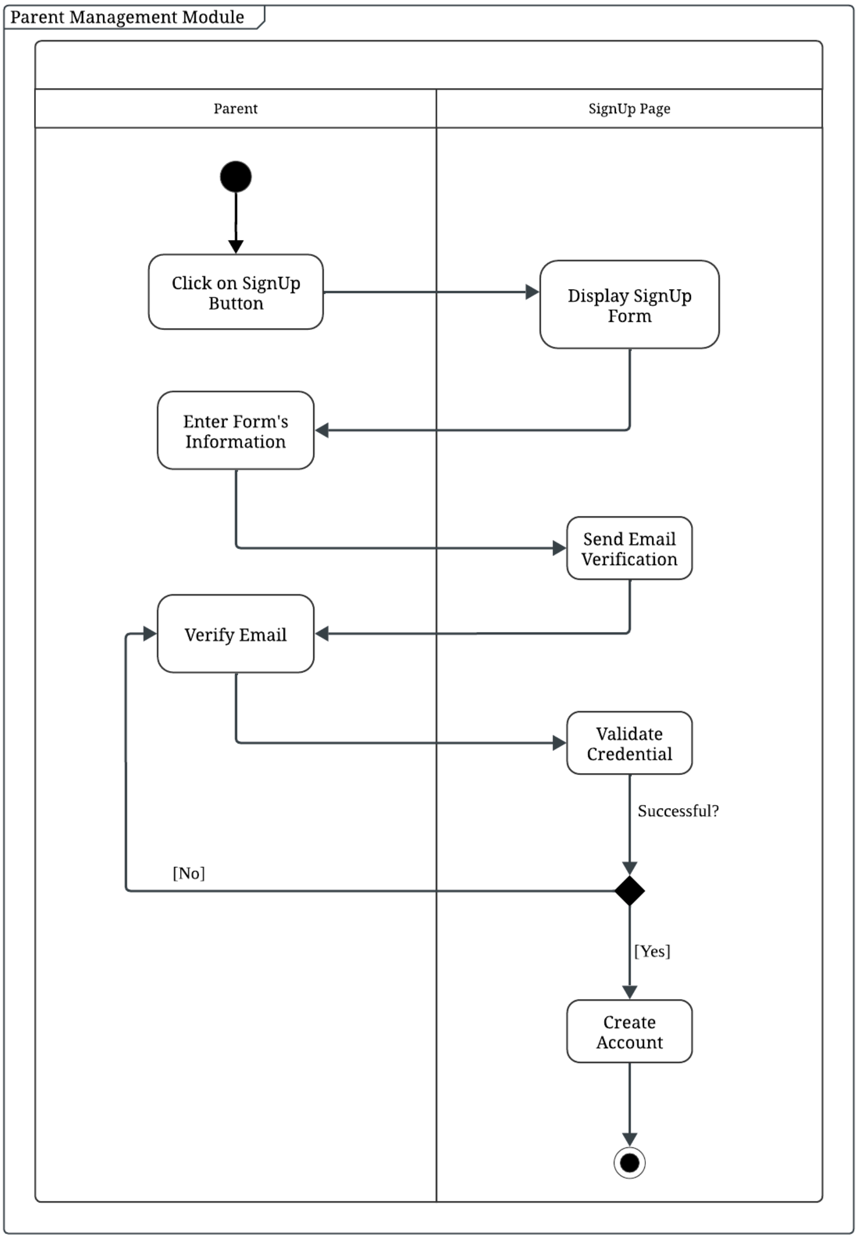


Figure 2.4.1.1: Activity Diagram for <<Parent SignUp>>

2.4.2 UC002: User Story Child Login Using PIN

User Story description of UC002

User story: Description for Child Login
ID: UC002
User Story Description As a child, I want to log in using a PIN so that I can securely access the app under parental supervision.
Flow of events: 1. The child opens the app and selects the "Child" option. 2. The system displays the Child Login Page, including a title "Child Login", an input field labeled "SECURE PIN" and a "LOGIN" button. 3. Child types the assigned PIN into the input field. 4. The system receives the PIN input and begins validation. 5. The system compares the entered PIN against the one stored in the database for the child's account: - If the PIN matches, the system logs the child into their account. - If the PIN is incorrect, the system triggers the Alternative Flow. 6. The system displays a mood tracking interface.
Alternative flow : 1. If the child enters an incorrect PIN: -The system displays an error message in red: "Incorrect PIN. Please try again." -The child is allowed to retry up to 5 times. 2. If the child re-enters the correct PIN within the allowed attempts, the system proceeds with the normal login flow.
Acceptance Criteria Postcondition: Child is logged in successfully. Precondition: Valid PIN is entered. Other Conditions: The system informs the parent of the login.
Exception flow: 1. If the child enters an incorrect PIN 5 times consecutively, the system temporarily locks their account. - The screen displays the message: "ACCOUNT LOCKED DUE TO MULTIPLE FAILED ATTEMPTS. PLEASE CONTACT YOUR PARENT".

Table 2.4.2: User Story Description for Child Login

Activity Diagram of UC002

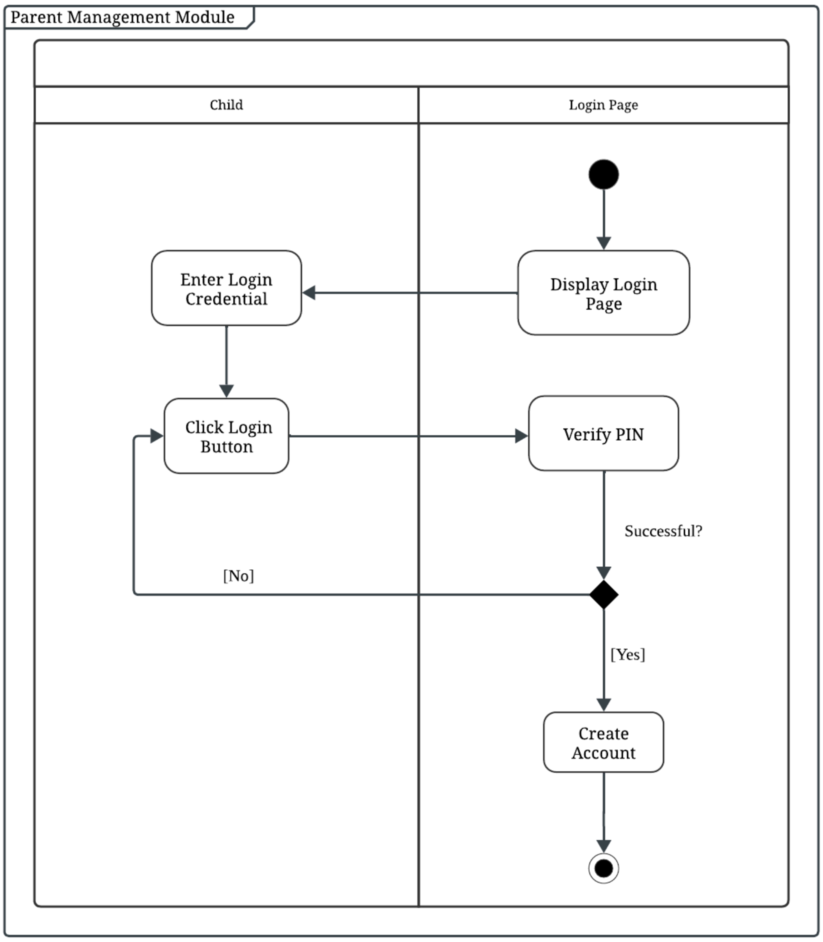


Figure 2.4.2: Activity Diagram for <<Child Login Using PIN>>

System Sequence Diagram of UC002

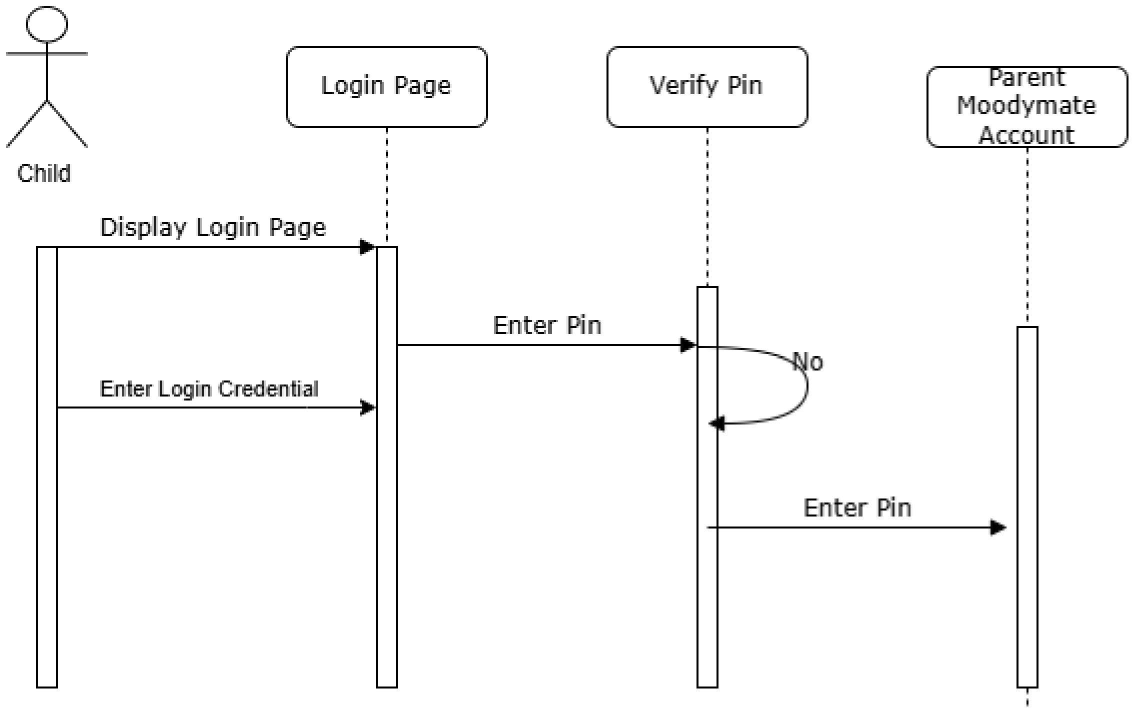


Figure 2.4.2: Sequence diagram for UC002

2.4.3 UC003: User Story Play Therapy Games

User Story description of UC003

User story:Play Therapy Games
ID: UC003
User Story Description **As a child,** **I want to play therapy games** **so that I can learn coping skills in an engaging and fun way.**
Flow of events: 1. Child Log in to the application using pin 2. Child choose mood before proceed to therapy game page 3. Child chooses a therapy game. 4. Child selects a therapy game. 5. System starts the selected game. 6. Child completes the game. 7. System displays the score 8. System send the score to parent.
Acceptance Criteria Postcondition: Game score is displayed and saved. Precondition: Child logs into the app successfully and logs their mood Other Conditions: The parent can view the scoreboard.
Exception flow: If the game crashes, the system saves the progress and let them continue playing

Table 2.4.3: User Story Description Play Therapy Games

Activity Diagram of UC003

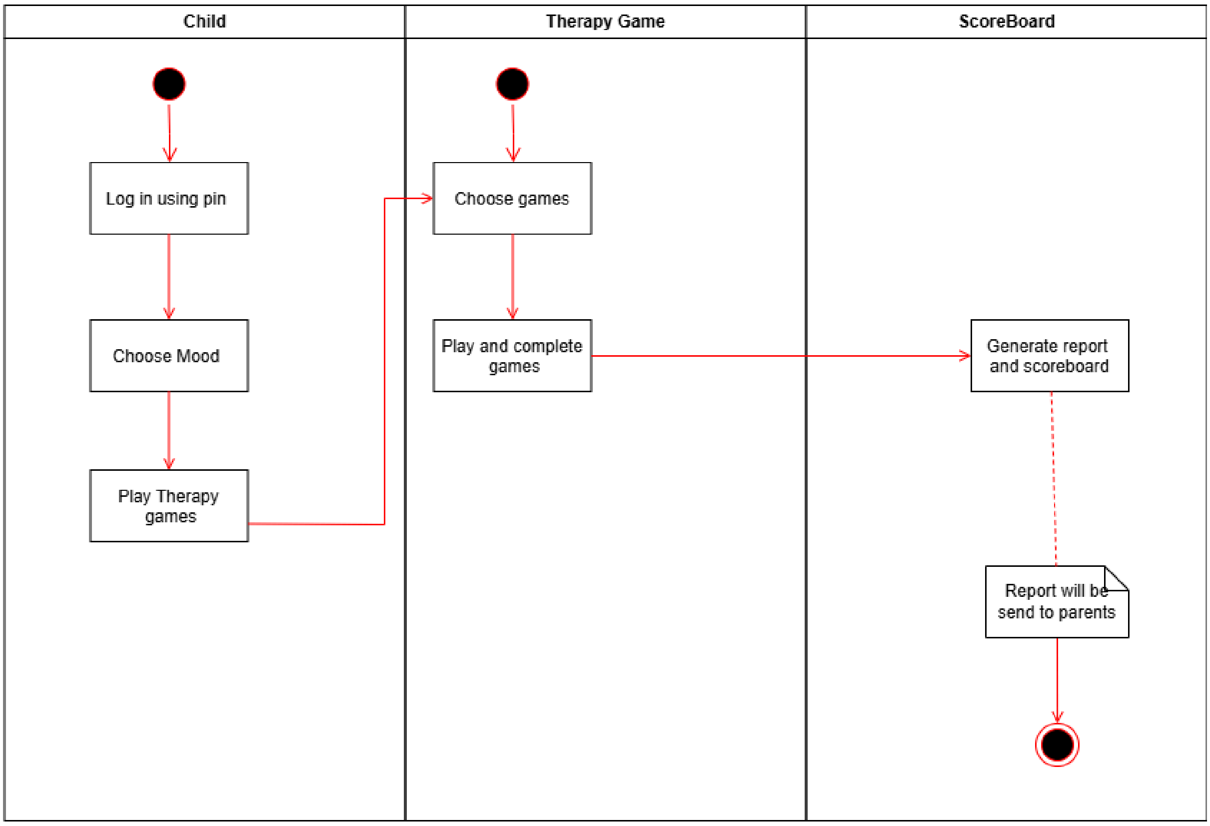


Figure 2.4.3: Activity diagram for UC003

System Sequence Diagram of UC003

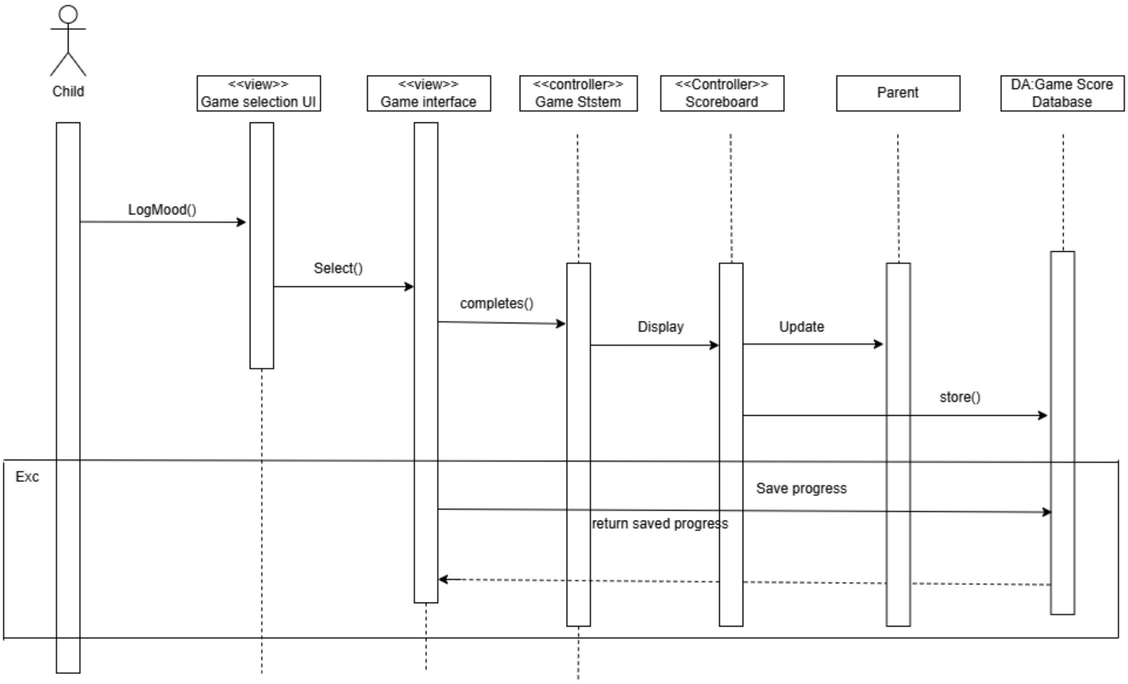


Figure 2.4.3: Sequence diagram for UC003

2.4.4 UC004 User Story Notifications for Child App Usage

User Story description of UC004

User story: Notifications for Child App Usage
ID: UC004
User Story Description **As a parent,** **I want to receive notifications when my child accesses the app** **so that I can monitor their usage.**
Flow of events: 1. Child logs into the app and the key in the pin, the parent will receive notification . 2. child choose mood , parent will receive notification 3. child play game, and game report will be send to parent 4. System sends a notification to the parent. 5. Parent views the notification.
Alternative flow :If the parent disables notifications, the system logs the child's login activity instead.
Acceptance Criteria Postcondition: Notification is sent and logged. Precondition: Parent has enabled notifications. Other Conditions: Notifications include date and time of access.
Exception flow: If the system fails to send a notification, it retries automatically

Table 2.4.4: User Story Description Play Therapy Games

Activity Diagram of UC004

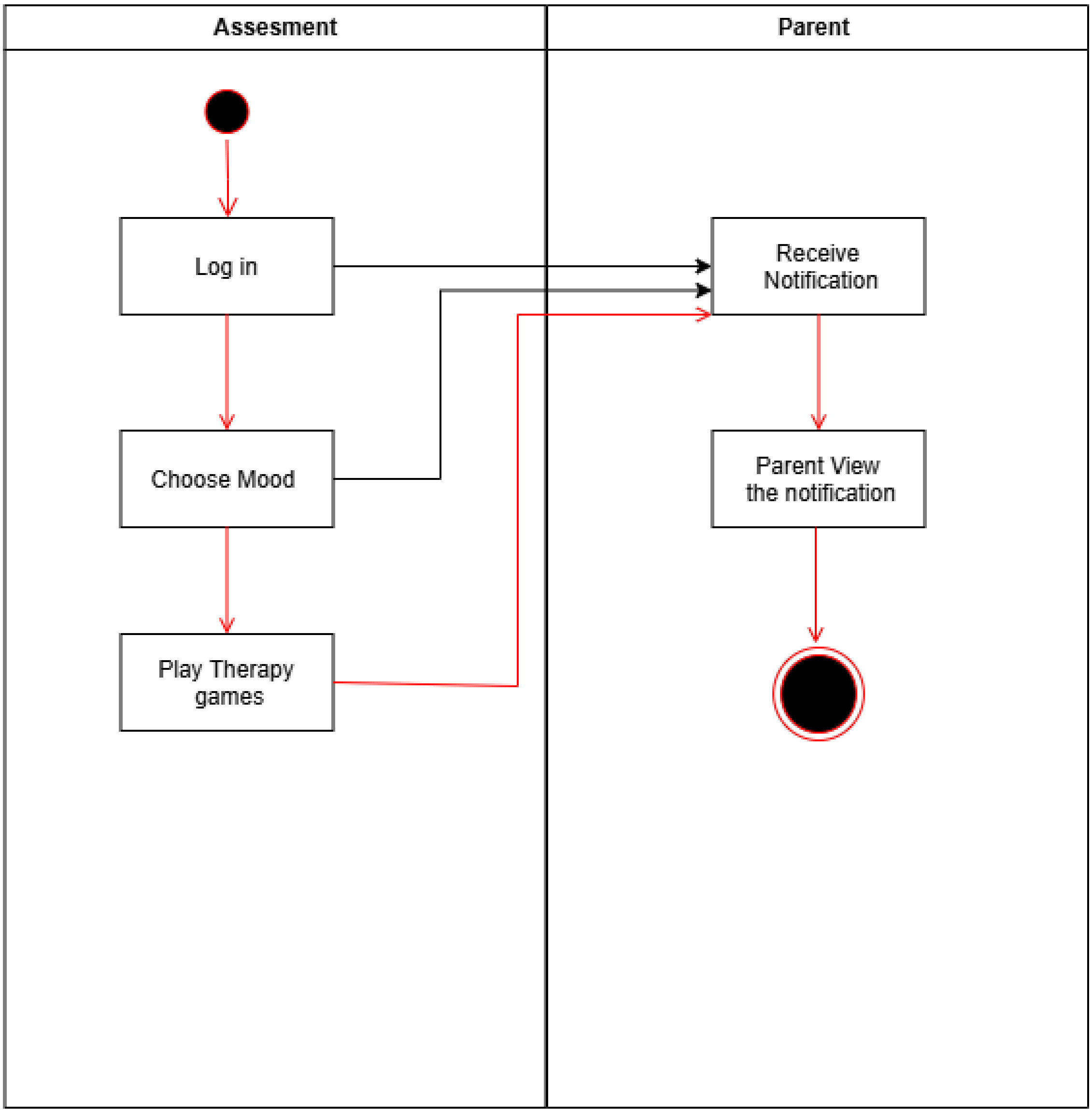


Figure 2.4.4 activity diagram UC004

System Sequence Diagram of UC004

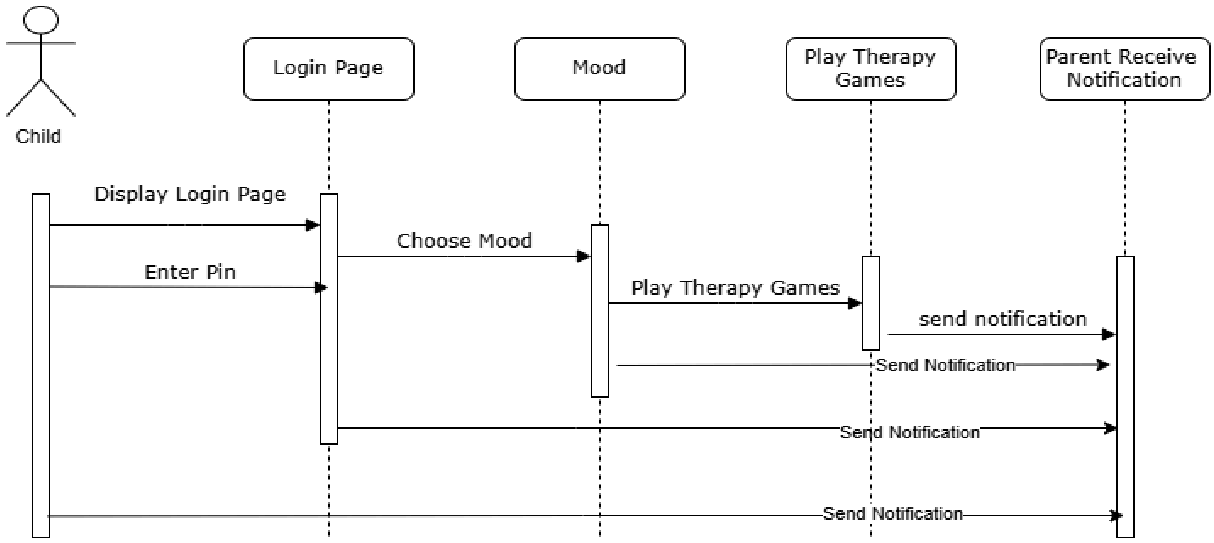


Figure 2.4.4: Sequence diagram for UC004

2.4.5 UC005: User Story Parent Assessment

User story: Parent Assessment
ID: UC005
User Story Description **As a parent,** **I want to complete a mental health assessment** **so that I can better understand my emotional state and its impact on my child.**
Flow of events: 1. Parents navigate to the “Parent assessment” page. 2. System displays the assessment questionnaire. 3. Parents complete and submit the questionnaire. 4. System analyzes the responses and provides feedback. 5. 5. System saves progress and allows the user to resume later if there exists mid-assessment.
Acceptance Criteria Postcondition: Feedback on the parent’s mental health is provided. Precondition: Parent completes the assessment questionnaire. Other Conditions: The feedback is stored securely for reference.
Exception flow: If a user leaves the page for a long time, the system will prompt the user to login again and previously entered data is saved as draft.

Table 2.4.5: User Story Description Parent assessment

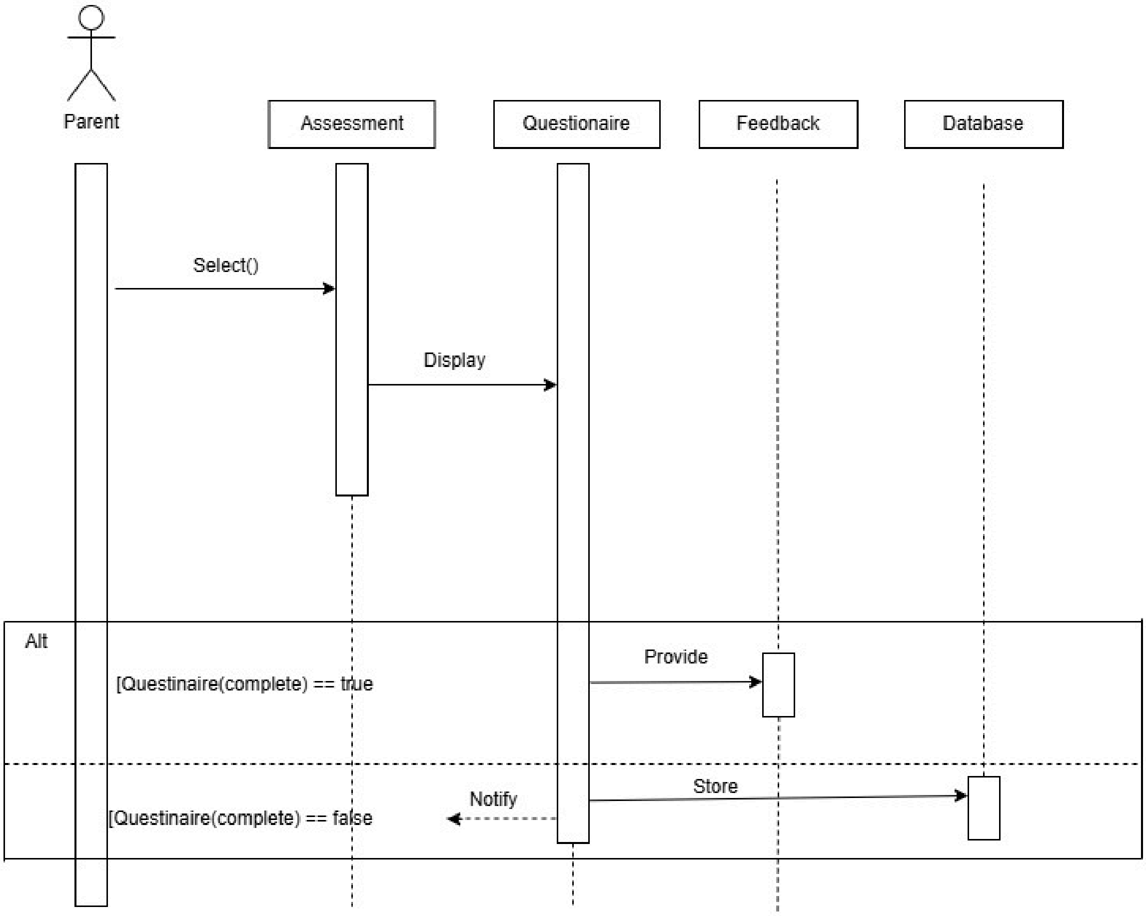


Figure 2.4.5: Sequence diagram for UC005

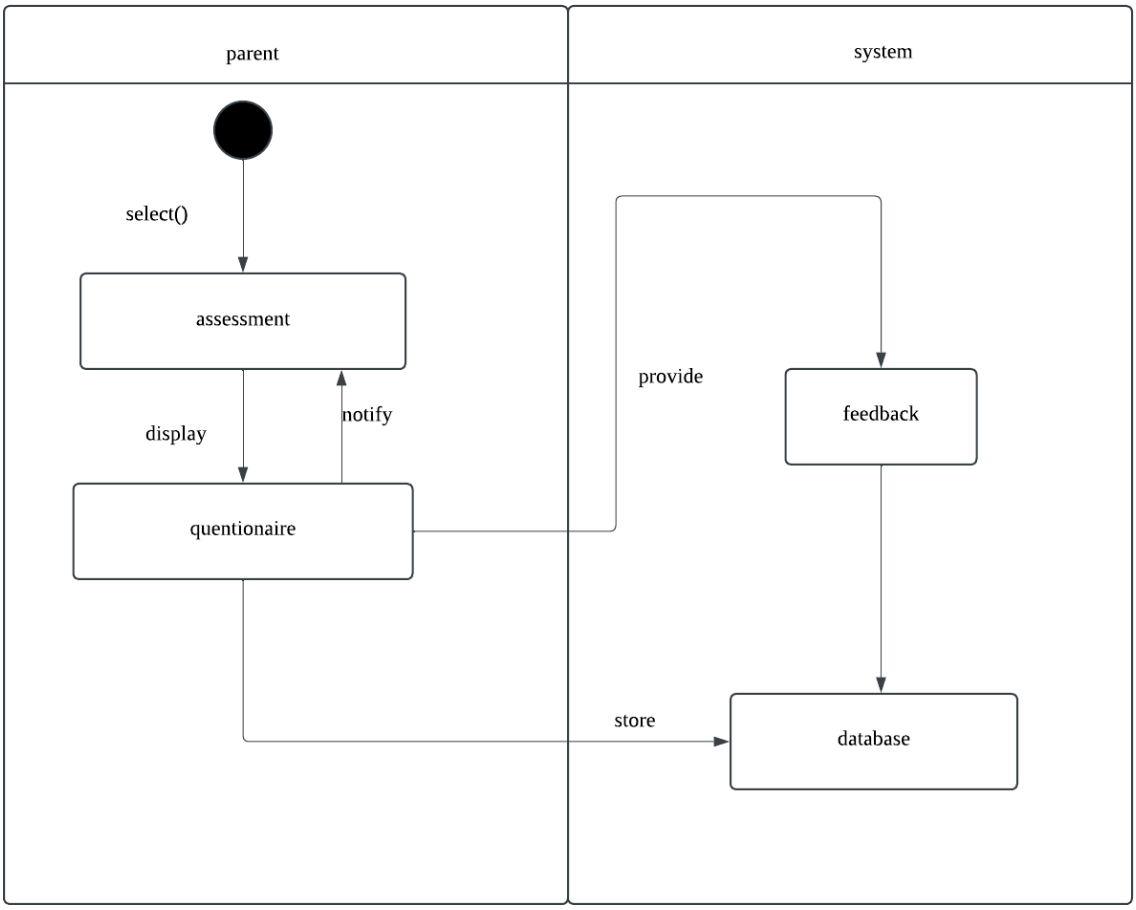


Figure 2.4.5: Activity diagram for UC005

2.4.6 UC006 User Story Parenting Resources

User story: Parenting Resources
ID: UC006
User Story Description As a parent, I want access to parenting resources so that I can learn techniques to support my child's mental health.
Flow of events: 1.Parent accesses the parenting resources page and is presented with a list of available resources. 2.Resources are displayed in a sequence, and each module is locked until the previous resource is completed. 3.Parent selects a resource to begin with. 4.System displays the resource content, which includes text. 5.After finishing the resource content, the parent is presented with a quiz based on the material covered in the resource. 6.The parent answers the quiz questions. 7.If the parent answers enough questions correctly, the resource is marked as completed, and the next resource becomes unlocked. 8.The system records the completion of the resource and updates the parent's progress. 9.The parent can now move on to the next unlocked resource.
Alternative flow : 1. If the parent cannot finish the quiz in one session, their progress is saved, and they can resume where they left off.
Acceptance Criteria Precondition: The parent has accessed the parenting resources section and is starting with the first resource in the sequence. Postcondition: The parent successfully completes the resource and quiz, and the next resource is unlocked. Other Conditions: The quiz is an integral part of unlocking each subsequent resource, and it must ensure the parent has absorbed key information.
Exception flow: 1. If module content fails to load (e.g., text is missing), the system retries loading the content.

Table 2.4.6: User Story Description Parenting resource

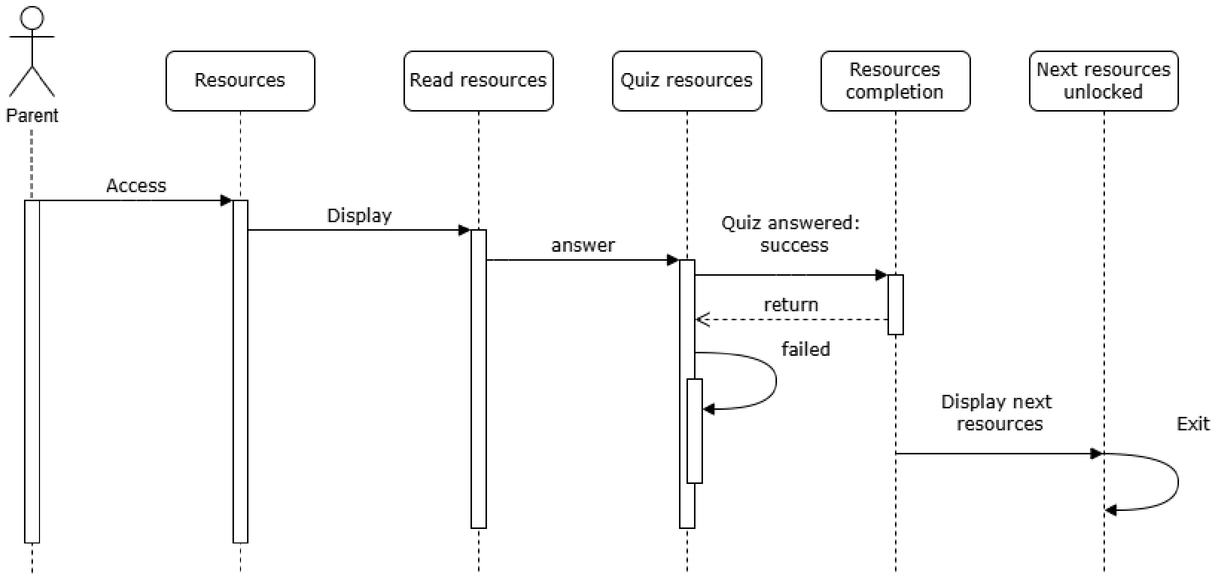


Figure 2.4.6: Sequence diagram for UC006

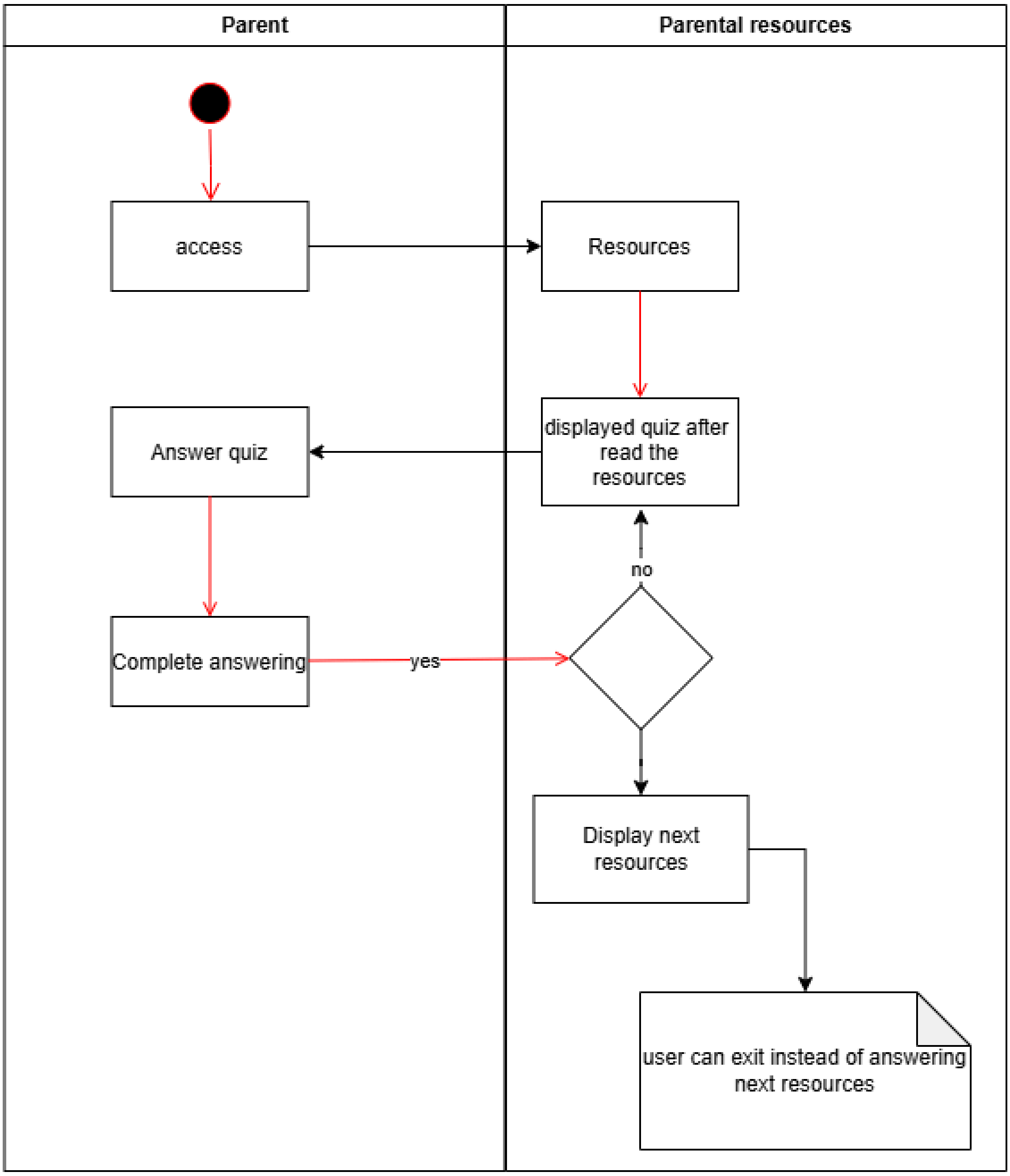


Figure 2.4.6: Activity diagram for UC006

2.4.7 UC007 User Story Child Mood Tracker

User story: Child Mood Tracker
ID: UC007
User Story Description **As a child,** **I want to log my mood daily** **so that I can understand my emotions better over time**
Flow of events: 1. Child navigates to the “Mood tracking” page. 2. System displays mood options (e.g., emojis or sliders). 3. Child selects a mood from the list and clicks submit.. 4. System stores the mood entry in the child’s profile. 5. System reminds users to log their mood if they skip a day.
Acceptance Criteria Postcondition: Mood entry is saved successfully. Precondition: Child accesses the app using their PIN. Other Conditions: Parent can view the child’s mood log.
Exception flow: If user select a mood and attempt to save it during a simulated server outage, system displays an error message indicating a submission failure and prompt user to retry again.

Table 2.4.7: User Story Description Child Mood Tracker

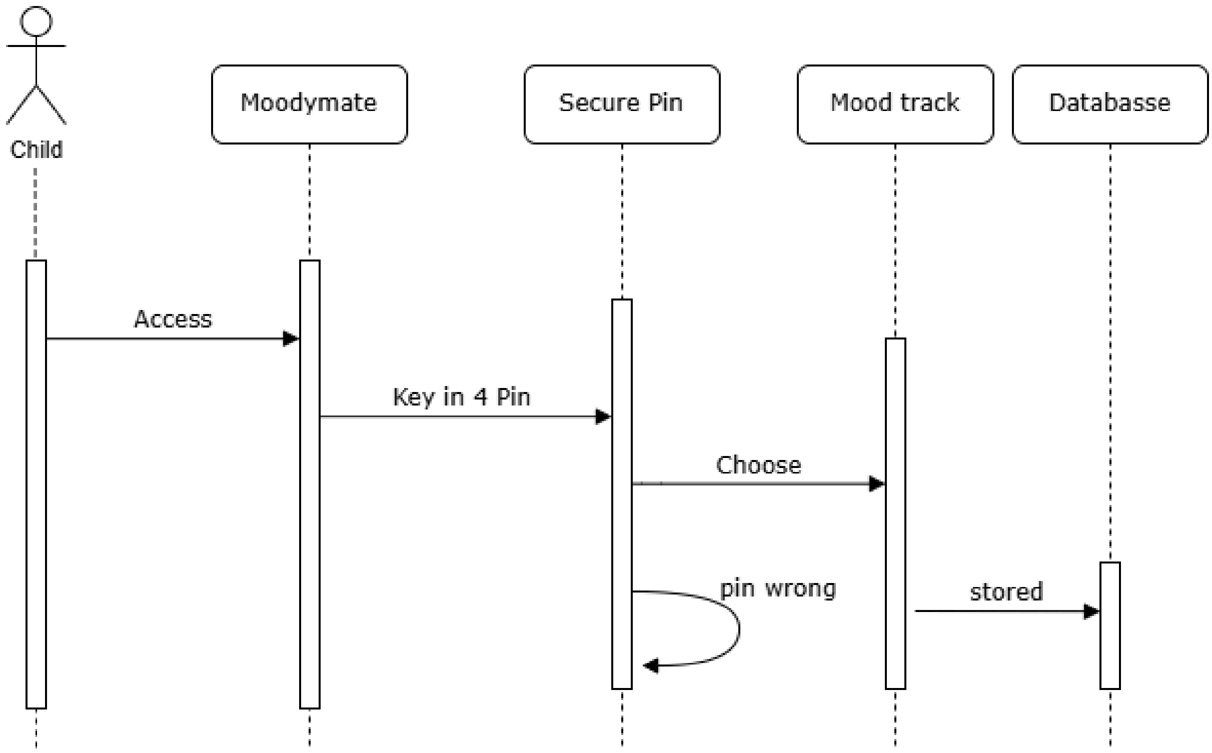


Figure 2.4.7: Sequence diagram for UC007

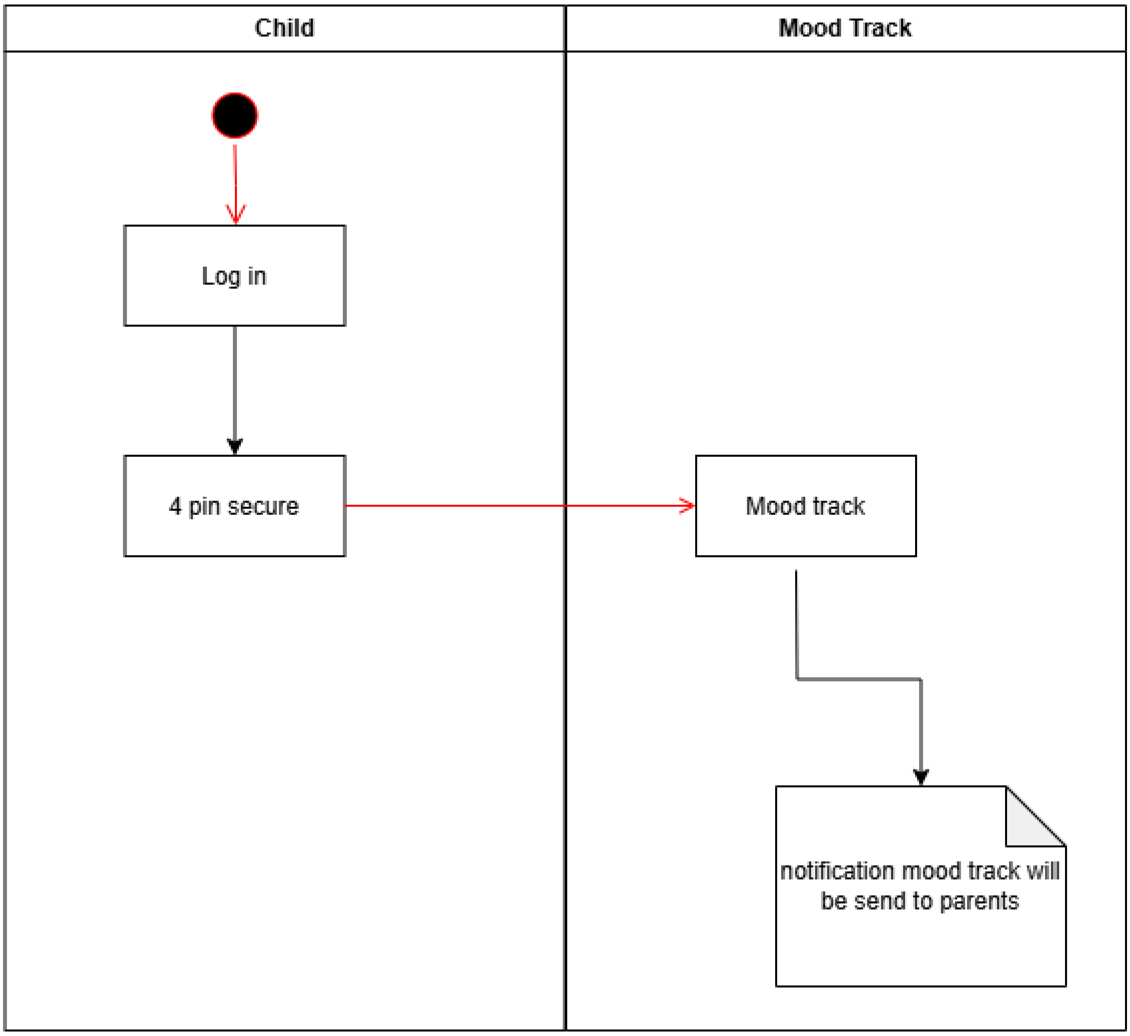


Figure 2.4.7: Activity diagram for UC007

2.4.8 UC008 User Story Alarm for Daily Goals

User story: Alarm for Daily Goals
ID: UC008
User Story Description **As a parent,** **I want to set an alarm for daily goals** **so that I can establish positive habits for my child.**
Flow of events: 1. Parent selects the alarm feature. 2. System prompts the parent to set a time and description for the alarm. 3. System saves and schedules the alarm. 4. Alarm notification triggers at the set time.
Alternative flow :If the parent doesn't set a specific time, the system suggests default time options.
Acceptance Criteria Postcondition: Alarm is triggered as scheduled. Precondition: Parent successfully sets an alarm. Other Conditions: Alarms can be edited or deleted by the parent.
Exception flow: If the alarm fails to trigger, the system logs the issue and alerts the administrator.

Table 2.4.8: User Story Description Alarm for Daily Goals

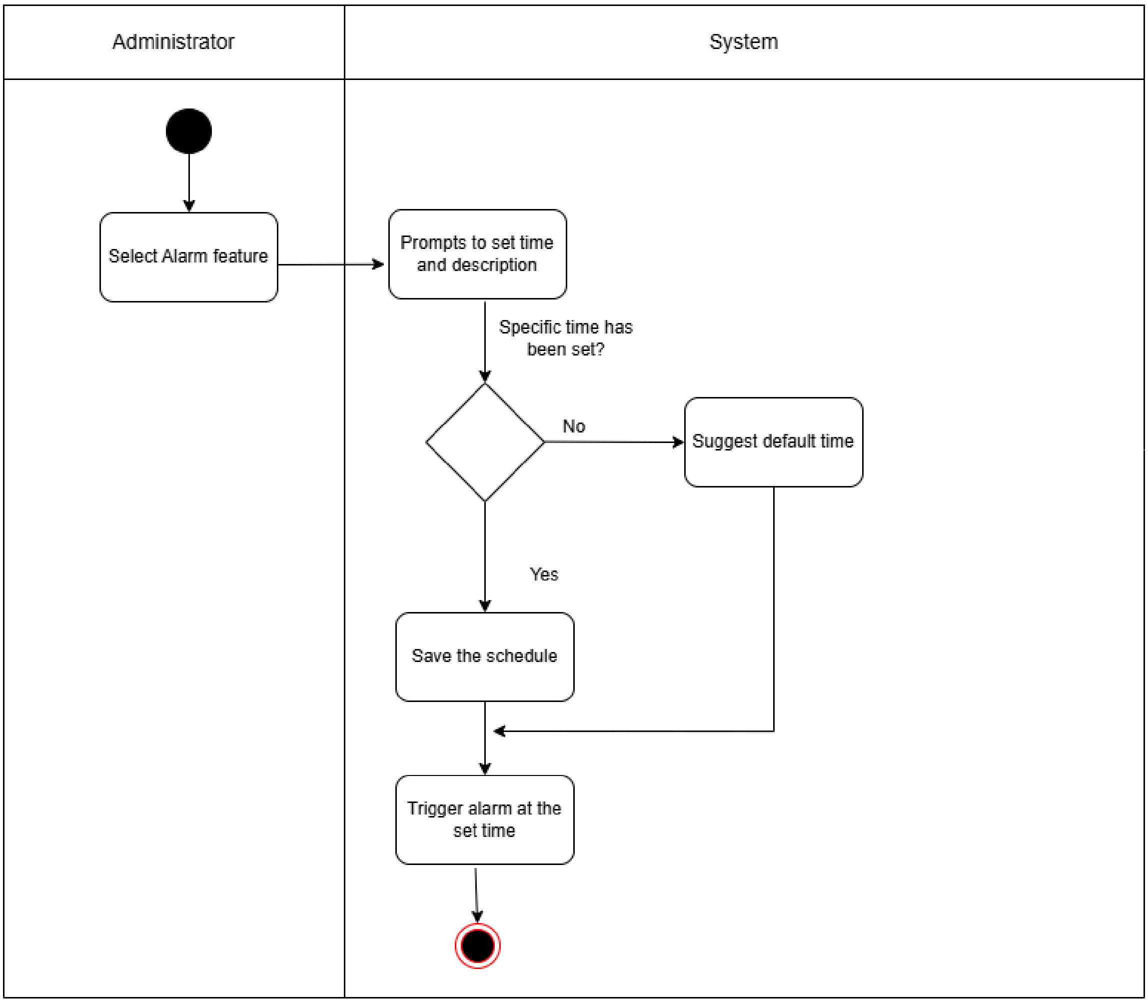


Figure 2.4.8 Activity diagram for UC008

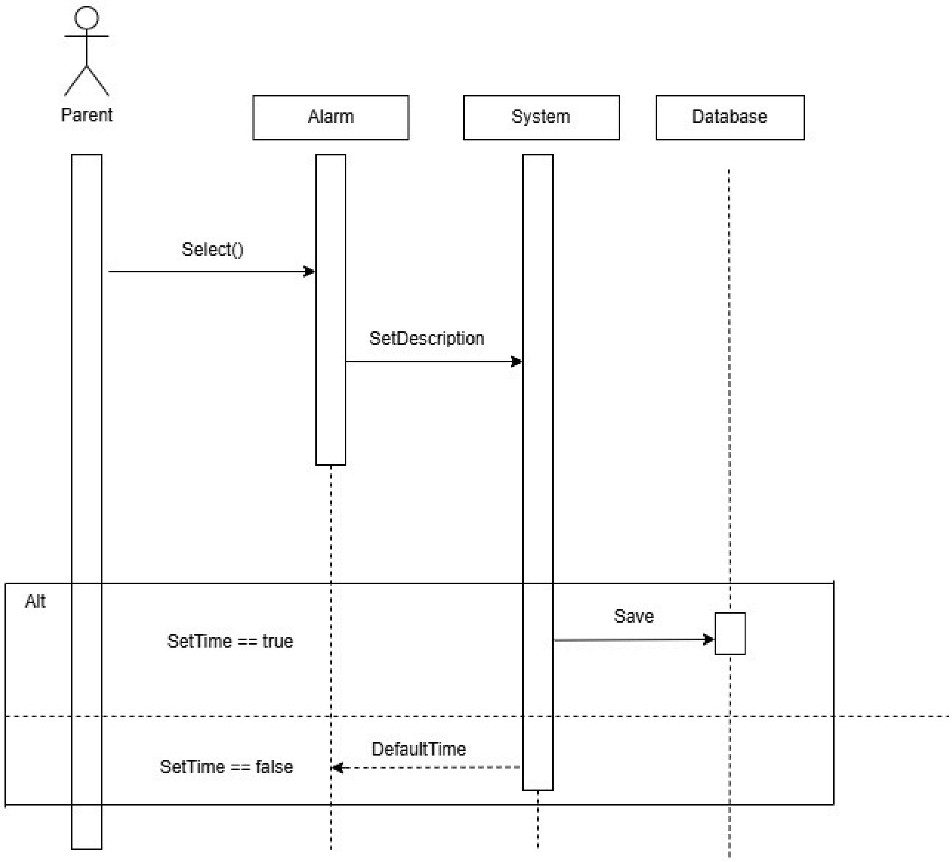


Figure 2.4.8: Sequence diagram for UC008

2.4.9 UC009 User Story : View Game Report based on Scoreboard

User story: View Game Report based on Scoreboard
ID: UC009
User Story Description **As a parent,** **I want to view my child's therapy game scores** **so that I can monitor their engagement and progress.**
Flow of events: 1. Child completes a therapy game. 2. System saves the game score. 3. Parents access the scoreboard feature. 4. System displays the child's game scores and progress chart.
Alternative flow :If no game has been completed, the system notifies the parent that no scores are available yet.
Acceptance Criteria Postcondition: Game scores are displayed to the parent. Precondition: Child plays therapy games. Other Conditions: Scores are updated in real-time after each game.
Exception flow: If the scoreboard fails to load, the system retries and notifies the parent of any errors.

Table 2.4.9: User Story DescriptionView Game Report based on Scoreboard

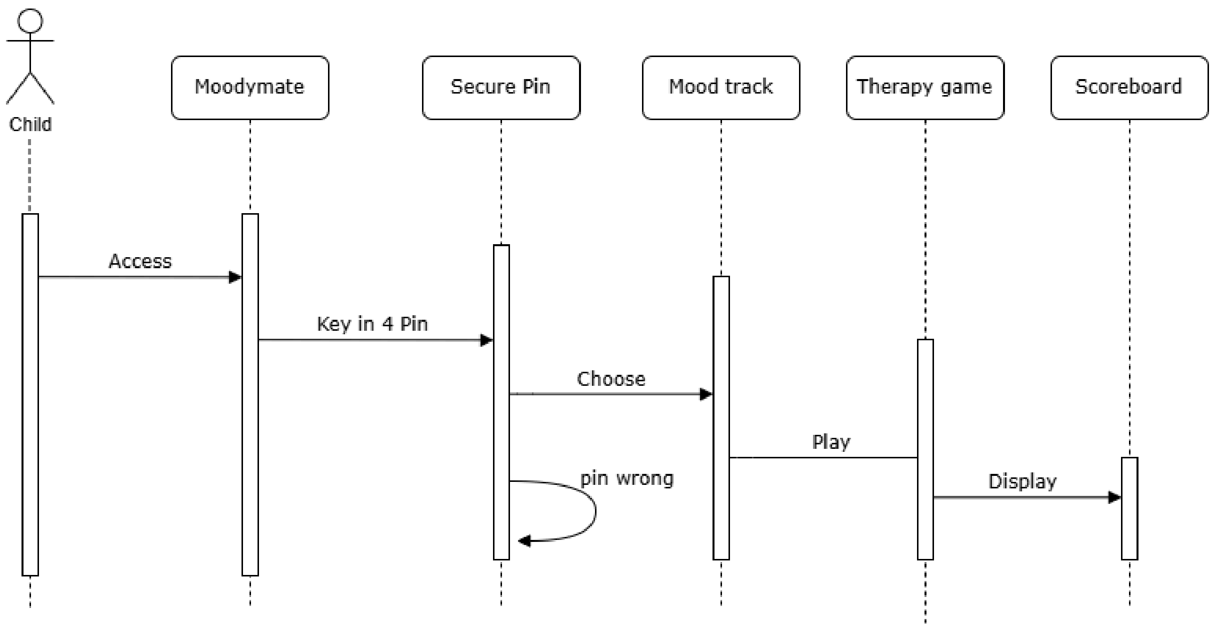


Figure 2.4.9: Sequence diagram for UC009

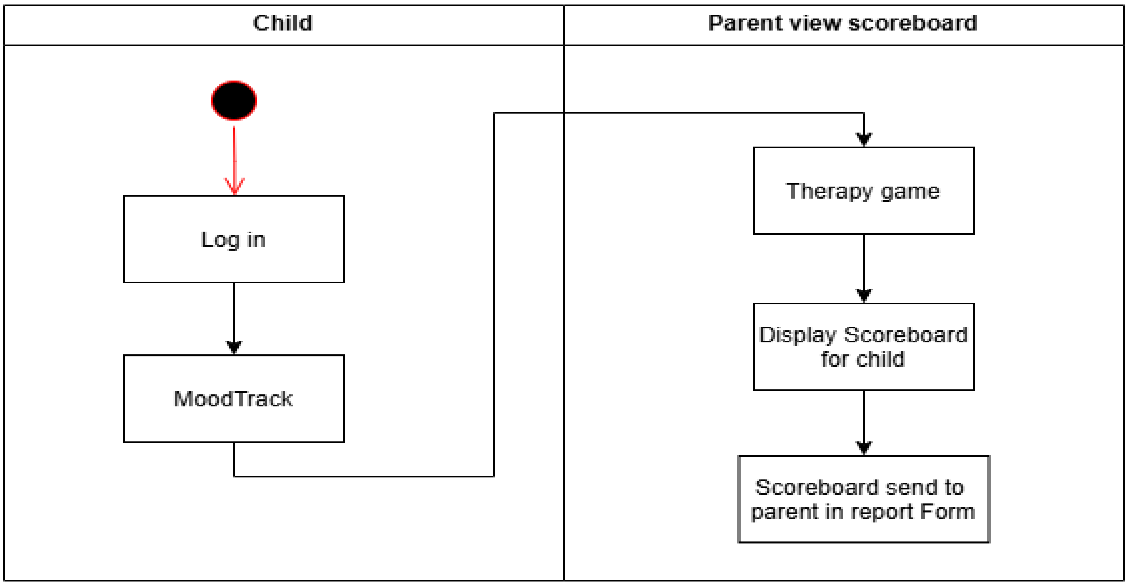


Figure 2.4.9: Activity diagram for UC009

2.4.10 UC010: User Story Access Online Therapy Videos

User story: Access Online Therapy Videos
ID: UC010
User Story Description **As a parent,** **I want access to online therapy videos** **so that I can follow guided sessions for mental health improvement.**
Flow of events: 1. Parent selects the therapy video feature. 2. System displays a list of therapy videos categorized by topic. 3. Parent selects a video to watch. 4. System streams the video and tracks progress.
Acceptance Criteria Postcondition: Video is streamed successfully, and progress is tracked. Precondition: Parent has an active internet connection. Other Conditions: Video completion is logged in the parent's profile.
Exception flow: If the video fails to play, the system offers troubleshooting tips or alternative videos.

Table 2.4.10: User Story Description Access Online Therapy Videos

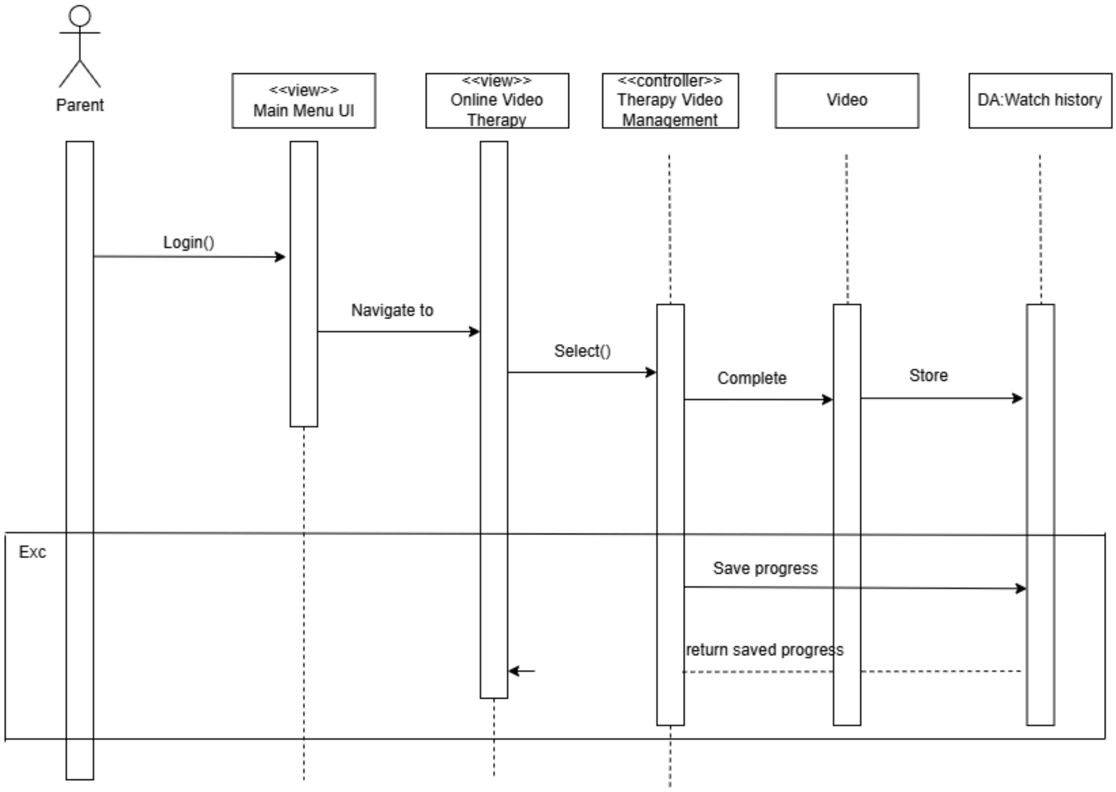


Figure 2.4.10: Sequence diagram for UC010

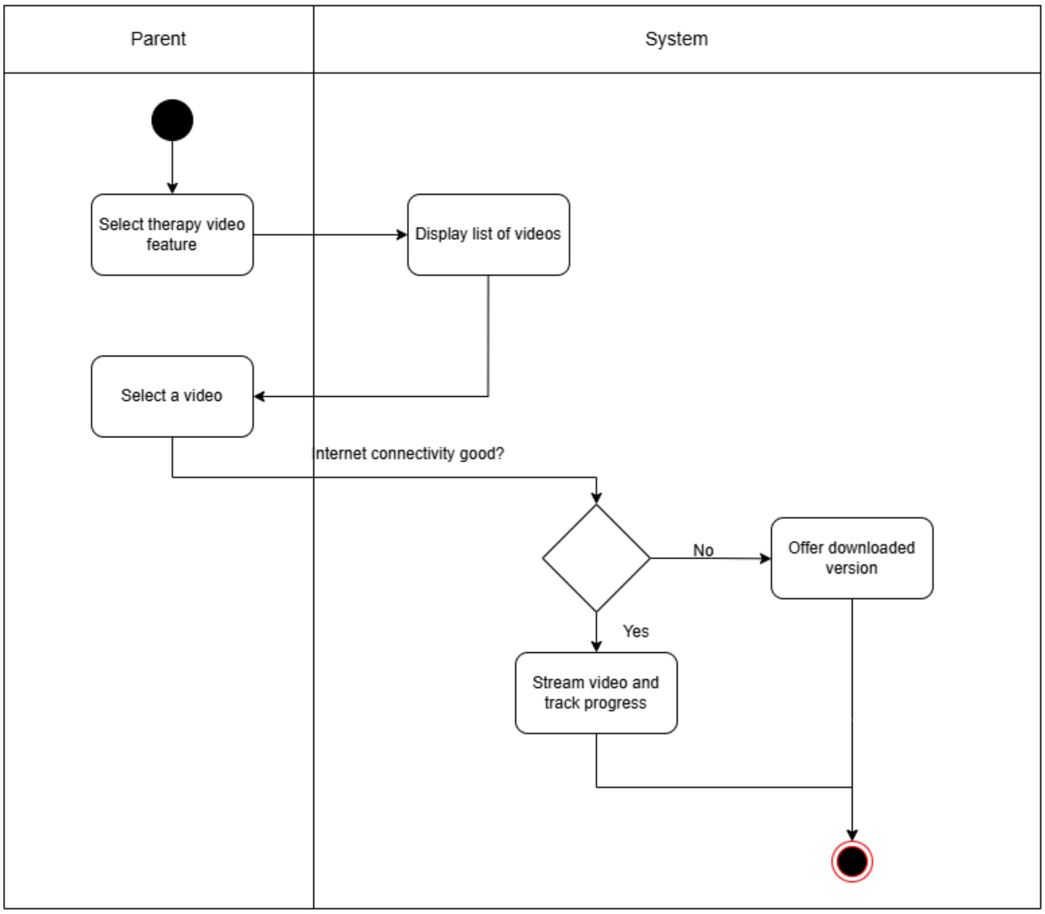


Figure 2.4.10 Activity diagram for UC010

2.4.11 UC011: User Story Profile Supervision

User story: Profile Supervision
ID: UC011
User Story Description As a parent, I want to monitor my child's profile activity so that I can ensure they are using the app safely and effectively.
Flow of events: 1. Parent logs into their account. 2. System displays the child's activity summary. 3. The system retrieves and displays relevant information about the child's activity, including mood logs and game scores. 3. Parent can review the displayed data, ensuring proper use and engagement.
Alternative flow :

1. If no activity has been logged for the child, the system will notify the parent and suggest actions such as encouraging the child to use the app, checking if there are any issues with app usage.
Acceptance Criteria Postcondition: Parent successfully reviews the child's activity summary. Precondition: Parent has access to the supervision module. Other Conditions: Activity data is updated in real time.
Exception flow: 1. If there is an issue loading the activity data, the system will either retry loading the data or notify the administrator of the failure.

Table 2.4.11: User Story Description Profile supervision

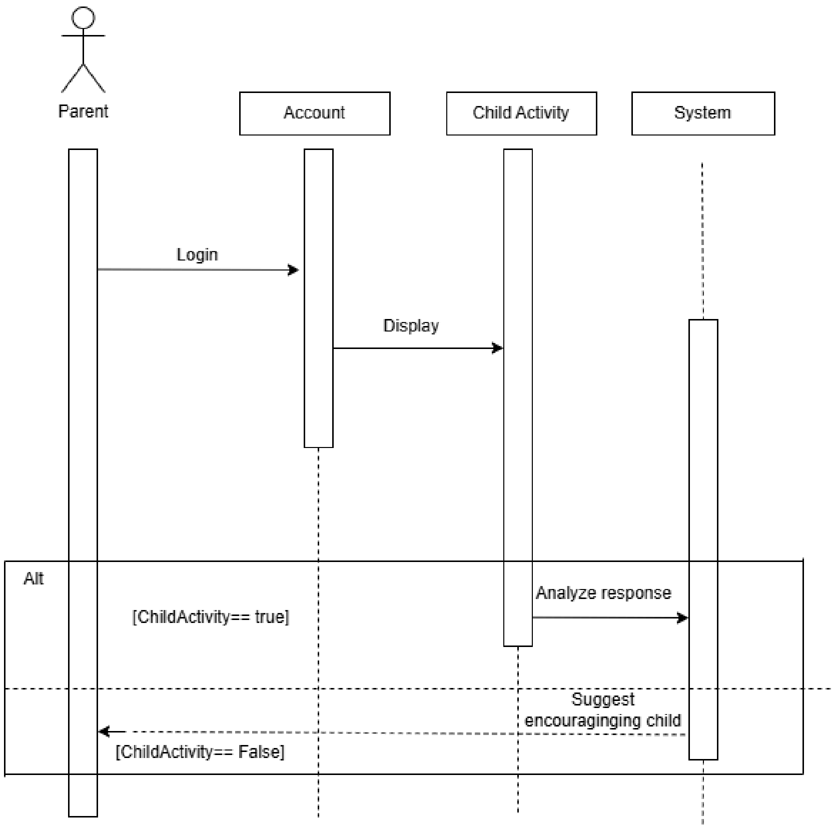


Figure 2.4.11: Sequence diagram for UC011

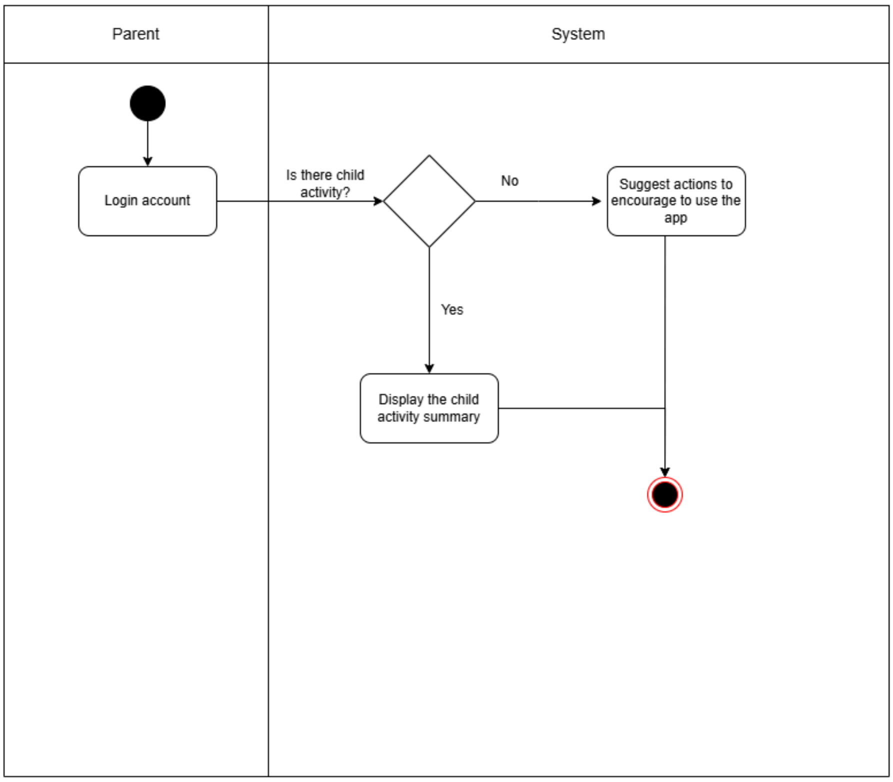


Figure 2.4.11 Activity diagram for UC011

2.5 Performance and Other Requirements

- <u>Software System Attributes</u>

  i) Usability

- -The interface of the system should be developed for the range of distance education users from 9 to 15 years old and parents of such children.

  ii) Reliability

- -Each component should be able to fail and recover within 5 minutes of a crash or data loss with little to no loss of the user's data.

  iii) Maintainability

- -The system must be quite simple or amenable to being updated with new features or to fix bugs, this should not take longer than 2 hours of update downtime.

iv) Portability

- -The system should operate on Windows, MacOS, iOS, and Android operating systems.

v) Compatibility

- -To work in parallel seamlessly with healthcare systems, wellness platforms, and teletherapy services, the system has to use APIs of similar standards.

Performance Requirements

i) Response Time

- -Performance: Mental health assessment results must be processed and displayed within 2 seconds since the final user session is done.
- -Parental monitoring dashboard should take no more than 3 seconds up to perusing on all devices, whether it is a PC, laptop, tablet, or phone.
-

ii) Throughput

- -It should be noted that the target audience of the system is the 500 concurrent users who perform the assessment or viewing the resources with no degradation in performance.

iii) Capacity

- -The system has to incorporate a database that encompasses profiles and mental health information for at least 10,000 users, although it should consider prospects of increasing the upper limit of the users to 100,000 users in the future.

iv) Availability

- -The system should be made available 99.9% availability to the users, and any drag down for maintenance should be done during off time.

Other Requirements

i) Security

- -All user data including the results of the assessment and the data about the user must be transmitted (using TLS) and securely.
- -This is because the back-end has sensitive information relying on multi-factor authentication (MFA) as the minimum requirement for any administrator.
- -In order to ensure that user's privacy is protected the system should be governed by the Personal Data Protection Act 2010 (PDPA) in Malaysia.

ii)Safety

- -There also needs to be prompt and obvious statements that the system isn't a substitute for a professional mental healthcare provider and users should seek one if symptoms worsen.
- -For the purpose of the assessments, the emergency contacts and hotlines should be provided and placed where the users displaying the signs of high risk are likely to notice.

- iii)Legal and Regulatory
- -The system needs to conform to local laws and regulate spare parts of the Malaysian PDPA that relates to handling of user sensitive data.
- -Information regarding mental health and all related resources must be moderated to guarantee the material's credibility and compliance with the Malaysian Clinical Practice Guidelines.

- iv) Environmental
- -The system should try to reduce its carbon footprint by using cloud services that will be supported by energy efficient data centers as well as renewable energy.

2.6 Design Constraints

1. **Platform compatibility:**

   The system must support both mobile and web platforms to ensure accessibility across devices.

2. **Age-specific design:**

   The user interface and functionality should be suitable for young Malaysians aged 9 to 15, with additional supervision tools for parents.

3. **Privacy and Security**:

   - · User data must be securely stored and comply with local data protection regulations due to the sensitive nature of mental health information.
   - · The system should include secure messaging for user communication.

4. **Integration Capabilities**:

   - The system should integrate with external platforms such as healthcare systems, educational portals, and employee wellness platforms.

5. **Scalability**:
   - The system must handle increasing user demand while maintaining performance and reliability.

6. **Low Cost for Users**:
   - The system must remain accessible, preferably free or with minimal costs, to encourage widespread use.

7. **Interactive and Educational Features**:
   - Features such as storytelling cards and learning modules must be engaging and educational, aligning with cognitive development principles.

8. **Parental Supervision**:
   - Tools for monitoring and insights should be non-intrusive but provide valuable data on children's mental health trends.

9. **Cultural Relevance**:
    - Content and features should be tailored to the cultural context and needs of Malaysian users.

3 Architectural Rationale

3.1 Architectural Style and Rationale

The MoodMate system is designed using a layered architecture to ensure a modular, scalable, and maintainable structure. This architecture divides the system into distinct layers, each with specific responsibilities. This separation of concerns enhances flexibility and supports future feature expansion while ensuring compliance with design constraints such as privacy, security, and platform compatibility.

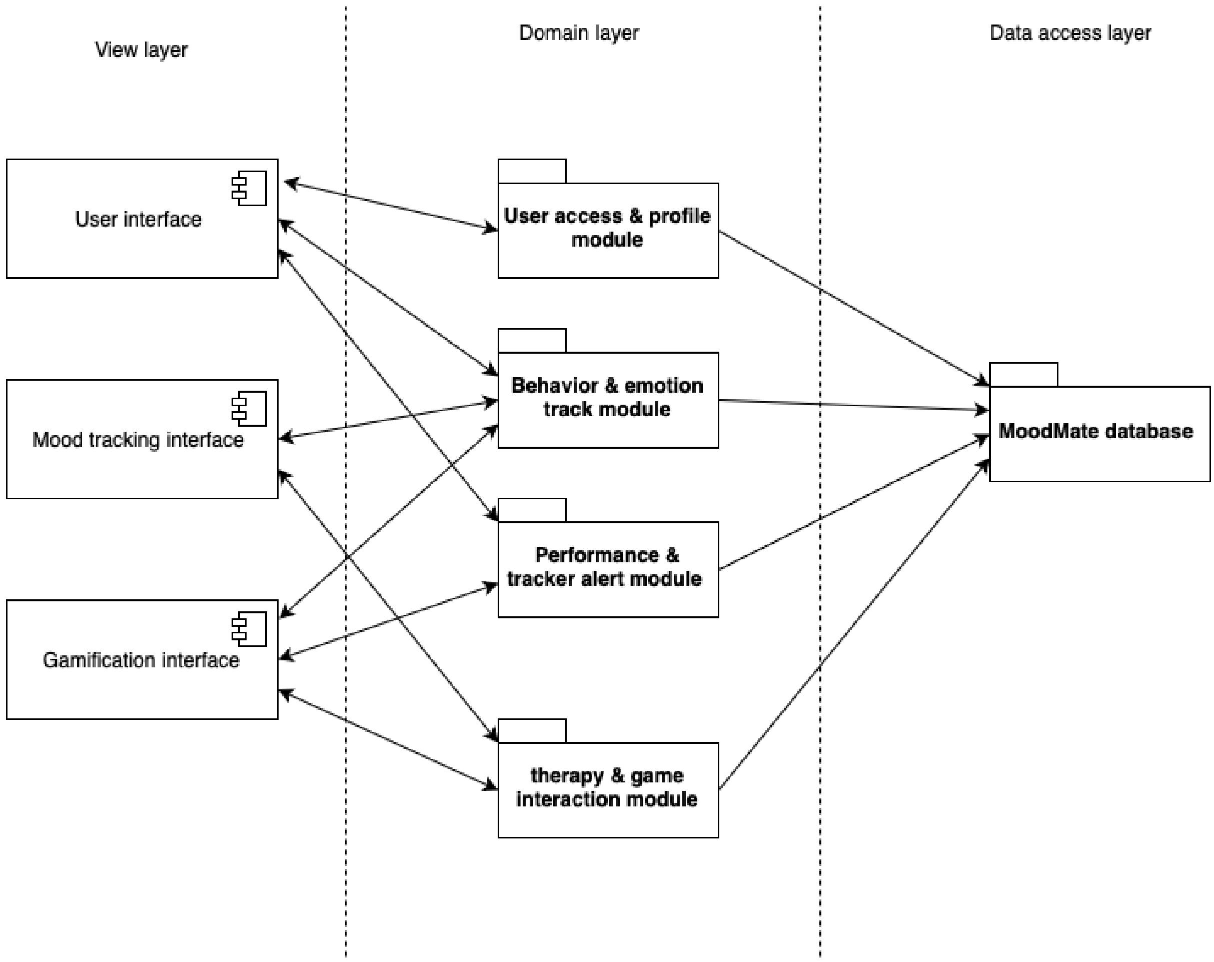
Rationale: Modularity: Each layer handles a specific set of functionalities, making the system easier to develop, test, and maintain.

Scalability: The architecture supports horizontal scalability by allowing layers like the data storage layer to be scaled independently as user demand grows.

Reusability: Components within each layer can be reused across different parts of the application.

Security: Sensitive operations, such as data handling, can be confined to specific layers, ensuring robust compliance with regulations like PDPA.

Platform Independence: By separating user interface concerns from business logic and data storage, the system can be implemented on both web and mobile platforms without affecting core functionalities.



4 Architectural Views

4.1 Use Case View

The MoodMate use case diagram illustrates the interactions between two primary actors: it exists between Parent and Child, encouraging activities for mental health, also tracks behavior and does therapy. The application allows the parents to open an account by entering their details and, in addition, their child's details but with an added measure of pin number to secure the account of the child. Parents are provided access to parenting resources, tools for tracking their child's progress, and notifications about app usage, enabling them to supervise profiles effectively and monitor daily goal achievements.

The child communicates with the system through games that involve basic operation, reporting of mood, and playing therapy games for enhanced mental health, and virtual therapy sessions. These interactions also try to create a playful atmosphere involved with children but, at the same time, document their emotional states for further use.

The system also includes surveys where parents give detailed responses to assess their child, including the child's emotional and behavioral health, apart from the child self-testing questionnaires for mental health. Daily advancement and interaction are indicators by features such as the game score board showing the performance of the child on therapy related activities. The core activities concern the roles of the parents and children mainly focusing on safety, support, and interaction. By integrating tools like alarms for daily goals, notifications, and therapy games, the MoodMate system ensures that both parents and children receive the resources and feedback they need to foster mental and emotional well-being. The system's design prioritizes data security and the seamless interaction of all features to enhance the overall user experience.

Figure 4.1: Use Case Diagram for <MoodMate>

4.2 Implementation View

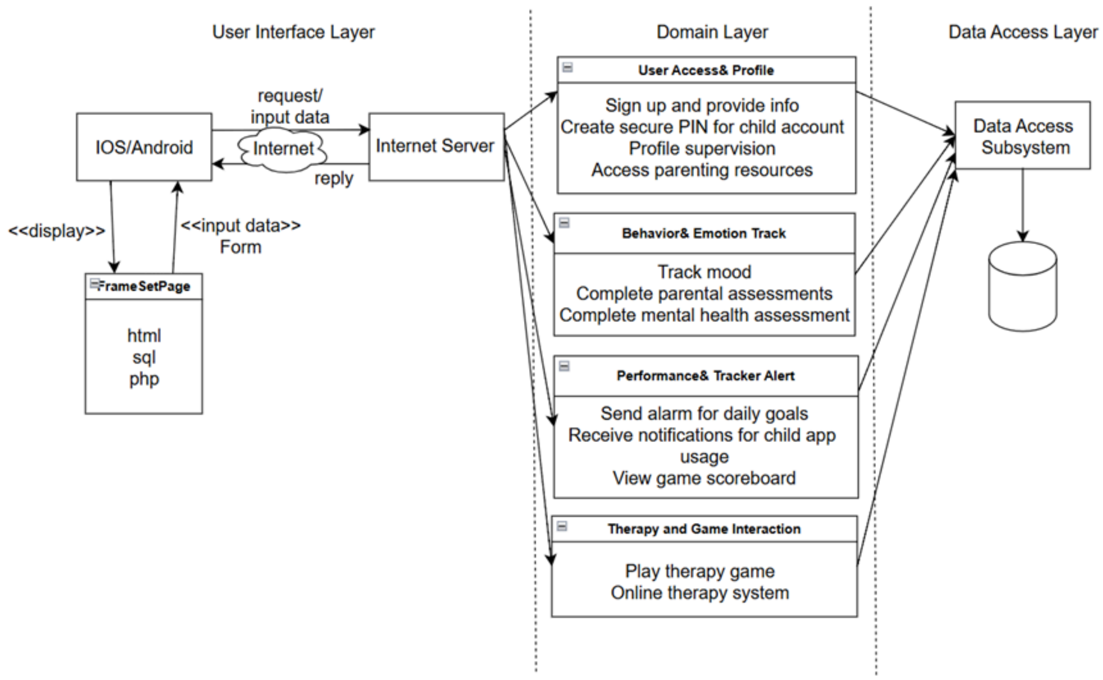


Figure 4.2.1: Three-Layer Architectural Design of <Moodmate>

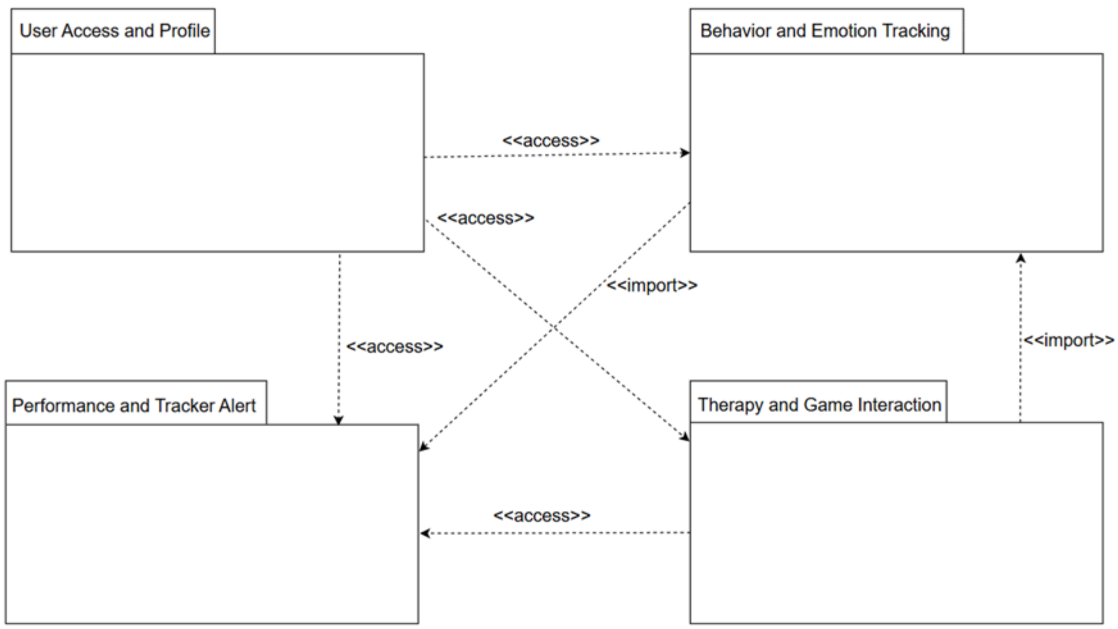


Figure 4.2.2: Package Diagram for <Moodmate>

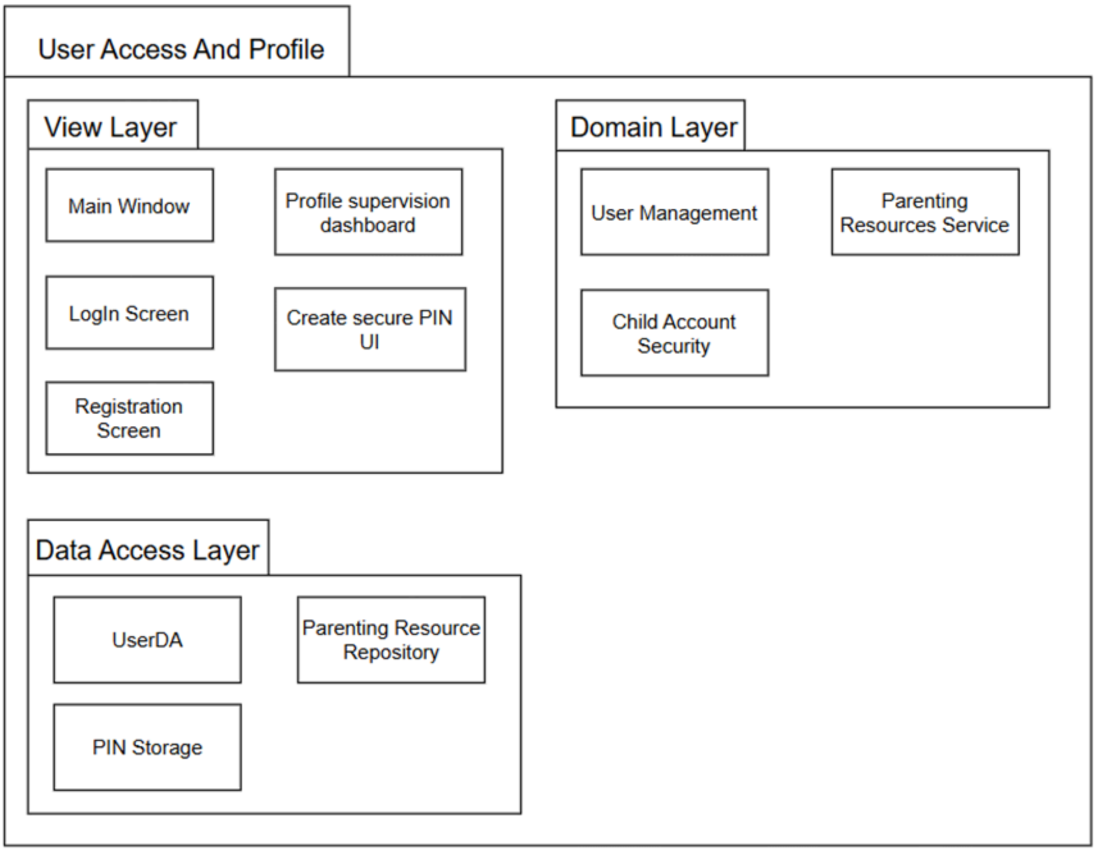


Figure 4.2.3: Package Diagram for <User Access And Profile> Subsystem

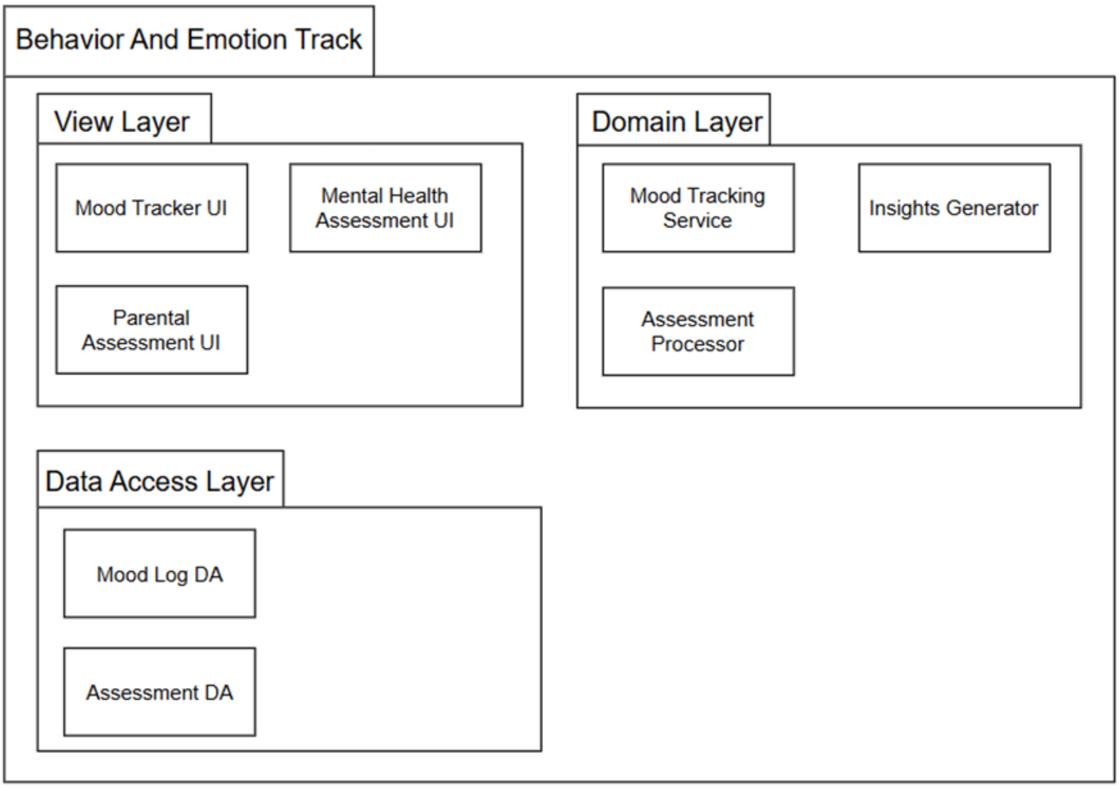


Figure 4.2.4: Package Diagram for <Behavior And Emotion Track > Subsystem

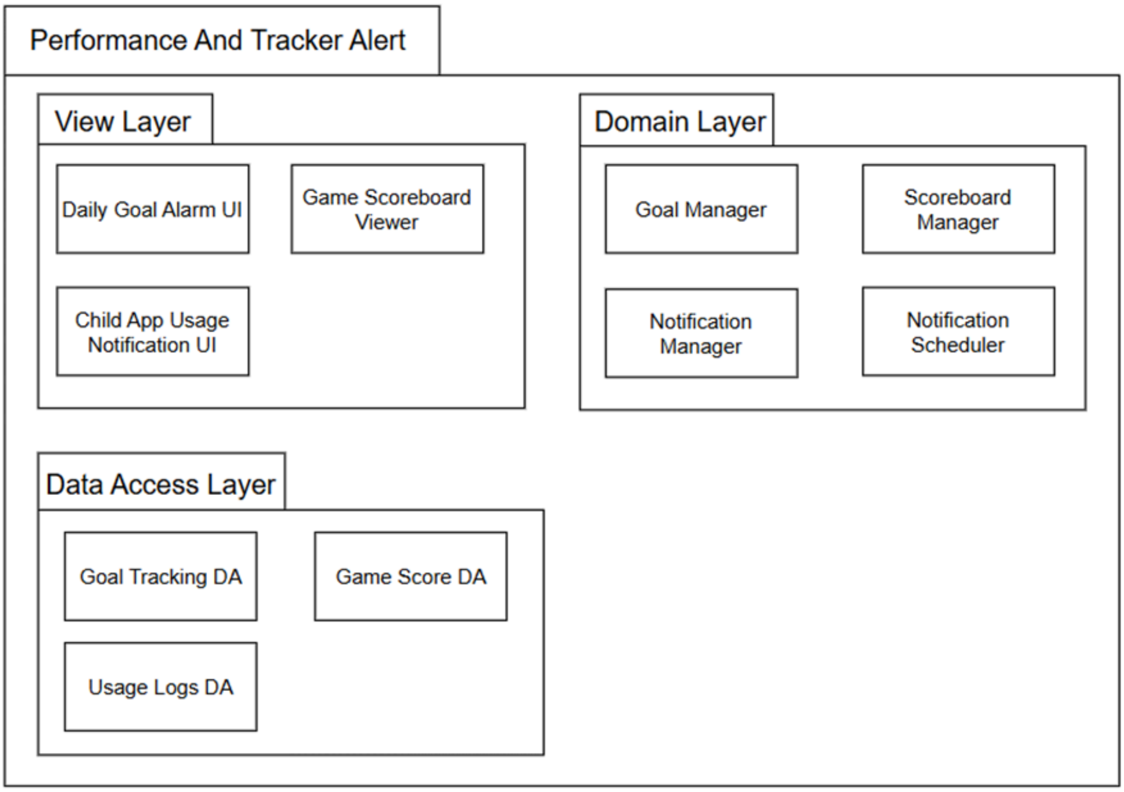


Figure 4.2.5: Package Diagram for <Performance And Tracker Alert> Subsystem

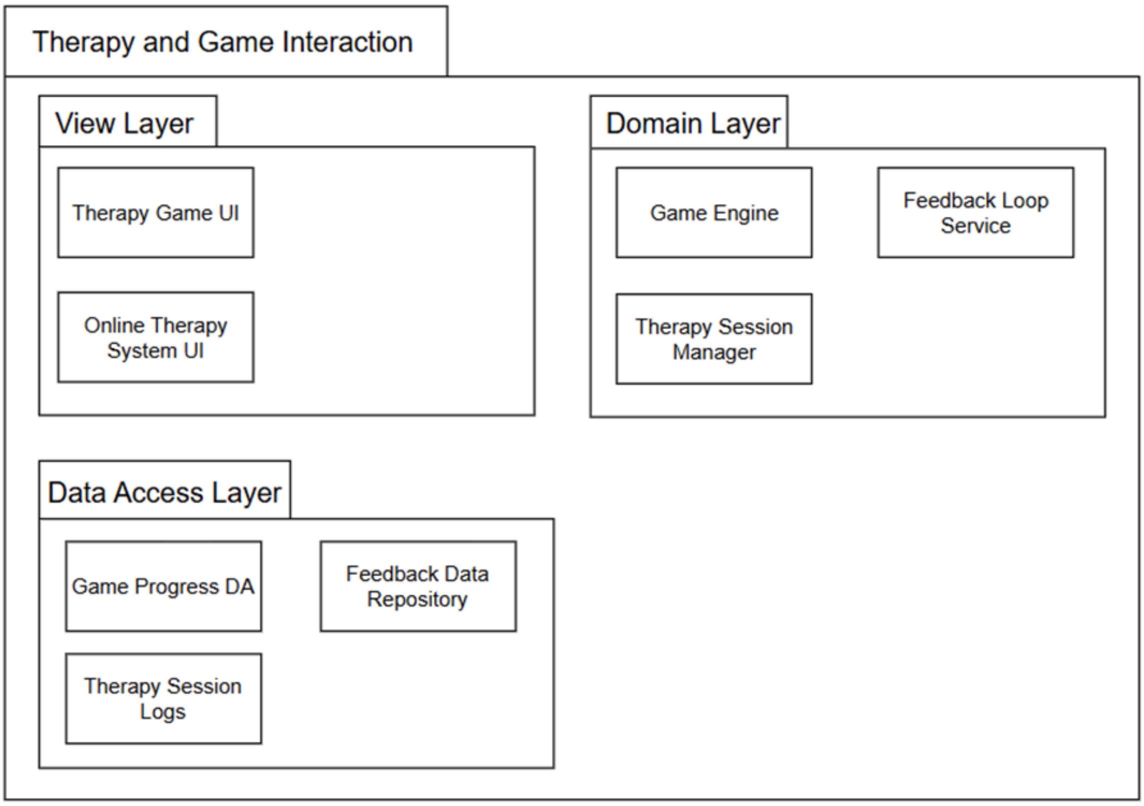
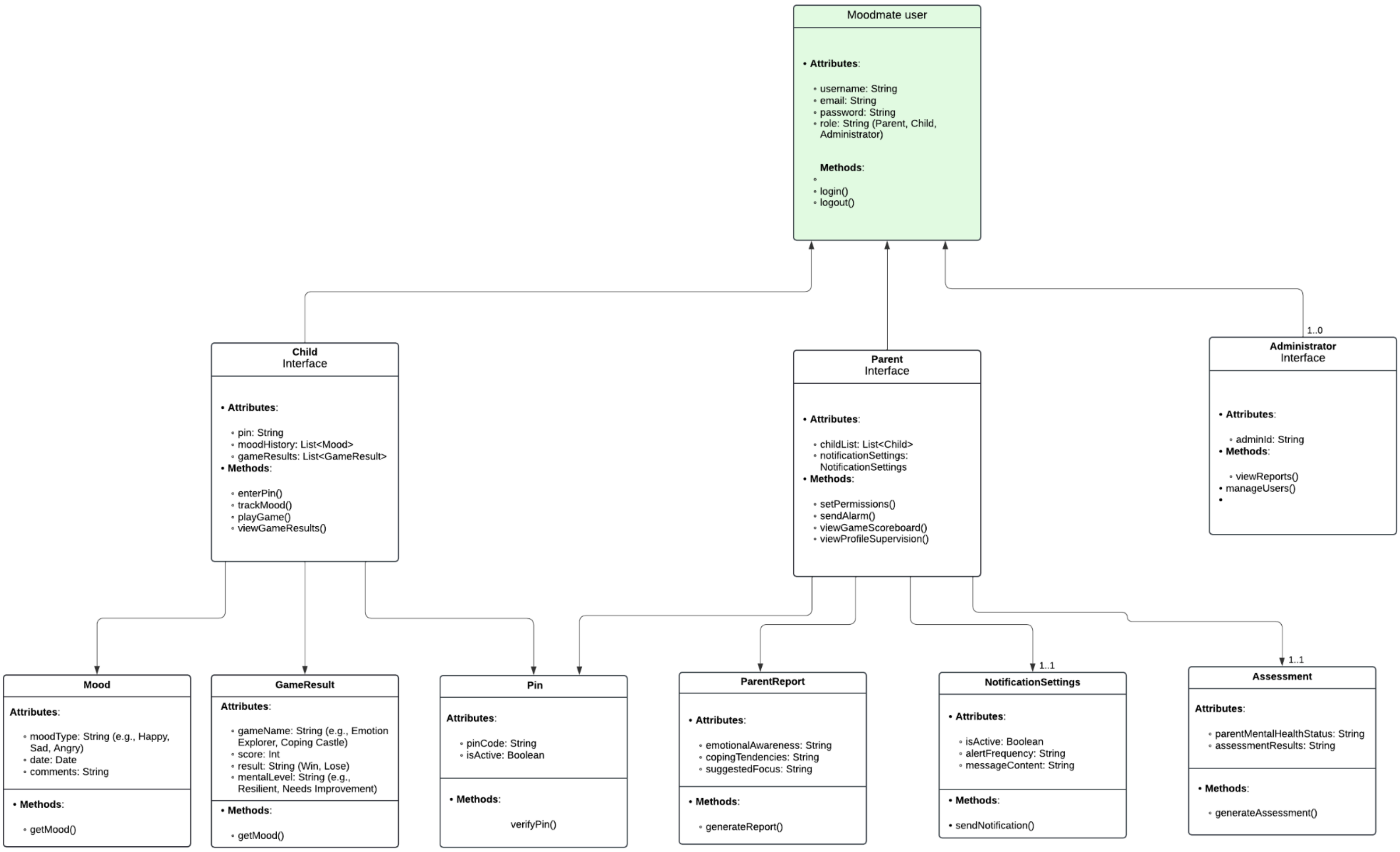


Figure 4.2.6: Package Diagram for <Therapy and Game Interaction > Subsystem

4.3 Logical View

This UML Class Diagram represents the whole use case and effectively outlines the core functionalities of MoodMate. Each interface (child, parent, administrator) has unique attributes and methods tailored to its role while interacting with shared entities like moods, reports, and assessments.



This UML Class Diagram represents the whole use case and effectively outlines the core functionalities of MoodMate. Each interface (child, parent, administrator) has unique attributes and methods tailored to its role while interacting with shared entities like moods, reports, and assessments.

Figure 4.3.1: Class diagram for all use case

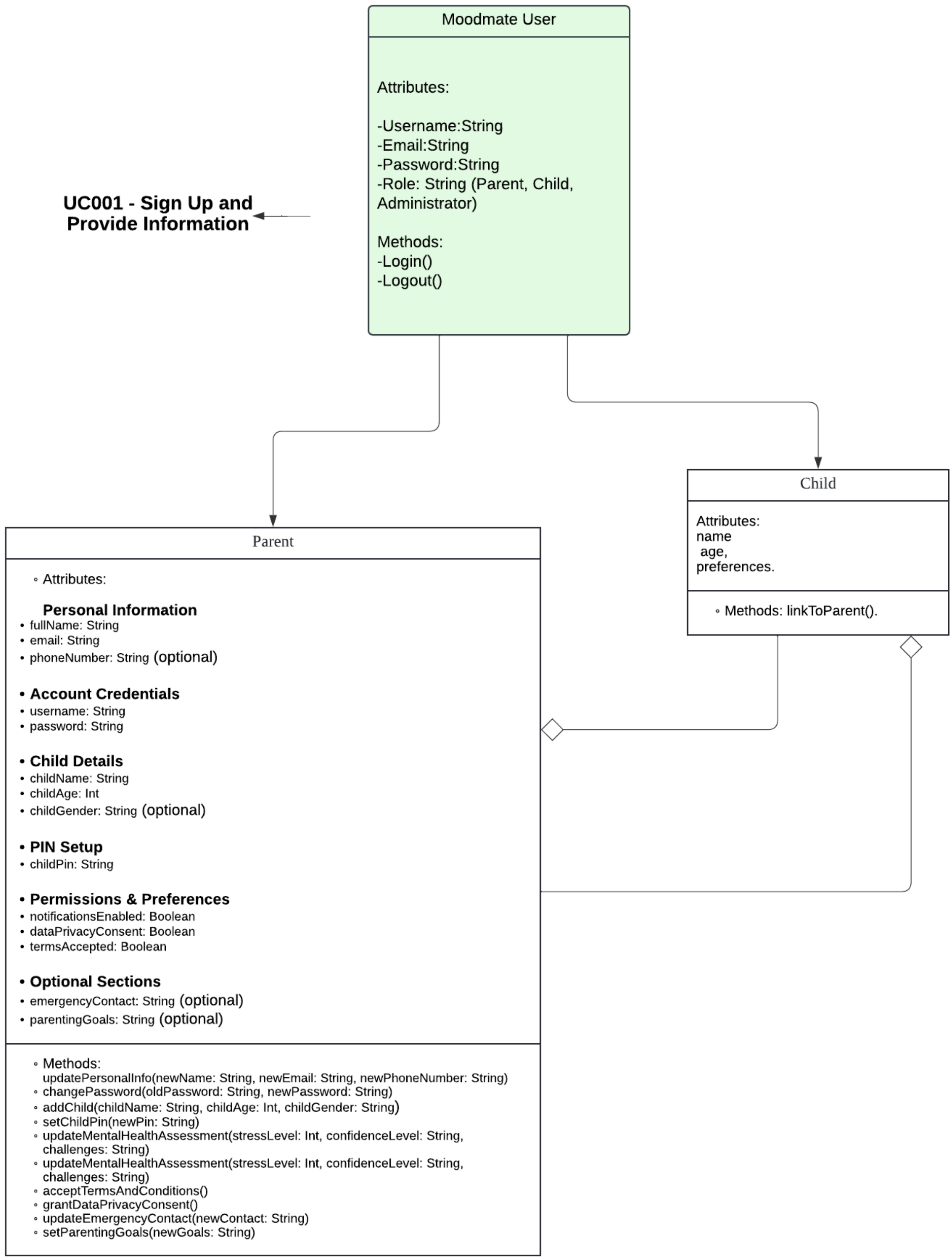


Figure 4.3.2: Class Diagram for uc001 sign up and provide information

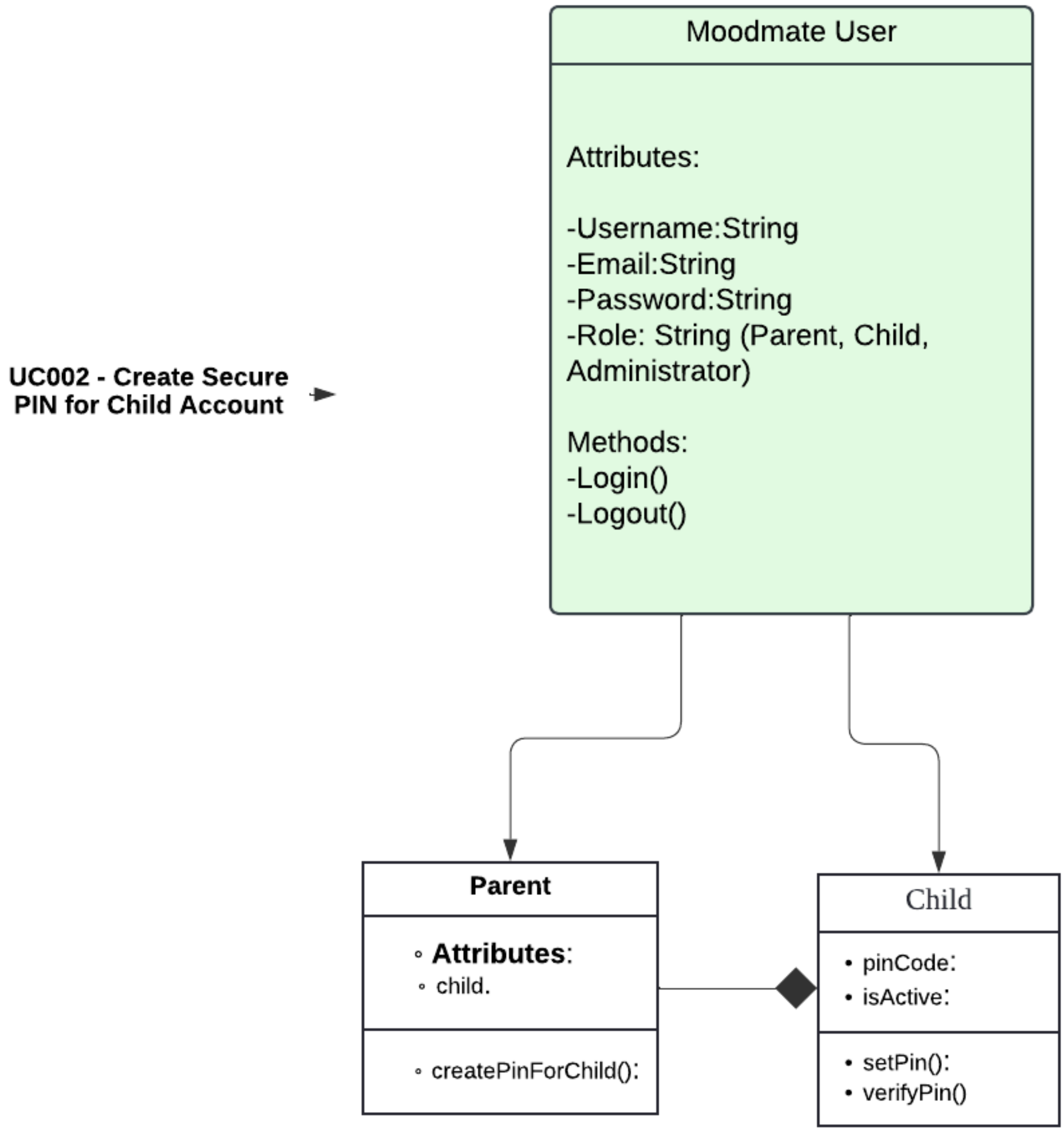


Figure 4.3.3: Class Diagram for UC002 Create Secure Pin For Child Account

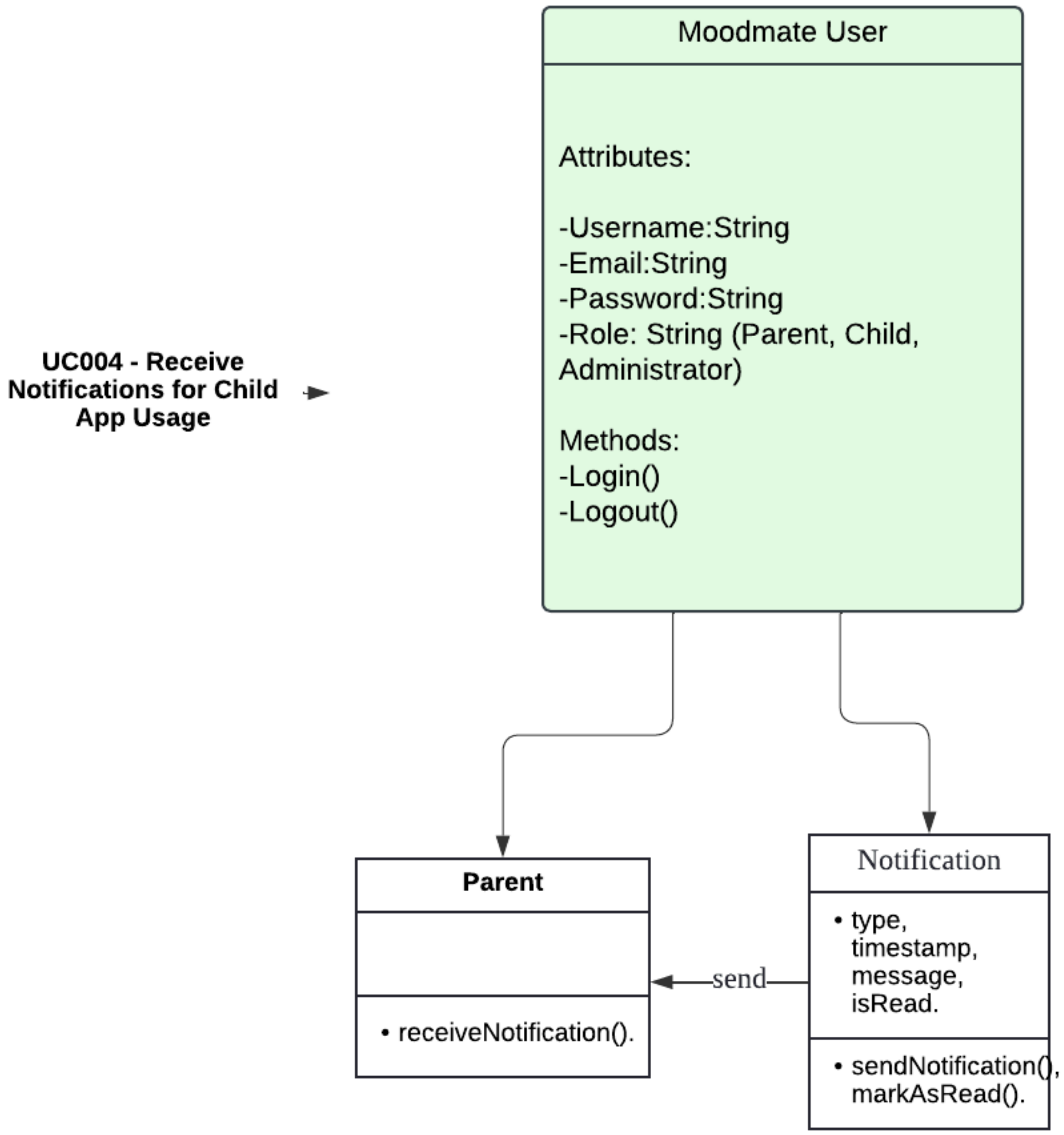


Figure 4.3.4: Class Diagram for UC004 Receive Notifications for Child App Usage

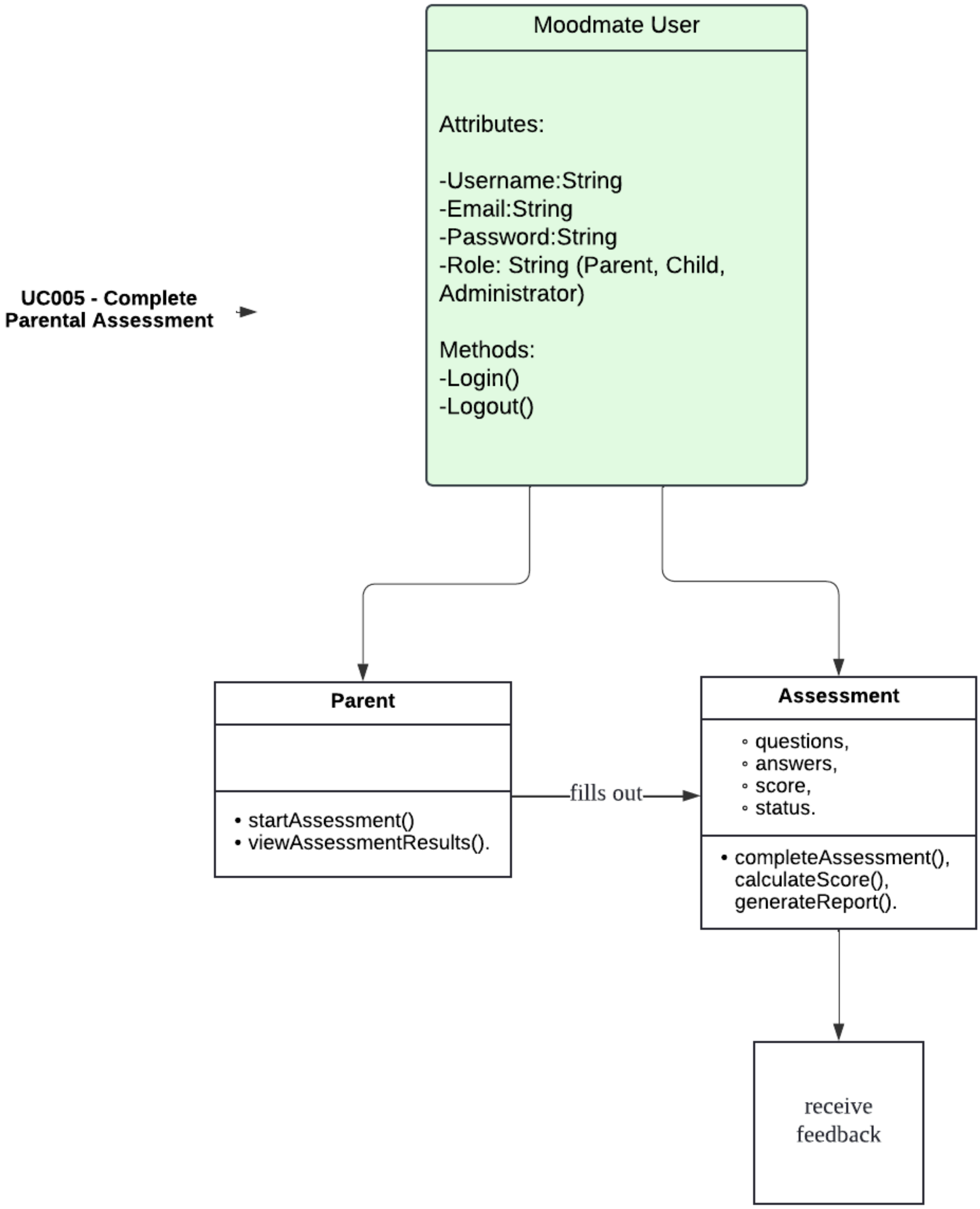


Figure 4.3.5: Class Diagram for UC005 - Complete Parental Assessment

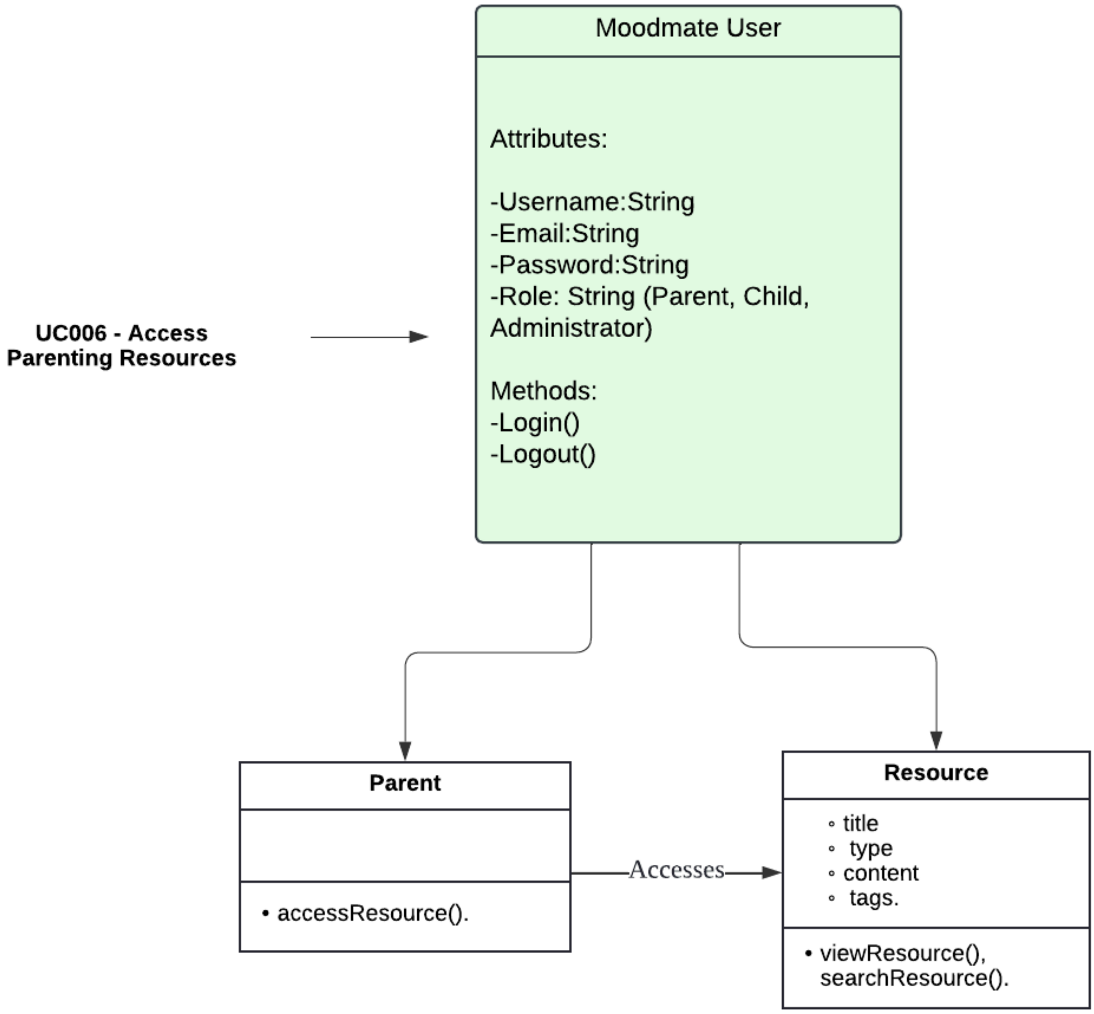


Figure 4.3.6: Class Diagram forUC006 - Access Parenting Resources

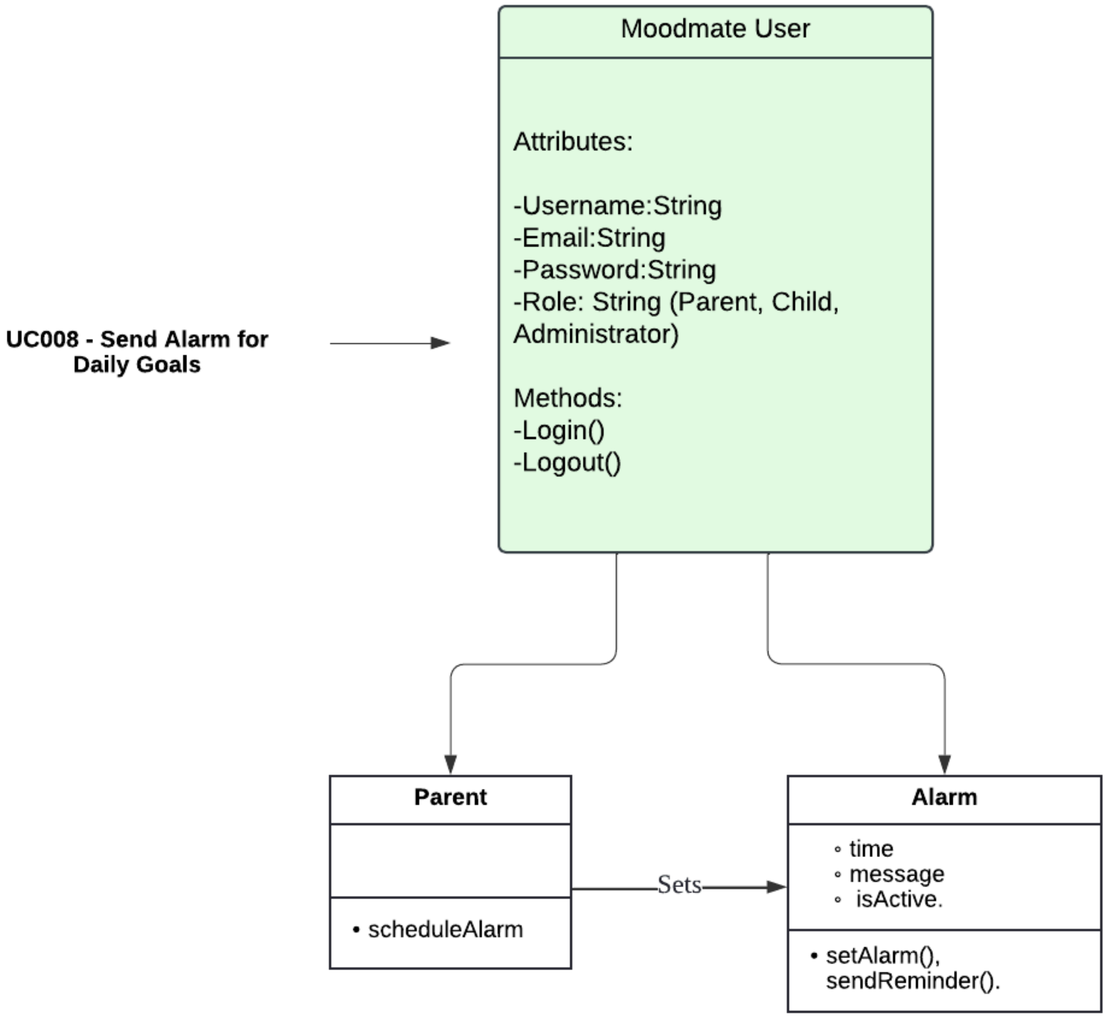


Figure 4.3.7: Class Diagram forUC008 - Send Alarm for Daily Goals

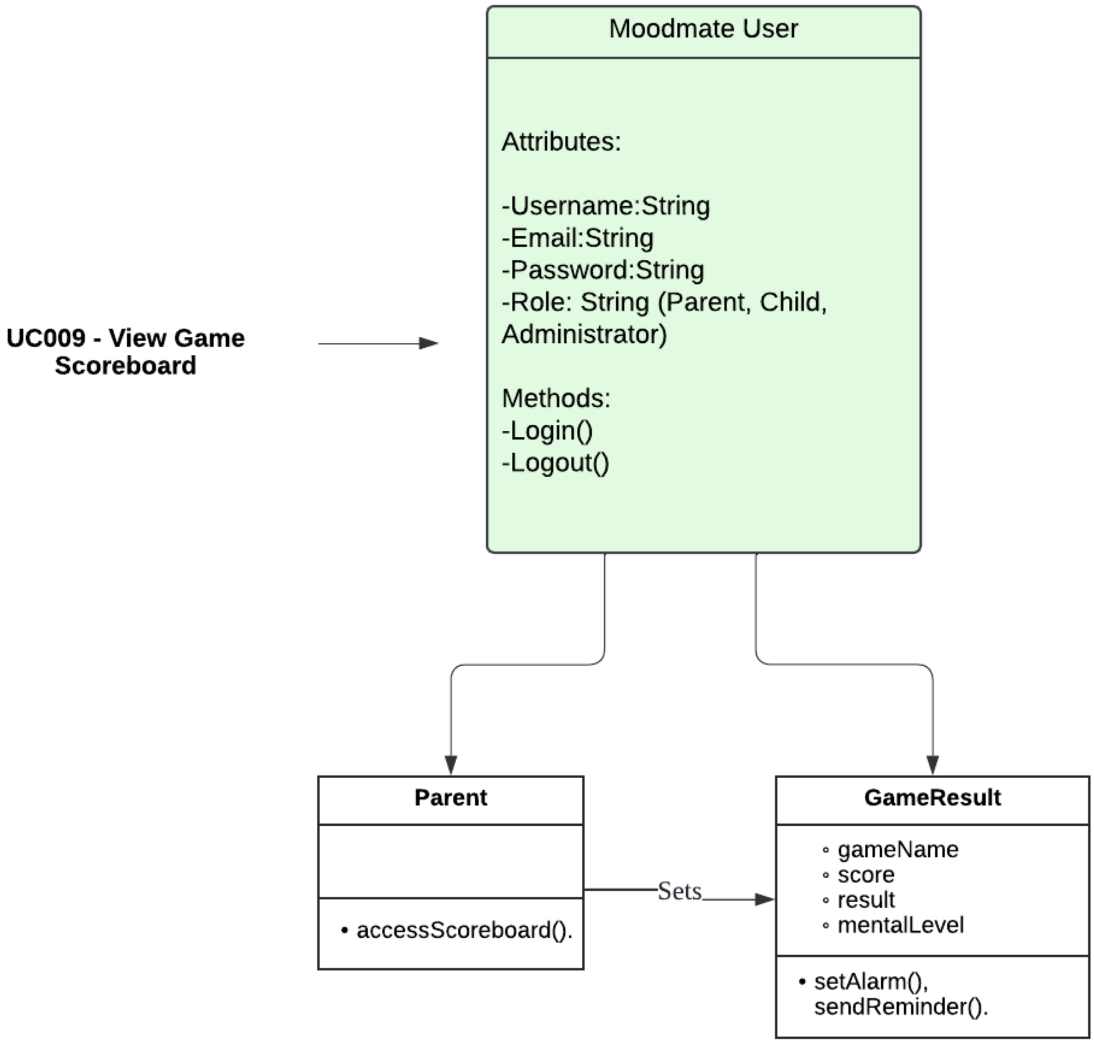


Figure 4.3.8: Class Diagram for UC009 - View Game Scoreboard

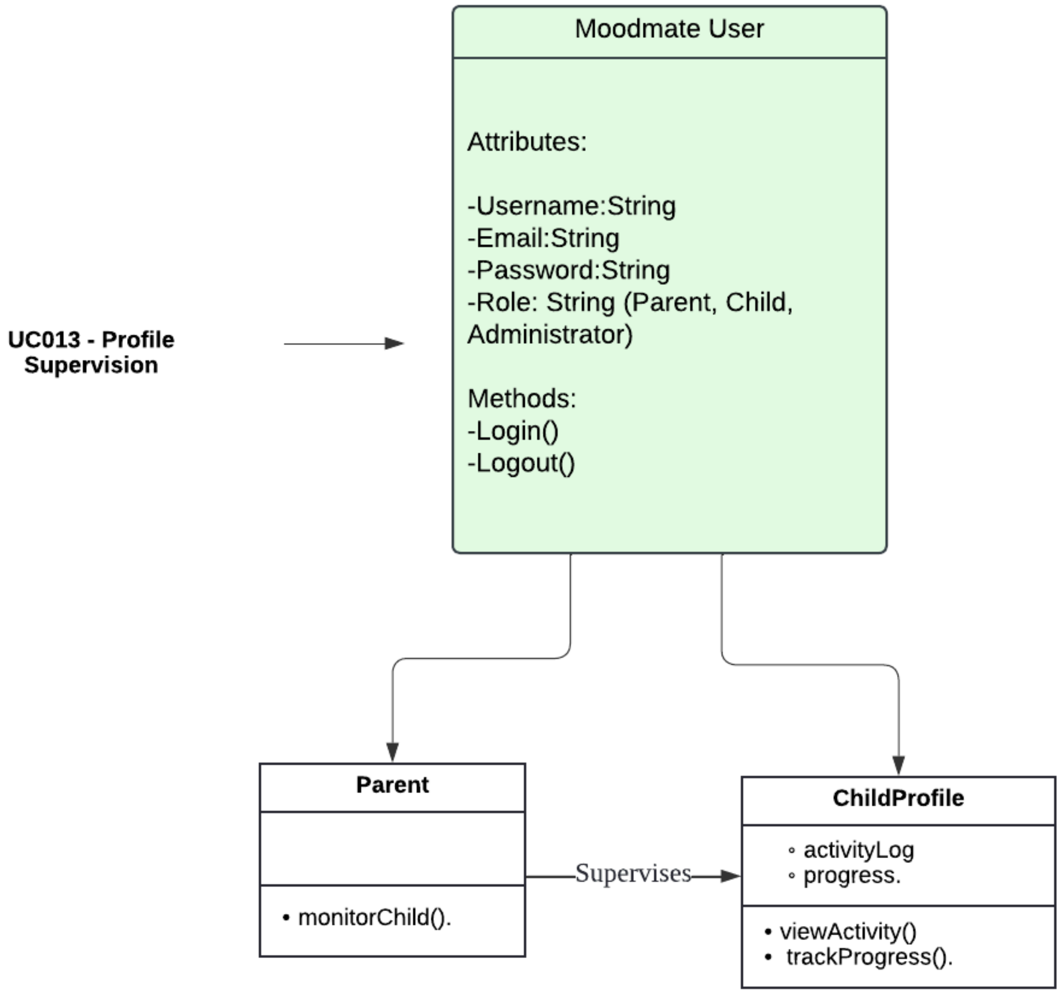


Figure 4.3.9: Class Diagram for UC013 - Profile Supervision

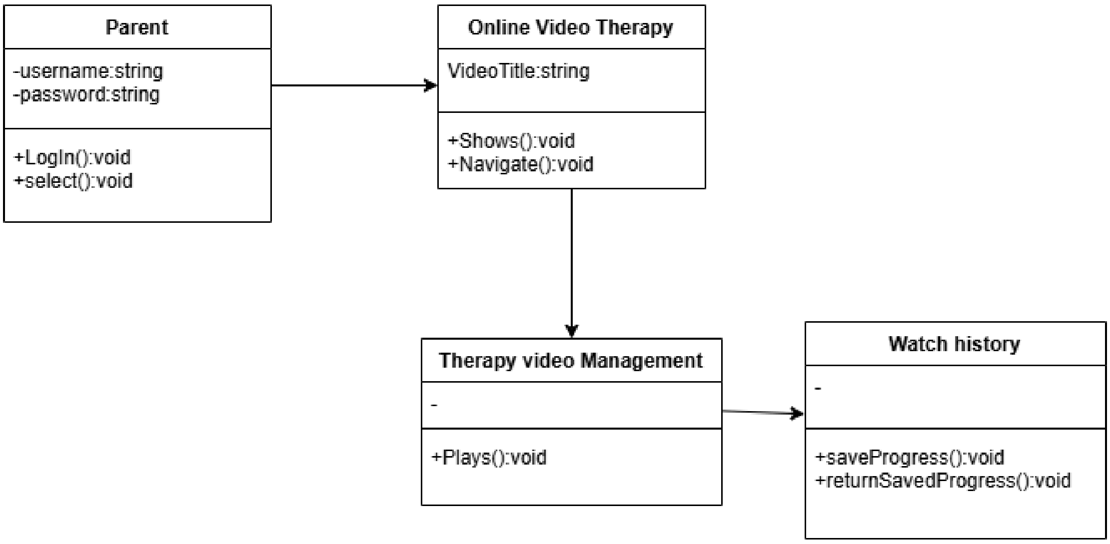


Figure 4.3.10: Class Diagram for UC010 - Access online therapy video

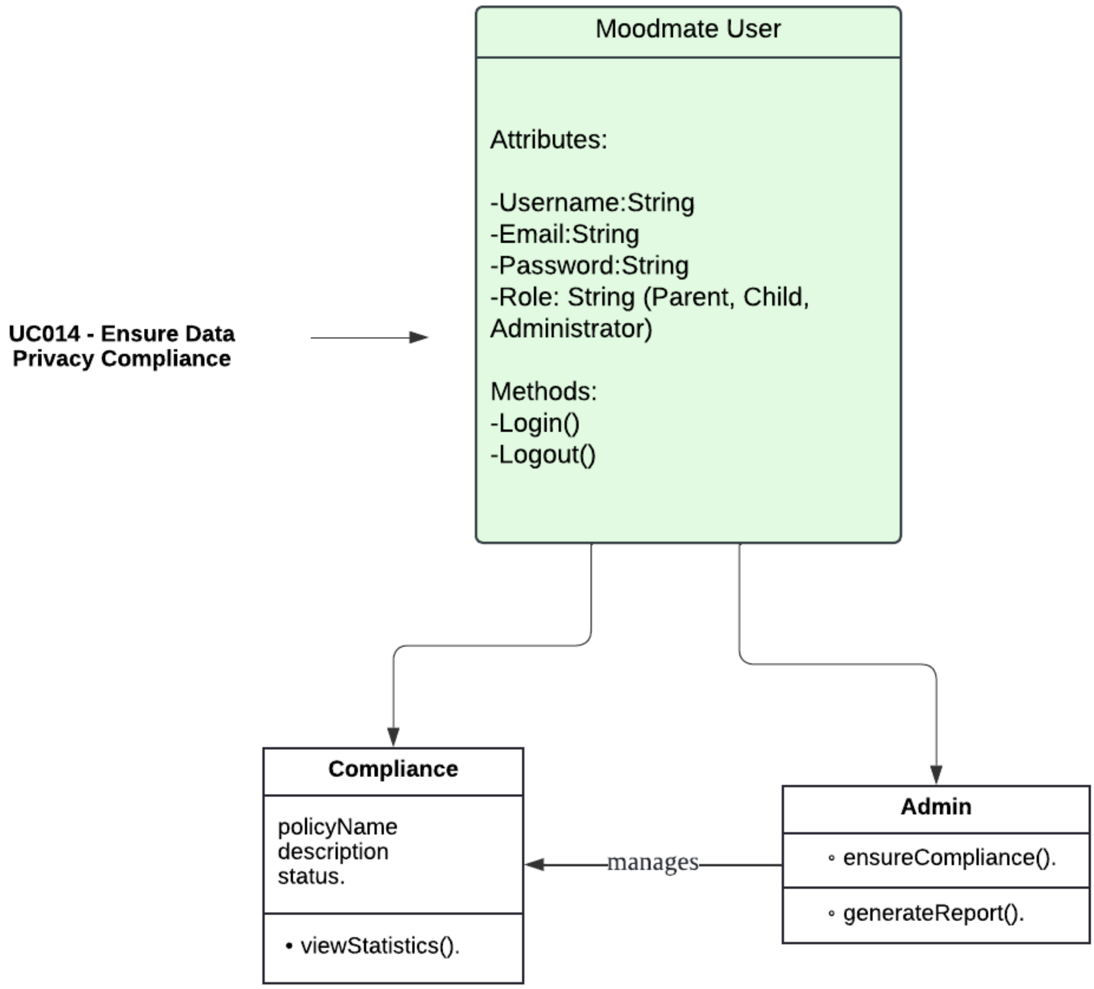


Figure 4.3.11: Class Diagram for UC014 - Ensure Data Privacy Compliance

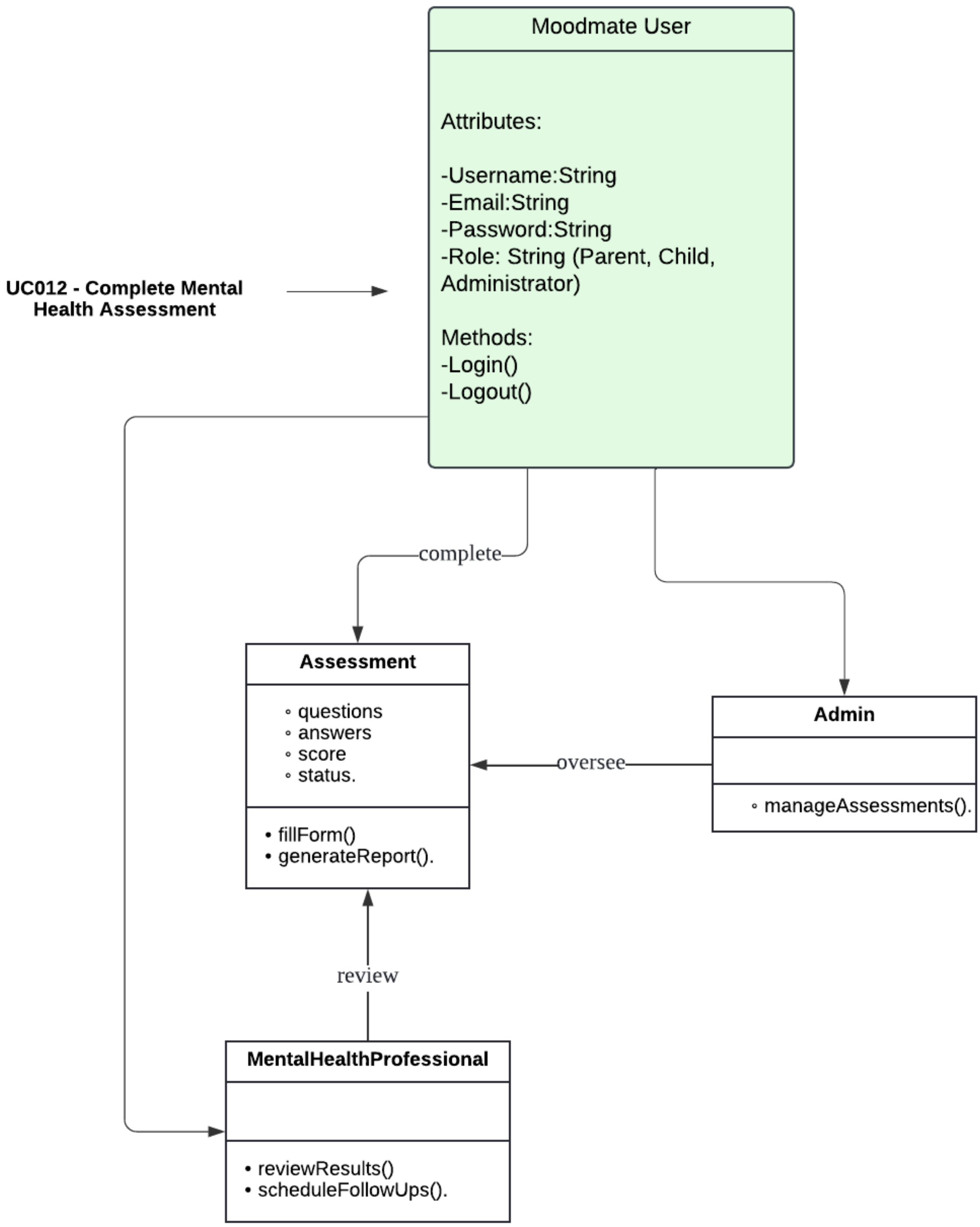


Figure 4.3.13: Class Diagram for UC012 - Complete Mental Health Assessment

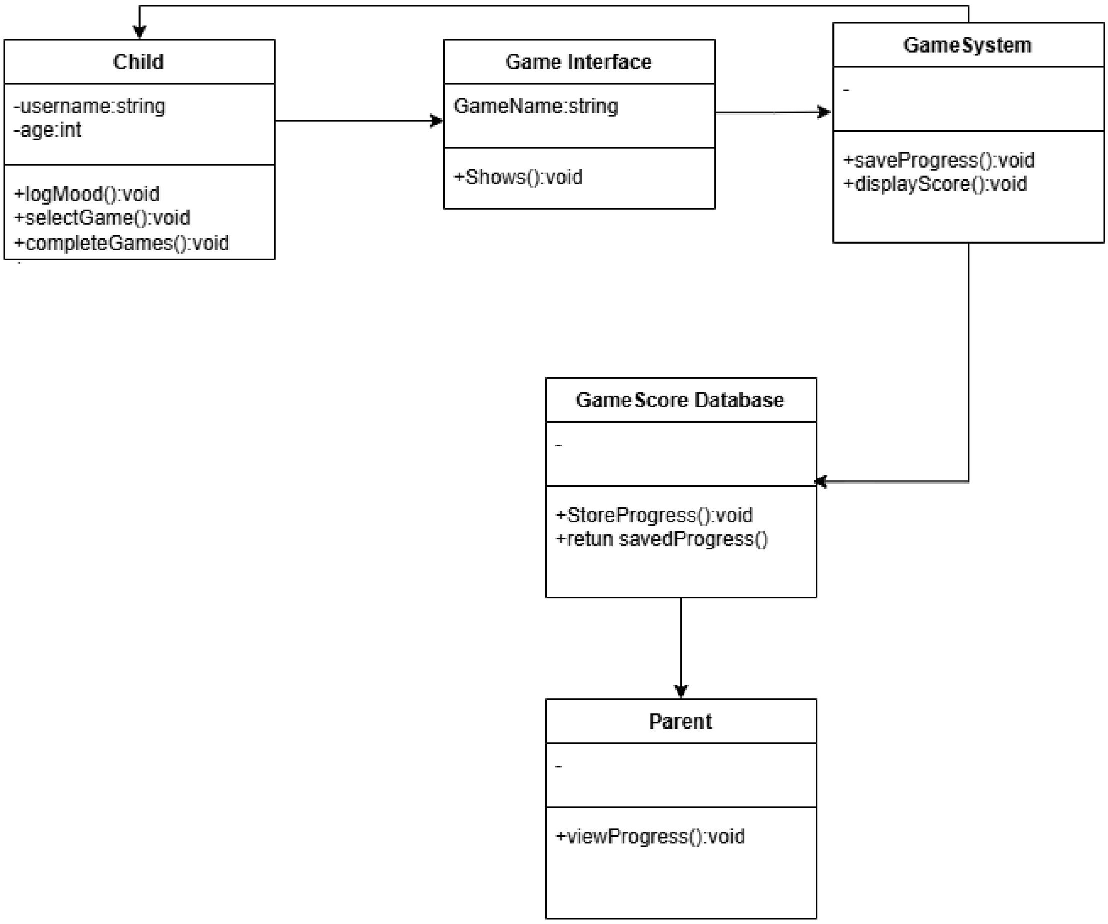


Figure 4.3.14: Class Diagram for UC003 - Play Therapy Games

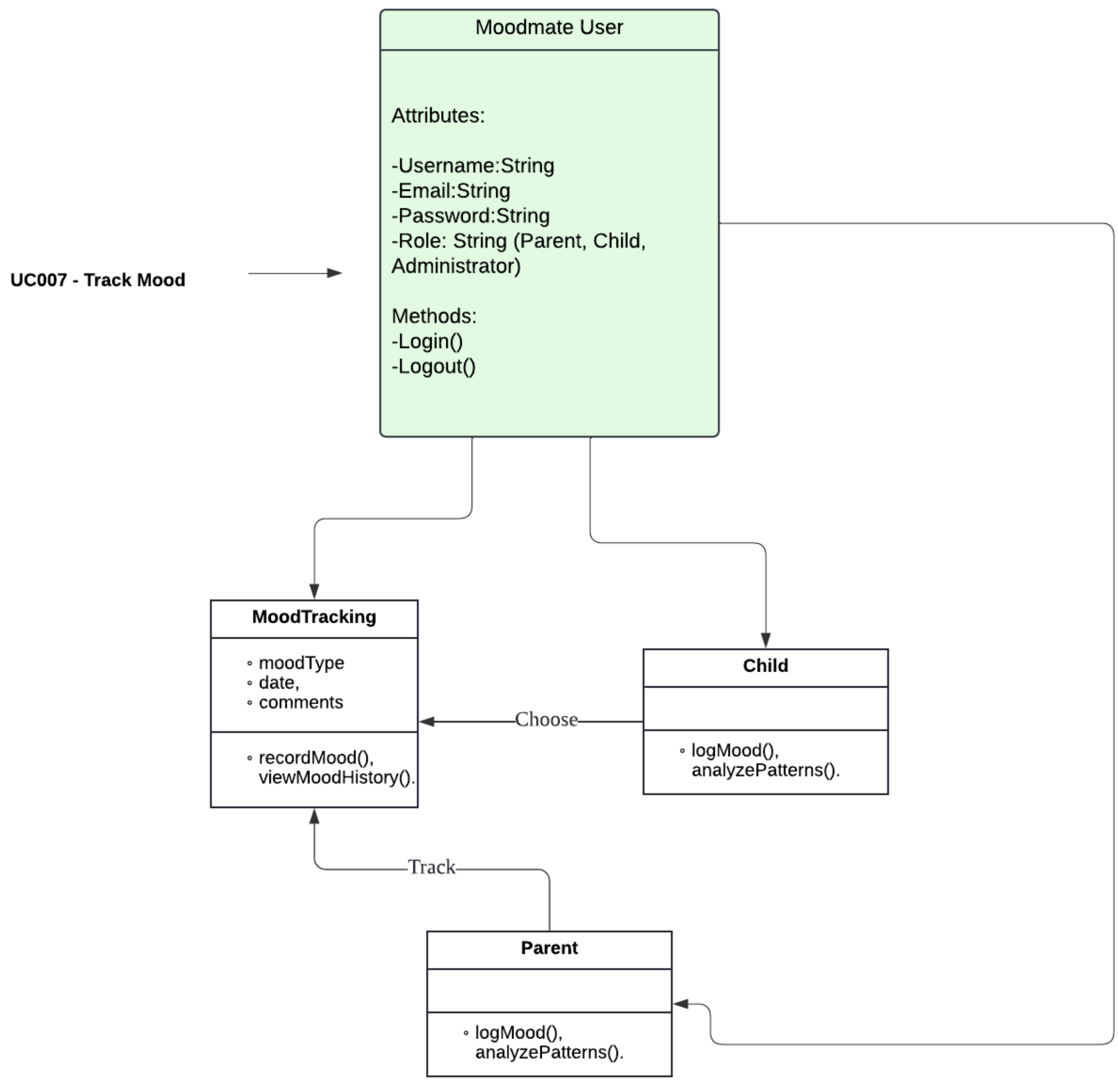


Figure 4.3.15: Class Diagram for UC007 - Track Mood

4.4 Process View

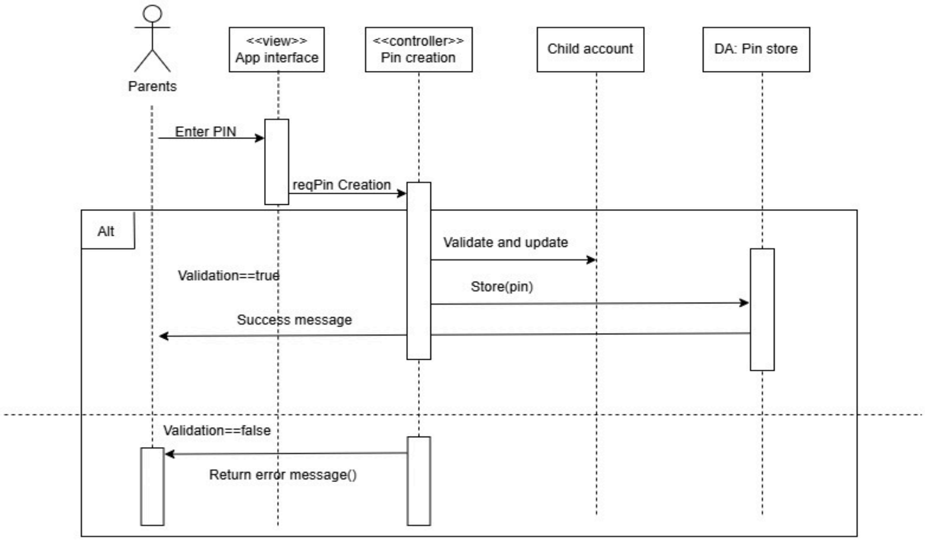


Figure 4.4.1 Process view of user access and Profile

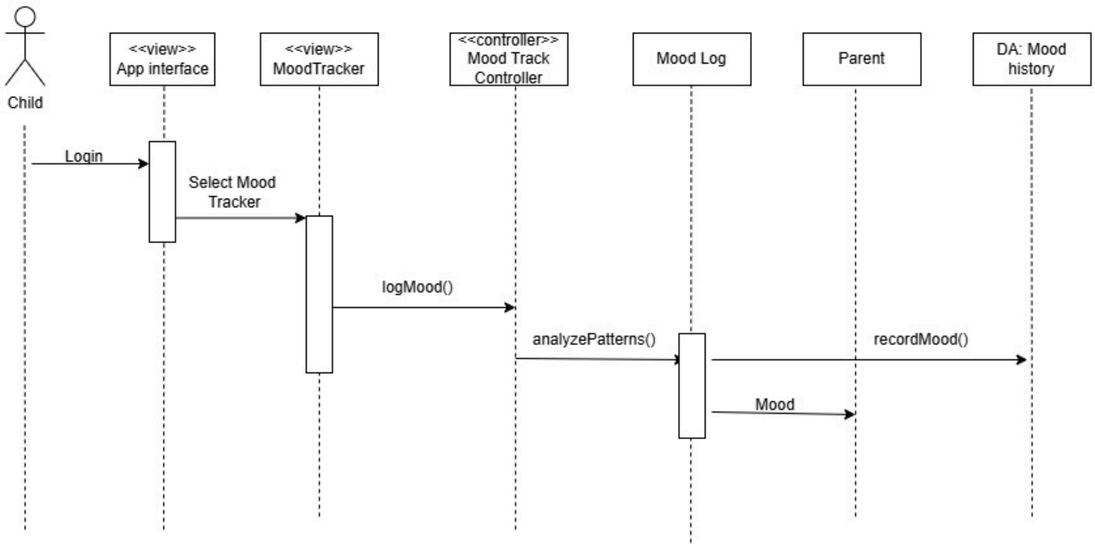


Figure4.4.2 Process view of behavior and Emotion tracker

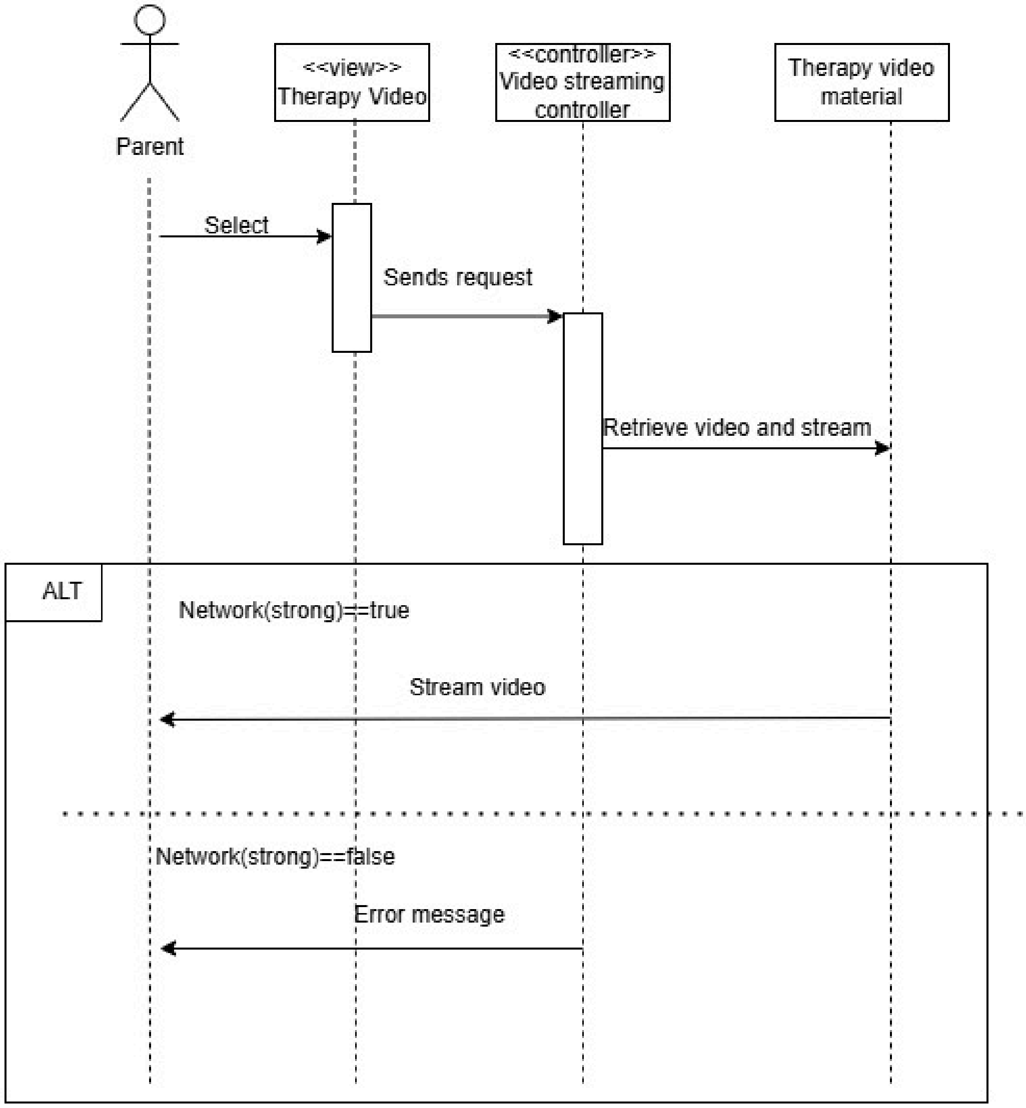
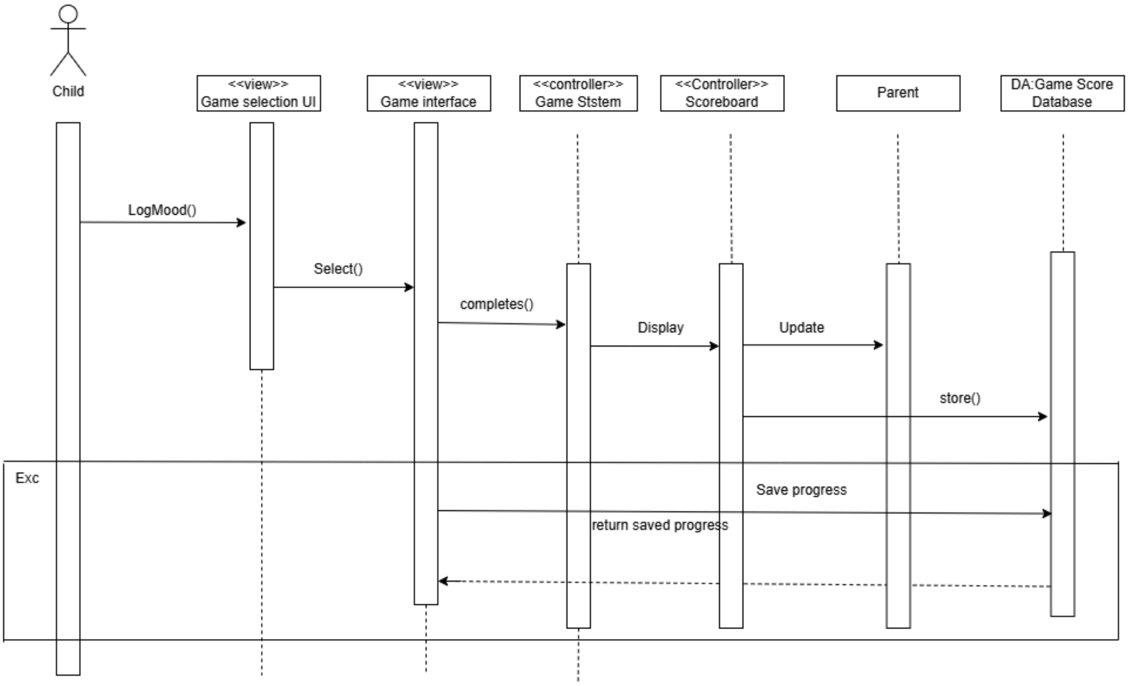
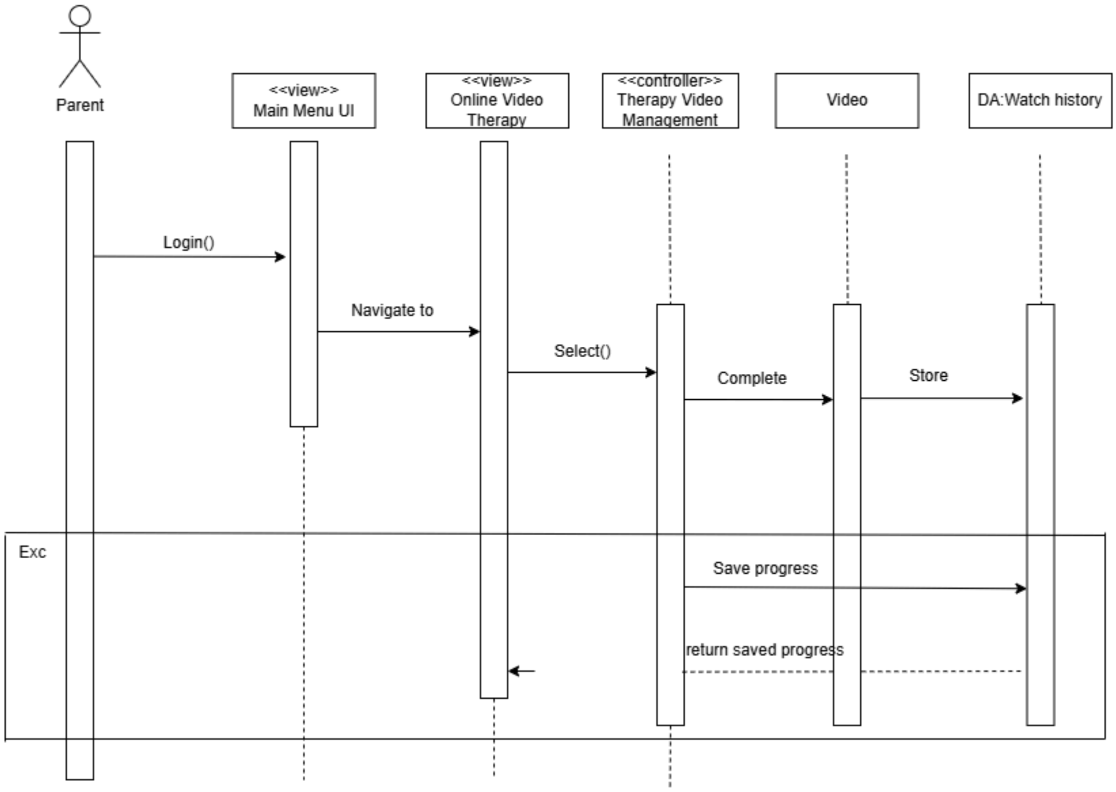


Figure4.4.3 process view of performance and tracker alert



process view of playing therapy games



Process view of Access online therapy video

4.5 Deployment View

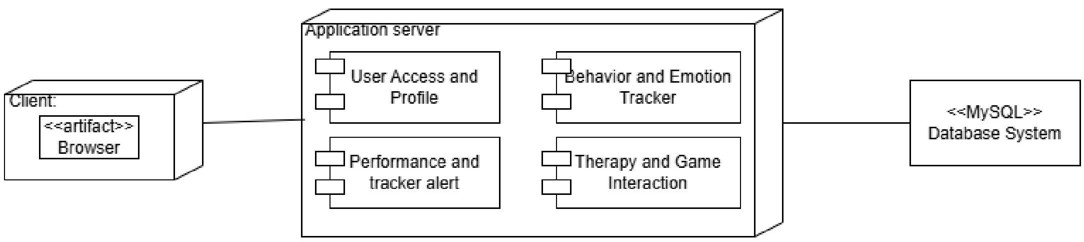


Figure 4.5.1 deployment view of MoodMate

5 Data Design

5.1 Data Dictionary

5.1.1 Entity: <User>

Attribute Name	Type	Description
UserID	string	Unique identifier for each user.
password	string	Password for user authentication.

Table 5.1.1 Example Data dictionary

5.1.2 Entity: <Parent>

Attribute Name	Type	Description
Name	string	Full name of the parent.
icNo	string	Identification number of the parent (e.g., national ID or passport number).
Gender	String	Gender of the parent
age	int	Age of the parent.
occupation	string	Current job or profession of the parent.
address	string	Residential address of the parent.
phoneNo	string	Contact phone number of the parent.
email	string	Email address of the parent.

Table 5.1.2 Data dictionary Parent

5.1.3 Entity: <Child>

Attribute Name	Type	Description
Name	string	Full name of the child.
icNo	string	Identification number of the child (e.g., national ID or passport number).
age	int	Age of the child.
gender	string	Gender of the child (e.g., male, female, non-binary).
address	string	Residential address of the child.
schoolAdd	string	Address of the school the child attends.
email	string	Email address of the child (if applicable).

Table 5.1.3 Data dictionary Child

5.1.4 Entity: <Parent Resource>

Attribute Name	Type	Description
title	string	Title of the educational resource.
type	string	Type of the resource (e.g., article, video, guide).
content	string	Detailed content or link to the educational resource.

Table 5.1.4 Data dictionary Resource

5.1.5 Entity: <AssessmentTool>

Attribute Name	Type	Description
toolName	string	Name of the assessment tool.
description	string	Description of the assessment tool.
results	string	Results generated by the assessment.

Table 5.1.5 Data dictionary Assessment Tool

5.1.6 Entity: <Mode Tracker>

Attribute Name	Type	Description
trackerID	string	Unique identifier for the mood tracker entry.
date	date	Date when the mood was recorded.
mood	string	Mood description (e.g., happy, sad, stressed).
notes	string	Additional notes or details about the mood.

Table 5.1.6 Data dictionary Mode Tracker

5.1.7 Entity: <Set alarm daily Goal >

Attribute Name	Type	Description
goal	string	The specific goal to be achieved.
description	string	Detailed description of the goal.
startDate	date	Date when the goal is set to start.
endDate	date	Deadline for achieving the goal.
status	string	Current status of the goal (e.g., in-progress, completed).

Table 5.1.7 Data dictionary set alarm daily goal

5.1.8 Entity: <Online Therapy Videos>

Attribute Name	Type	Description
VideosName	string	Unique identifier for the Online therapy Videos
date	date	The date when the therapy session takes place.
time	time	The time when the therapy session starts.
duration	int	The duration of the therapy session, measured in minutes.

Table 5.1.8 Data dictionary Online therapy videos

5.1.9 Entity: <Therapy Games>

Attribute Name	Type	Description
gameName	**string**	**Name of the game.**
description	**string**	**Detailed description of the game.**
score	**int**	**Points or score achieved in the game.**

Table 5.1.9 Data dictionary Therapy Games

6. User Interface Design

MoodMate is a user-friendly system designed to help parents monitor and support their child's mental health. Parents begin by registering and setting up profiles for themselves and their children, which include uploading personal details and profile pictures. Once logged in, they access a dashboard that provides an overview of the child's mood trends, activities, and progress.

Parents can supervise their child's profile to review mood updates, track activity logs, and monitor game progress. They can set daily or weekly goals tailored to their child's needs, such as completing games or practicing mindfulness, with customizable deadlines and priority levels. Progress toward these goals is displayed visually, helping parents understand their child's achievements and areas for improvement.

The system includes a library of mental-health-focused games that children can play through a PIN-protected interface. Notifications and alerts keep parents updated on their child's activity and progress, and detailed reports summarize trends and achievements over time. Feedback is provided through engaging visuals, animations, and messages, ensuring a supportive and interactive experience for both parents and children.

MoodMate's secure, engaging design and its focus on both tracking and support make it a valuable tool for fostering children's mental well-being.

6.1 Screen Images

Figure 6.1: Interface for the Application

On this interface, We design the front interface as the above. There will be a Child button and a parent button for both roles access into the application. This Application designed with colorful to have the child attraction on using this application

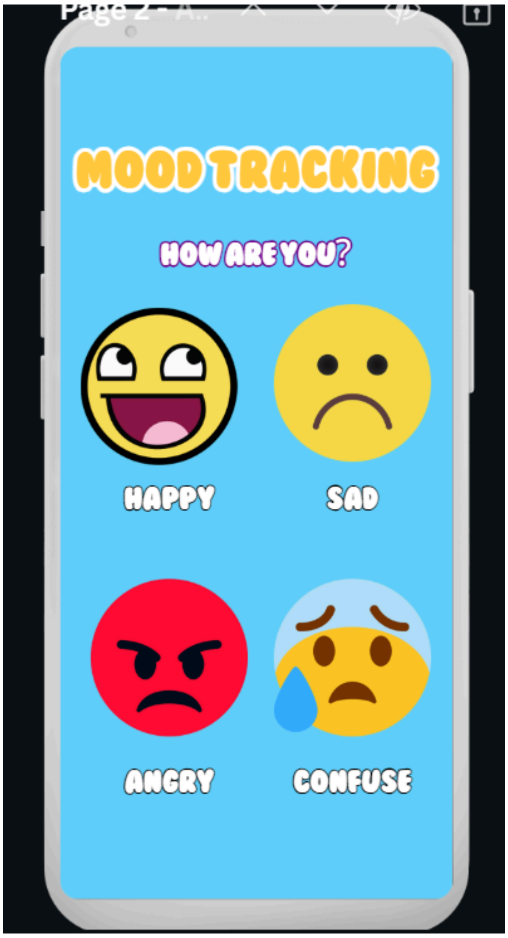


Figure 6.2: Interface for Child View< Mood Tracking>

After the user clicks on the child button, it will open the child interface and the first page will be Mood Tracking. Children must choose their current mood . There will be 4 moods button ; happy sad , angry and confused, for children to choose . If the child role clicks on one of the mood buttons, it will send the data to the parent and will receive a notification.

Figure 6.3: Interface for Child View< Playing Therapy games>

After children choose their mood, they are able to play the psychiatric games. There will be 8 kinds of games ;

1. Emotion Matching Games

- **Description: Games that teach children to recognize and understand emotions by matching facial expressions, scenarios, or emojis with the correct emotion.**
- **Example: A game where children match an animated character's face to words like "happy," "sad," or "angry."**
- **Therapeutic Benefit: Improves emotional literacy and empathy.**

---

2. Cognitive Behavioral Therapy (CBT) Puzzle Games

- **Description: Games based on CBT principles where children complete tasks that challenge negative thoughts and reinforce positive thinking.**
- **Example: Completing a "thought map" puzzle that replaces negative thoughts with positive affirmations.**

3. Relaxation and Breathing Games

- **Description: Interactive games that teach mindfulness and breathing techniques through visuals and sounds.**
- **Example: Guiding a balloon through an obstacle course by breathing in and out, following on-screen prompts.**
- **Therapeutic Benefit: Reduces anxiety and teaches relaxation skills.**

---

4. Problem-Solving Adventure Games

- **Description: Role-playing games where children solve challenges by making decisions and navigating scenarios.**
- **Example: Helping a character find their way out of a maze by solving riddles about emotions or healthy coping mechanisms.**
- **Therapeutic Benefit: Encourages problem-solving and critical thinking under stress.**

---

5. Social Skills Simulation Games

- **Description: Games that simulate real-life social situations, encouraging children to practice communication, conflict resolution, and teamwork.**
- **Example: Choosing the best response in scenarios like sharing toys or resolving disagreements.**
- **Therapeutic Benefit: Enhances social and interpersonal skills.**

---

6. Storytelling and Role-Playing Games

- **Description: Games that allow children to create characters and stories, exploring their emotions through narratives.**
- **Example: Building a story about a superhero overcoming fears or challenges.**
- **Therapeutic Benefit: Helps children process their emotions and boosts creativity.**

---

7. Memory and Focus Games

- **Description: Games designed to improve concentration, memory, and attention span through engaging activities.**
- **Example: A card-matching game with a theme of coping strategies or emotional words.**
- **Therapeutic Benefit: Strengthens cognitive functioning and helps with ADHD symptoms.**

8. Virtual Art Therapy Games

- **Description: Digital games where children can draw, paint, or create art while exploring their feelings.**
- **Example: A virtual drawing board where children can color a "mood tree" representing their current emotions.**
- **Therapeutic Benefit: Encourages self-expression and emotional exploration.**

This psychiatric game will determine and generate a result on the mental condition of the children.

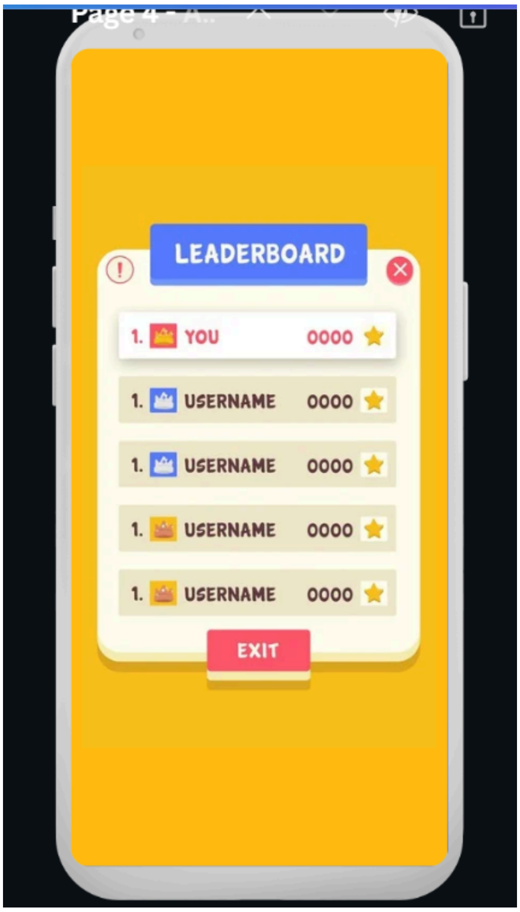


Figure 6.4: Interface for Child View< Leaderboard/Scoreboard>

The Leaderboard/Scoreboard interface is a visually engaging and user-friendly feature designed to encourage children through fun and motivating gameplay. It provides an interactive display that ranks players based on their game scores, fostering a sense of healthy competition and accomplishment. The interface prominently features the "LEADERBOARD" title at the top of the screen, using bold and colorful fonts for clarity and appeal. Below the title, a list of players is

displayed in ranked order, with the current child user being highlighted as "YOU" to ensure clear identification of their position. Each entry on the leaderboard shows the ranking number, username, score, and small decorative icons such as crowns or stars, which add a playful element to the experience.

The interface also includes a clearly labeled "EXIT" button at the bottom of the screen, allowing users to return to the main menu or exit the leaderboard view effortlessly. This button is designed in red to make it stand out, ensuring accessibility. The overall design incorporates a bright yellow background to attract attention and maintain engagement, while the leaderboard itself is styled as a popup window with rounded edges and a contrasting white background to draw focus.

The functionality of this feature is centered around motivating children to track their progress and improve their performance in a rewarding way. The leaderboard updates dynamically in real-time to reflect the latest scores, highlighting any changes in rankings to maintain excitement. It is designed to be intuitive for children aged 8–12, featuring large, readable text and simple navigation. Beyond its entertaining aspects, this interface has therapeutic benefits by promoting a healthy sense of competition, boosting self-confidence through the recognition of achievements, and encouraging goal-setting and perseverance. Additionally, the leaderboard serves as a tool to reinforce positive behavior and progress, aligning with the overall objectives of the application. Potential enhancements could include personalized motivational messages, filters for displaying specific rankings, or options for players to customize their usernames and avatars for a more personalized experience.

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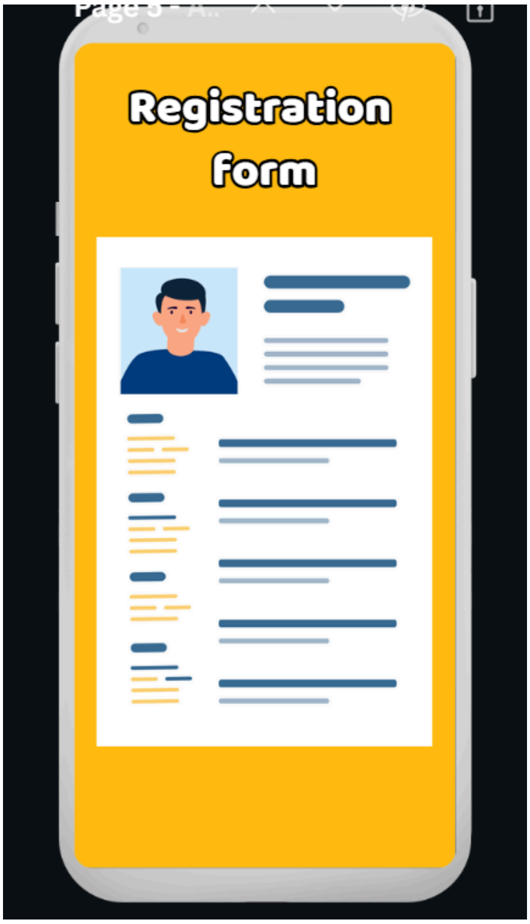

Figure 6.5: Interface for Parent View<Registration Form>

The end of the child interface, Now this is the parent interface .Based on the interface above shows registration form . Every parent must fill in everything that is required in the form. The information will be strictly stored safely in the database.

Figure 6.6: Interface for Parent View<Parent Assessment>

After the parent role has completely filled the registration form, they must answer the parent assessment. Parent assessment is required in this application software development with the aim to track the main causes of their children having mental health issues . If the Assessment shows the bad result, we know that their mental health issues were inherited by their parents.

Figure 6.6: Interface for Parent View<Parent Patient screening>

The Parent Patient Screening result interface is designed to provide parents with a comprehensive overview of their child's mental health status based on the Depression, Anxiety, and Stress Scale (DASS) assessment. The interface presents the results in a structured and visually engaging manner, ensuring ease of understanding for parents who may not be familiar with clinical terminologies.

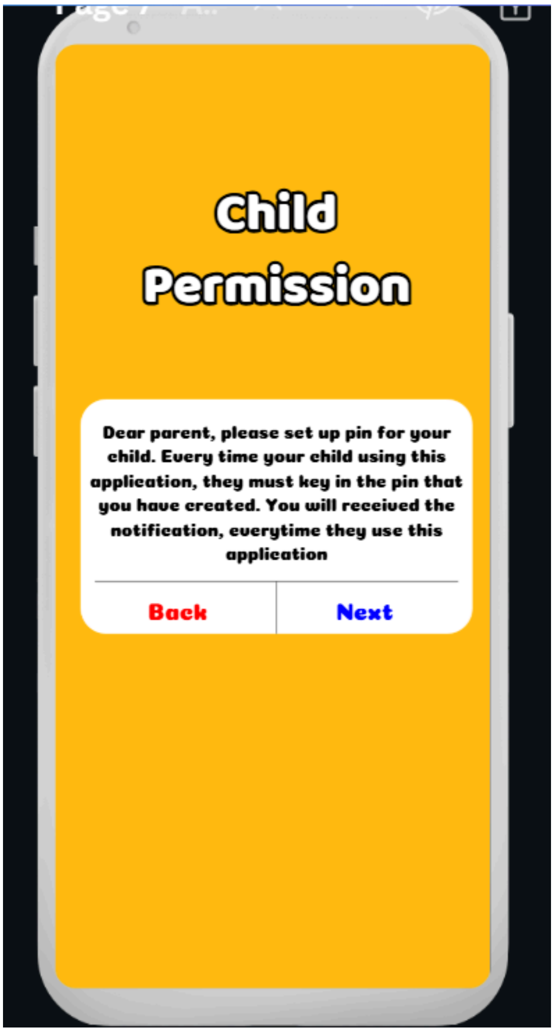

Figure 6.7: Interface for Parent View<Child Permission>

After they are done with assessment, this application will bring the user to child permission. As we can see, the pop up message will show up to notify the parent about the importance of setting up the pin for children.

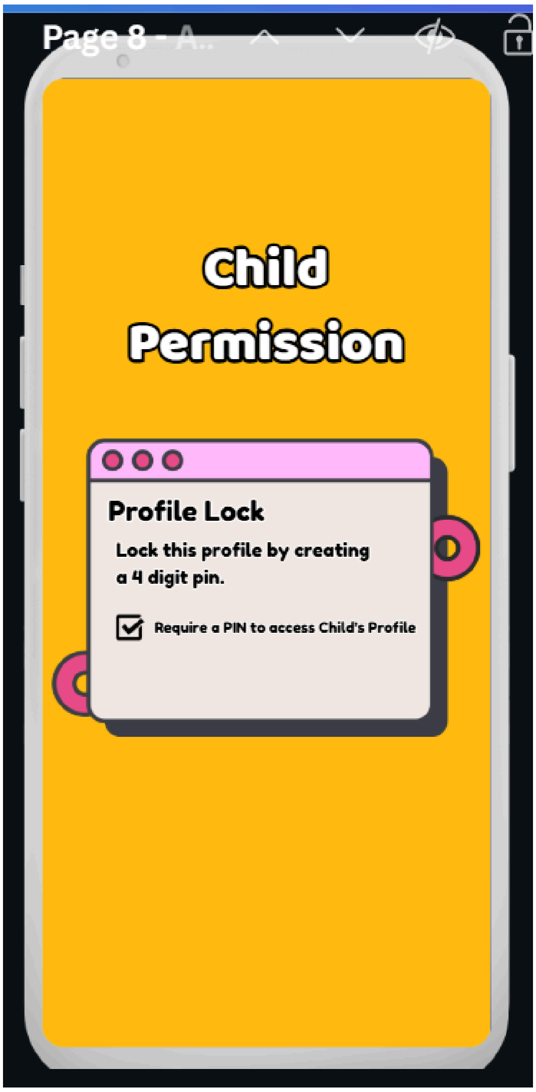

Figure 6.8: Interface for Parent View<Child Permission; creating the pin>

This interface shows parents must tick the checkbox as they will agree to put a 4 pin secure password.

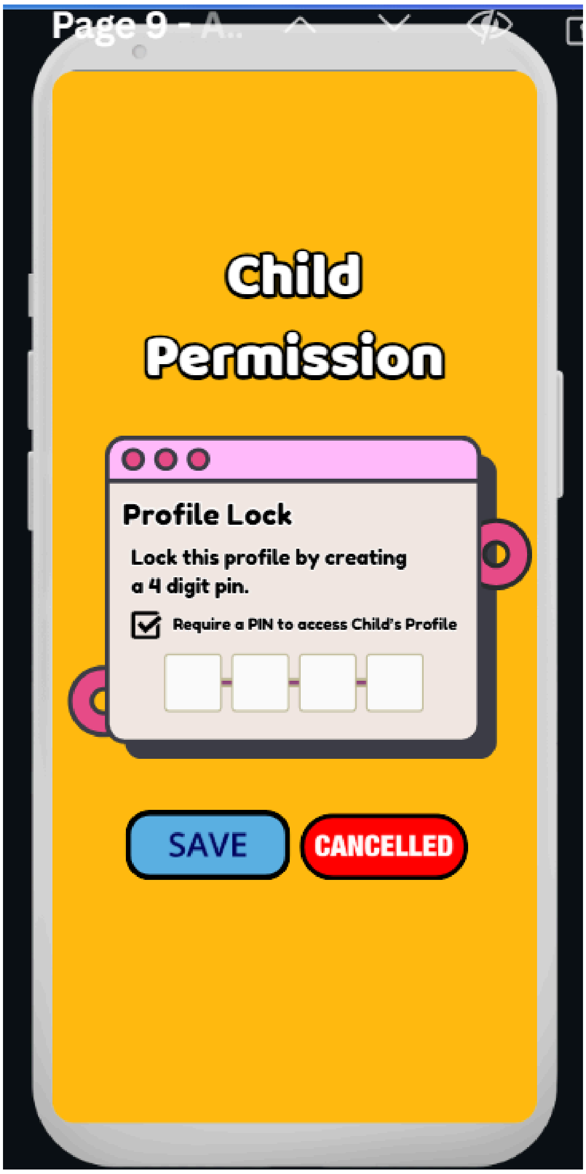

Figure 6.9: Interface for Parent View<Child Permission; creating the pin>

Parents must set the 4 pin secure lock for their child account . Click save button to proceed or cancel button to cancel the 4 pin secure lock.

Figure 6.10: Interface for Parent View<Child Permission>

The interface shows your child is ready, which means , the interface for the child is ready to be used. The button ok shows, the user must click to proceed to the next page.

Figure 6.11: Interface for Parent View<Menu>

In this interface, parents are able to choose the menu . The menu consist mood tracking , profile, online therapy video, module parenting for mental health, profile supervision, set a goal , alarm, notification setting

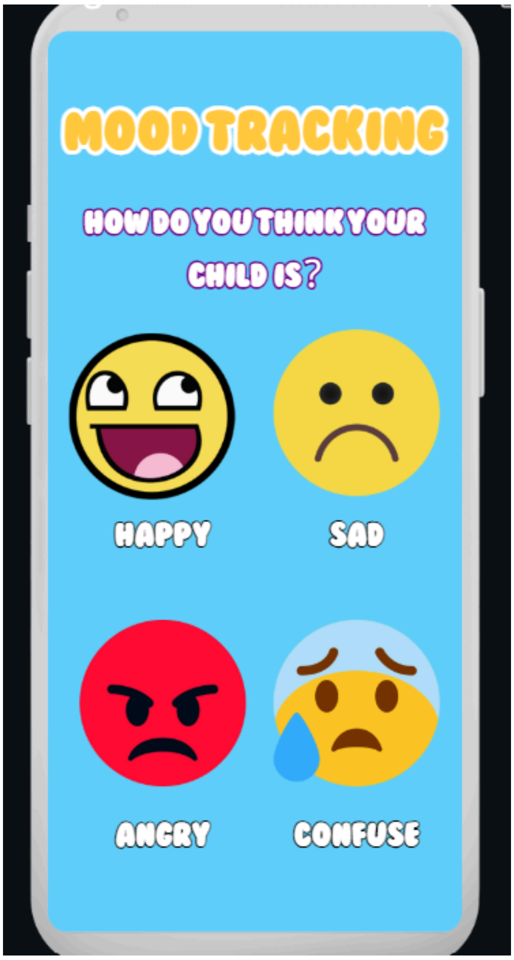

Figure 6.12: Interface for Parent View<Parent choose mood for children>

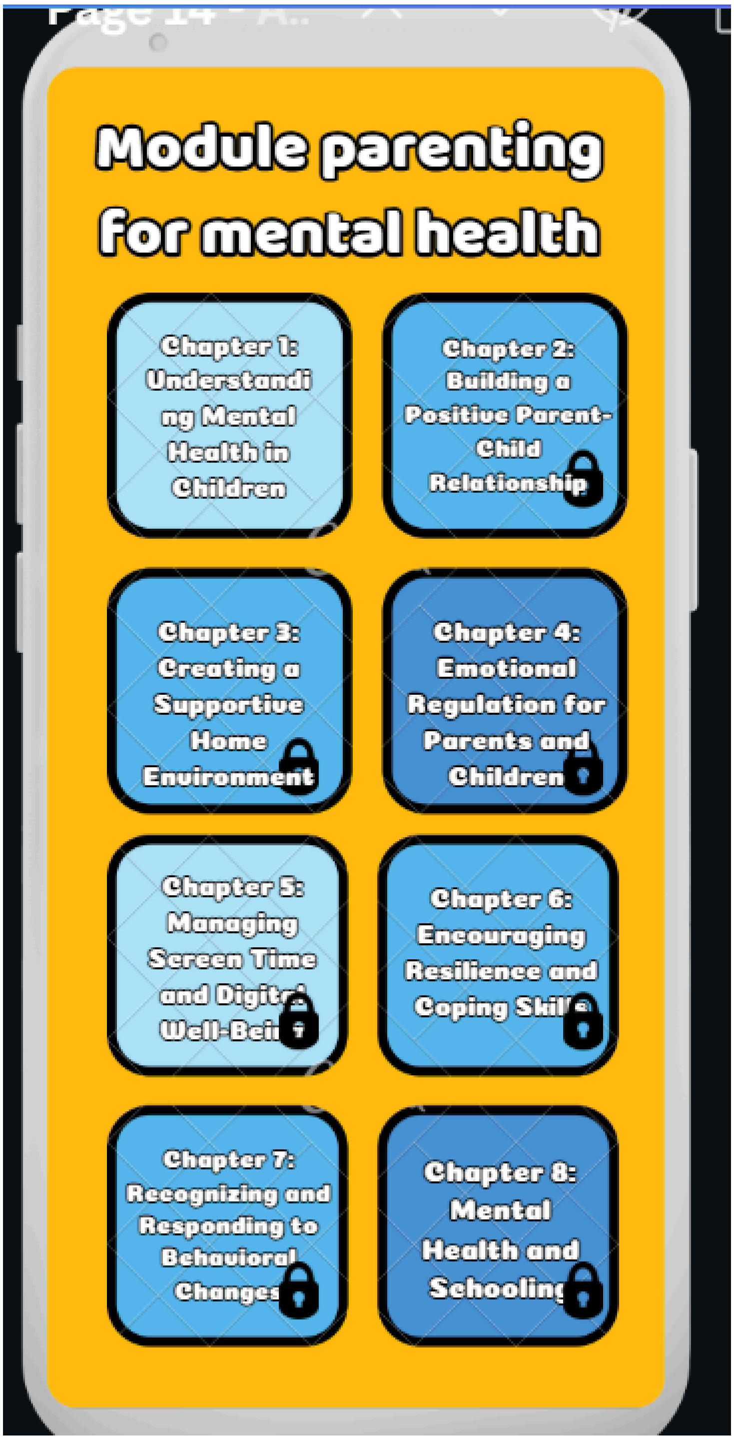

Figure 6.13: Interface for Parent View<Module parenting for mental health>

In this part , parents must choose a Chapter based on the sequence, the chapter is locked and the only way to unlock the module is, answering the quiz inside the chapter. The quiz was installed in this application to make sure the parent knows well about what they have learned inside the chapter.

Chapter 1: Understanding Mental Health in Children

- Overview of mental health and its importance.
- Common challenges and developmental milestones.
- Recognizing signs of mental health concerns.

Chapter 2: Building a Positive Parent-Child Relationship

- Effective communication strategies.
- The role of active listening and empathy.
- Fostering trust and emotional security.

Chapter 3: Creating a Supportive Home Environment

- Importance of routines and stability.
- Designing a safe and nurturing space.
- Encouraging family activities to strengthen bonds.

Chapter 4: Emotional Regulation for Parents and Children

- Teaching children to identify and express emotions.
- Managing stress and emotional outbursts.
- Techniques like mindfulness and breathing exercises.

Chapter 5: Managing Screen Time and Digital Well-Being

- Effects of excessive screen time on mental health.
- Setting healthy boundaries and digital habits.
- Promoting positive online interactions.

Chapter 6: Encouraging Resilience and Coping Skills

- Helping children deal with failure and setbacks.
- Teaching problem-solving and decision-making.
- Building self-esteem and confidence.

Chapter 7: Recognizing and Responding to Behavioral Changes

- Identifying triggers and patterns in behavior.
- Differentiating between normal and concerning behaviors.
- Approaching professional help when needed.

Chapter 8: Mental Health and Schooling

- Balancing academics with emotional well-being.
- Collaborating with teachers and school counselors.
- Recognizing signs of bullying or peer pressure.

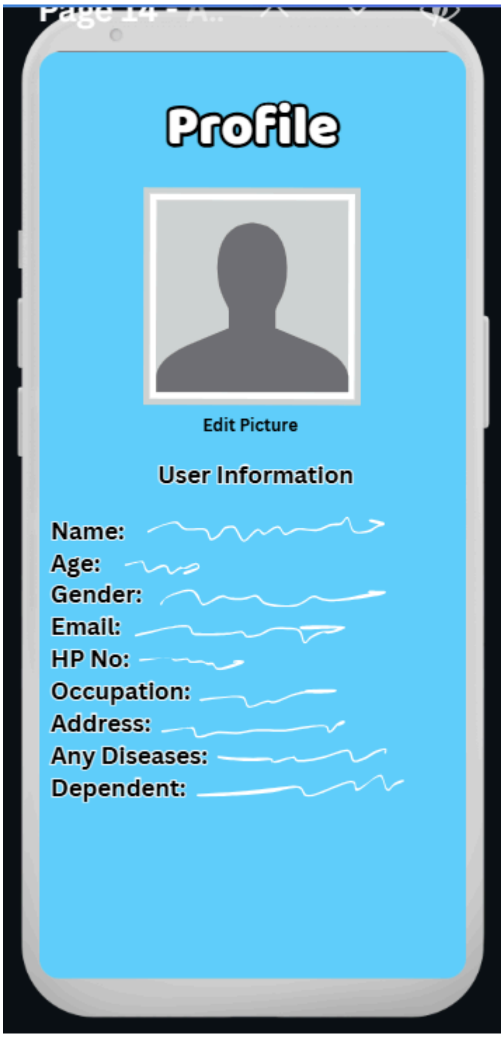

Figure 6.14: Interface for Parent View<Profile; User information>

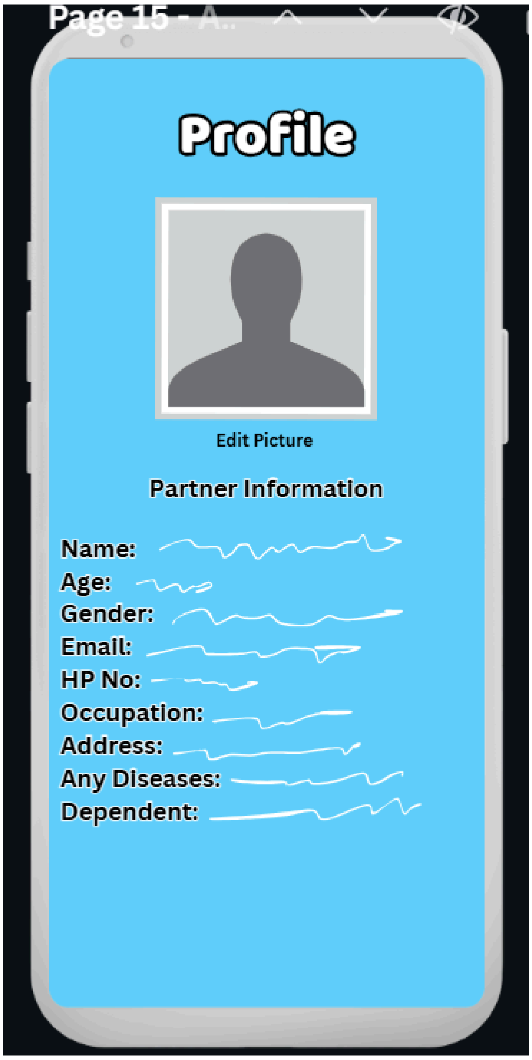

Figure 6.15: Interface for Parent View<Profile;Partner information>

Figure 6.16: Interface for Parent View<Profile;Child information>

Figure 6.17: Interface for Parent View<Online therapy video>

In this part , parents can choose a video . The video will help parents on how to handle their kids, on how to understand their child.

Session 1: Introduction to Child Mental Health

- Understanding the importance of mental health in childhood.
- Recognizing early signs of mental health issues.
- Common misconceptions and myths.

Session 2: Building a Strong Parent-Child Connection

- Effective communication techniques.
- The role of empathy and active listening.
- Creating trust and emotional security.

Session 3: Managing Emotions Together

- Teaching children to recognize and express emotions.
- Strategies for parents to model emotional regulation.
- Mindfulness and calming techniques for families.

Session 4: Dealing with Anxiety and Stress in Children

- Identifying triggers of anxiety and stress.
- Coping mechanisms for children.
- When to seek professional help.

Session 5: Addressing Behavioral Challenges

- Understanding the reasons behind behavioral changes.
- Positive reinforcement techniques.
- Setting boundaries without conflict.

Session 6: Creating a Balanced Routine

- Importance of structure and consistency.
- Balancing academic, social, and recreational activities.
- Limiting screen time and promoting healthy habits.

Session 7: Supporting Mental Health in School Settings

- How to work with teachers and counselors.
- Addressing bullying and peer pressure.
- Encouraging positive social interactions.

Session 8: Nurturing Resilience and Confidence

- Helping children bounce back from setbacks.
- Activities to build self-esteem.
- Teaching problem-solving and adaptability.

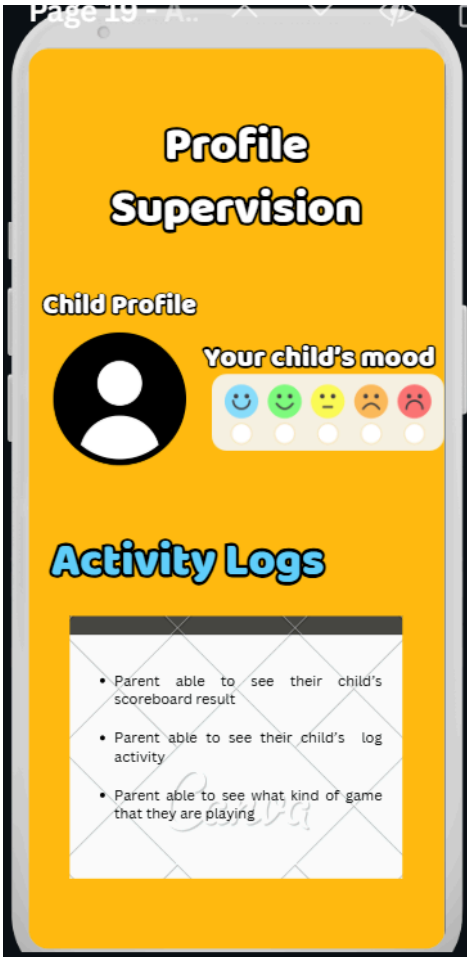

Figure 6.18: Interface for Parent View<Profile supervision>

The "Profile Supervision" section allows parents to view their child's profile, including a visual representation of their mood using emojis or icons that reflect different emotional states, such as happiness, sadness, or neutrality. This provides parents with an easy way to understand their child's emotional trends.

The "Activity Logs" section gives parents insights into their child's interactions within the app. They can access details like the scoreboard results, which might reflect progress in educational or therapeutic games, and logs of their child's activities. It also shows the types of games the child has been engaging with, ensuring they align with the app's objectives of supporting mental well-being.

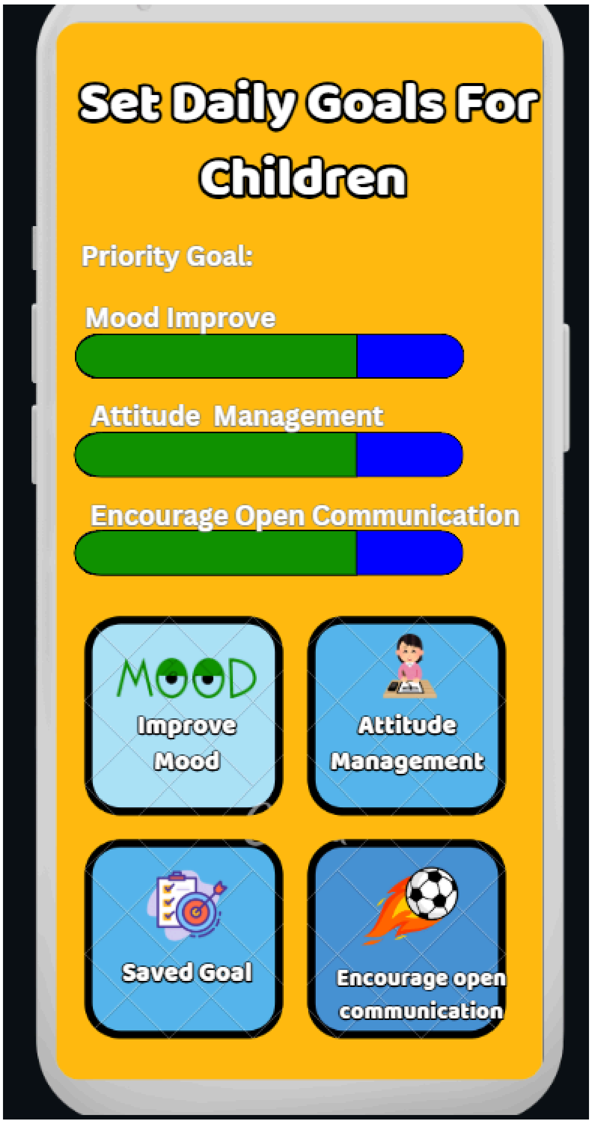

Figure 6.19: Interface for Parent View<Set Daily Goal>

The interface , "Set Daily Goals for Children," highlights the purpose of the feature. Below it, a section labeled "Priority Goal" lists three key focus areas for the child's development: improving mood, managing attitudes, and encouraging open communication. Each of these goals is paired with a progress bar, which visually represents the child's achievements or level of engagement with these objectives.

The lower part of the interface contains four buttons, each corresponding to a specific goal or feature. The "Improve Mood" button likely leads to activities or tools that help uplift the child's mood, such as games or relaxation exercises. The "Attitude Management" button might provide resources to guide parents in handling their child's behavior or teaching self-regulation strategies. The "Saved Goal" button seems to allow parents to revisit previously set or accomplished goals for continuity and tracking progress over time. The "Encourage Open Communication" button likely offers activities or techniques to foster conversations and strengthen the parent-child relationship.

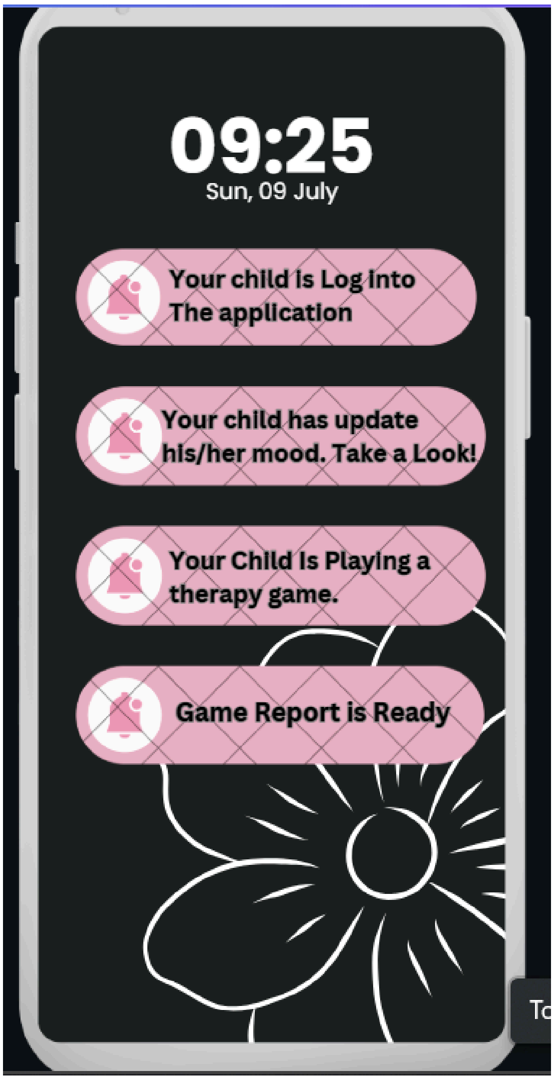

Figure 6.20: Interface for Parent View<Alarm>

This interface represents a notification screen within a parenting app, designed to keep parents updated on their child's mental health activities and interactions with the application. The interface combines functionality and design to deliver real-time insights.

At the top, the current time and date are displayed prominently, helping parents stay organized and aware of when specific updates occur. Below this, a series of notifications informs the parent about the child's activities within the app:

1. "Your child is logged into the application"
   This notification alerts the parent when the child has accessed the app, ensuring transparency and awareness of their usage.
2. "Your child has updated his/her mood. Take a look!"
   This message highlights that the child has recorded or updated their mood. It encourages parents to review the entry, which could provide insight into the child's current emotional state.

3. "Your child is playing a therapy game"
   This notification indicates that the child is engaged in a therapeutic game, which could be aimed at improving mental well-being, building resilience, or enhancing problem-solving skills.
4. "Game report is ready"
   This update informs the parent that a detailed report of the child's performance or engagement in the therapeutic game is available. This report may include metrics on progress, areas of improvement, or the child's mood during the activity.

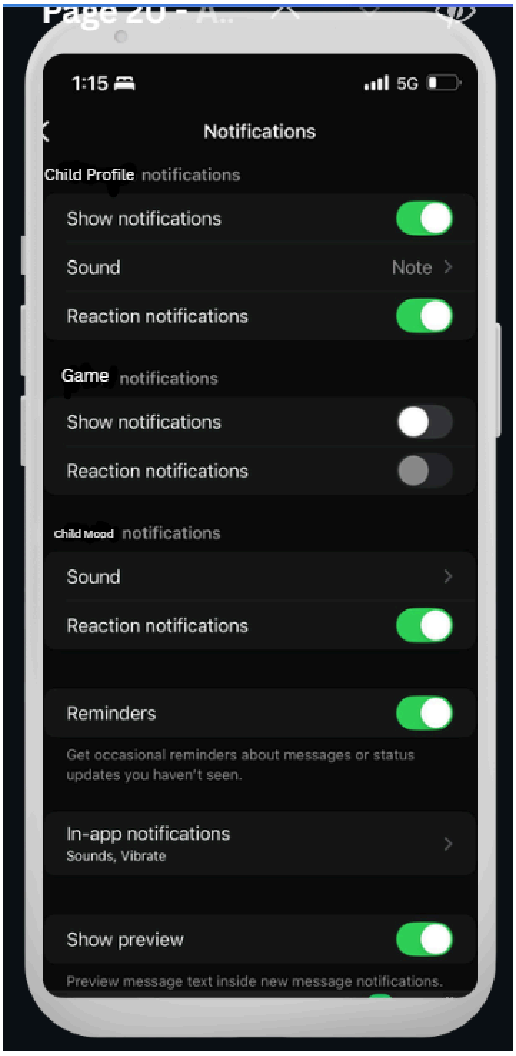

Figure 6.20: Interface for Parent View<Notification Setting>

On this part parent able to set their own notification in term to received the output from their children

7 . Traceability

Sprint#	Subsystem/Package	User Story ID
Sprint 1	User access and profile subsystem	UC001
		UC002
		UC006
		UC011
Sprint 2	Behavior and emotion subsystem	UC005
		UC007
Sprint 3	Performance and tracker alert subsystem	UC004
		UC008
		UC009
Sprint 4	Therapy and game interaction subsystem	UC003
		UC010

Table 7: traceability

8. Test Cases

8.1 TC001: Test < User Access and Profile> Subsystem: < Sign up and provide info (UC001)>

This test contains the following test cases:

(a) TC001_01: Test <User Registration Scenario (SD001)>

8.1.1 TC001_01: Test < User Registration Scenario (SD001)>

This test contains the following scenarios:

(a) TC001_01_01: Test normal scenario of Parent Sign-Up (SD001)

(b) TC001_01_02: Test alternate scenario of Parent Sign-Up (email verification fails)(SD001)

(c) TC001_01_03: Test exception scenario of Parent Sign-Up (missing required fields) (SD001)

DESCRIPTION	VERSION	TEST DATE
Test the scenario for user registration.	1.0	12-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the parent can successfully register by providing the required personal and child-related information.	Mark	

Test Case ID **(TC001_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the sign-up page	Button “SIGN UP” in the WELCOME BACK interface.	The sign-up page is displayed with fields for Name, Email, Password, Child's Name, and Child's Age**.**			Bernice Lou Min Yun	
2	Enter valid name, email, password, and child-related details	Example: Name: Bernice E-mail: bernicelou@gmail.com Password: Pass1234! Child Name: Emily Age: 6	The system accepts the inputs, validates the email format and password strength, and enables the **Sign Up** button.			Bernice Lou Min Yun	

3	Submit the form by clicking "Sign Up"	Button “SIGN UP” in CREATE ACCOUNT interface.	A message is displayed: "Please check the confirmation email."			Bernice Lou Min Yun	
4	Verify the email by clicking the confirmation link	Verification link from email	The system validates the link, creates the account, and redirects the user to the parent assessment.			Bernice Lou Min Yun	

Test Case ID **(TC001_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the sign-up page	Botton "SIGN UP" in WELCOME BACK interface.	The sign-up page is displayed with fields for Name, Email, Password, Child's Name, and Child's Age.			Bernice Lou Min Yun	
2	Enter valid name, email, password, and child-related details	Example: Name: Bernice E-mail: bernicelou@gmail.com Password: Pass1234! Child Name: Emily Age: 6	The system accepts the inputs, validates the email format and password strength, and enables the **Sign Up** button.			Bernice Lou Min Yun	

3	Submit the form by clicking "Sign Up"	Button “SIGN UP” in CREATE ACCOUNT interface.	A message is displayed: "Please check the confirmation email."			Bernice Lou Min Yun	
4	Email verification fails (user does not click the link or link expires)	Verification link from email	The system displays an error message: "Verification failed. Please try again.".			Bernice Lou Min Yun	
5	User requests to resend the confirmation email		A message is displayed: "Please check the confirmation email."			Bernice Lou Min Yun	

Test Case ID **(TC001_01_03)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the sign-up page	Botton “SIGN UP” in WELCOME BACK interface.	The sign-up page is displayed with fields for Name, Email, Password, Child's Name, and Child's Age.			Bernice Lou Min Yun	
2	Enter incomplete information (e.g., missing email or child’s age)	Example: Name: Bernice E-mail: bernicelou@gmail.com Password: Pass1234! Child Name: Emily Age:	The system displays an error message below the missing field: "PLEASE FILLED IN THIS FIELD."			Bernice Lou Min Yun	

3	Submit the form by clicking "Sign Up"	Button “SIGN UP” in CREATE ACCOUNT interface.	Account creation is prevented until all required fields are filled correctly.			Bernice Lou Min Yun	

8.2 TC002: Test < User Access and Profile> Subsystem: < Create secure PIN for child account (UC002)>

This test contains the following test cases:

(a) TC002_01: Test <Child Login Using PIN (SD002)>

8.2.1 TC002_01: Test <Child Login Using PIN (SD002)>

This test contains the following scenarios:

(a) TC002_01_01: Test <normal scenario of Child Login Using PIN(SD002)>
(b) TC002_01_02: Test <alternate scenario of Child Login (incorrect PIN)(SD002)>
(c) TC002_01_03: Test <exception scenario of Child Login (account lock due to multiple failed attempts)(SD002)>

DESCRIPTION	VERSION	TEST DATE
Test scenario of Child Login Using PIN	1.0	12-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the child can successfully log in using the PIN provided by the parent.	Mark	

Test Case ID **(TC002_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the child login page	Botton”Child”	The Child Login Page is displayed with the title "Child Login," a SECURE PIN field, and a "LOGIN" button.			Bernice Lou Min Yun	

2	Enter valid PIN provided by the parent	PIN: 1234	The entered PIN is accepted, and the system proceeds to log in the child.			Bernice Lou Min Yun	
3	Log in to the child account.	Button "LOGIN"	The child is logged into their account, and the system displays the Mood Tracking interface.			Bernice Lou Min Yun	

Test Case ID **(TC002_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the child login page.	Botton"Child"	The Child Login Page is displayed with the title "Child Login," a SECURE PIN field, and a "LOGIN" button.			Bernice Lou Min Yun	
2	Enter an incorrect PIN.	PIN: 0000	System displays an error message: "Incorrect PIN."			Bernice Lou Min Yun	

3	Retry with a correct PIN.	PIN: 1234	The entered PIN is accepted, and the system proceeds to log in the child.				

Test Case ID **(TC002_01_03)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the child login page.	Botton" Child"	The Child Login Page is displayed with the title "Child Login," a SECURE PIN field, and a "LOGIN" button.			Bernice Lou Min Yun	
2	Enter an incorrect PIN.	PIN: 0000	The system displays an error message in red: "INCORRECT PIN.			Bernice Lou Min Yun	

3	Enter incorrect PIN repeatedly	PIN: 0000 (5 times)	The system locks the account temporarily and displays: "ACCOUNT LOCKED DUE TO MULTIPLE FAILED ATTEMPEDS. PLEASE CONTACT YOUR PARENT."				

8.3 TC003: Test < User Access and Profile> Subsystem: < Access parenting resources (UC006)>

This test contains the following test cases:

(a) TC003_01: Test <Parent Access to Parenting Modules (SD003)>

8.3.1 TC003_01: Test <Parent Access to Parenting Modules (SD003)>

This test contains the following scenarios:

(a) TC003_01_01: Test <normal scenario of Accessing Parenting Modules(SD003)>

(b) TC003_01_02: Test <alternate scenario of Accessing Parenting Modules (progress saved)(SD003)>

(c) TC003_01_03: Test <exception scenario of Accessing Parenting Modules (content fails to load)(SD003)>

DESCRIPTION	VERSION	TEST DATE
Test the scenario for parents accessing and completing parenting modules.	1.0	12-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the parent can successfully access, complete, and resume parenting modules.	Mark	

Test Case ID **(TC003_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the parenting resources page.	Button "Parenting Resources" in MENU.	Parenting Resources Page is displayed with modules listed in sequence.			Bernice Lou Min Yun	

2	Select a locked parenting resource.	Resource: Supporting Mental Health	Resource content is displayed correctly.			Bernice Lou Min Yun	
3	Go through the resource		Resource progress is recorded and marked as complete.			Bernice Lou Min Yun	

4	Take the quiz at the end of the resource.	Answer questions	Parent answers correctly and the resource is completed. Next resource is unlocked .			Bernice Lou Min Yun	
5	System logs the completion of the resource.		The progress is logged.				

Test Case ID **(TC003_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments

1	Navigate to the parenting resources page.	Button “Parenting Resources” in MENU.	Parenting Resources Page is displayed with modules listed in sequence.			Bernice Lou Min Yun	
2	Select a locked parenting resource.	Resource: Supporting Mental Health	Resource content is displayed correctly.			Bernice Lou Min Yun	

3	Exit the resource before completion		System saves the current progress.			Bernice Lou Min Yun	
4	Re-access the resource	Resource: Supporting Mental Health	System resumes the resource from where it was left.			Bernice Lou Min Yun	

Test Case ID **(TC003_01_03)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to the parenting resources page.	Button “Parenting Resources” in MENU.	Parenting Resources Page is displayed with modules listed in sequence.			Bernice Lou Min Yun	
2	Select a locked parenting resource.	Resource: Supporting Mental Health	System attempts to load the resource content.			Bernice Lou Min Yun	

3	Resource content fails to load.	Error: Content loading issue	System displays error message: "Content failed to load. Please try again.".				

8.4 TC004: Test < User Access and Profile> Subsystem: < Profile supervision (UC011)>

This test contains the following test cases:

(a) TC004_01: Test <Access to Child Profile Activity Summary (SD004)>

8.4.1 TC004_01: Test <Access to Child Activity Summary (SD004)>

This test contains the following scenarios:

(a) TC004_01_01: Test <normal scenario of accessing child activity summary(SD004)>

(b) TC004_01_02: Test <alternate scenario of no logged activity(SD004)>

(c) TC004_01_03: Test <exception scenario of missing activity data(SD004)>

DESCRIPTION	VERSION	TEST DATE
Test scenario of accessing child activity summary.	1.0	12-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the parent can access the child's activity summary, which includes mood logs, game scores, and usage time.	Mark	

Test Case ID **(TC004_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Parent logs into their account	Button "Sign In" in the WELCOME BACK interface.	Parent account is accessed successfully			Bernice Lou Min Yun	

2	Navigate to the supervision module.	Button “Profile Supervision” in the MENU interface.	Child’s activity summary (child feeling and activity logs) is displayed correctly.			Bernice Lou Min Yun	

3	Review the activity summary	Activities: how your child feels and activity logs.	Parents can review the activity data that is complete, accurate, and up-to-date.			Bernice Lou Min Yun	

Test Case ID **(TC004_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Parent logs into their account	Button “Sign In” in the WELCOME BACK interface.	Parent account is accessed successfully.			Bernice Lou Min Yun	

2	Navigate to the supervision module	Button “Profile Supervision” in the MENU interface.	System displays message: "No activity logged. Encourage child to use the app.".			Bernice Lou Min Yun	

Test Case ID **(TC004_01_03)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Parent logs into their account	Button “Sign In” in the WELCOME BACK interface.	Parent account is accessed successfully			Bernice Lou Min Yun	
2	Navigate to the supervision module	Button “Profile Supervision” in the MENU interface.	System retries to load data or displays an error notification: "Data unavailable".			Bernice Lou Min Yun	

8.5 TC005: Test <Behavior and Emotion Tracking> Subsystem: <Complete Parental Assessments (UC005)>

This test contains the following test cases:

(a) TC005_01: Test <Parental Assessments Scenario (SD005)>

8.5.1 TC005_01: Test <Parental Assessments Scenario (SD005)>

This test contains the following scenarios:

(a) TC005_01_01: Test <normal scenario of parent assessments (SD005)>
(b) TC005_01_02: Test <exception scenario of parent assessments (SD005)>

DESCRIPTION	VERSION	TEST DATE
Test the functionality of completing parental assessments to track a child's behavior and emotions.	1.0	12-Jan-2024
TEST OBJECTIVE	**AUTHOR**	**REVIEWER**

Verify that users (parents) can complete and submit the parental assessment successfully.	**Karen Yam Vei Xin**	

Test Case ID **(TC005_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to "Parental Assessments" page		Page is loaded successfully			Karen Yam Vei Xin	
2	Fill in the assessment questionnaire	Answers to each question	Data is saved correctly			Karen Yam Vei Xin	
3	Submit the completed assessment	Click "Submit" button	Assessment is submitted			Karen Yam Vei Xin	

Test Case ID **(TC005_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to "Parental Assessments" page		Page is loaded successfully			Karen Yam Vei Xin	
2	Leave the page idle for a long time (simulate timeout)		The system prompts the user to log in again, and previously entered data is saved as a draft			Karen Yam Vei Xin	

8.6 TC006: Test <Behavior and Emotion Tracking> Subsystem: <Track Mood (UC007)>

This test contains the following test cases:

(a) TC006_01: Test <Mood Tracking Scenario (SD006)>

8.6.1 TC006_01: Test <Mood Tracking Scenario (SD006)>

This test contains the following scenarios:

(a) TC006_01_01: Test <normal scenario of mood tracking (SD006)>
(b) TC006_01_02: Test <exception scenario of mood tracking (SD006)>

DESCRIPTION	VERSION	TEST DATE
Test the functionality to allow users to track their mood.	1.0	12-Jan-2024
TEST OBJECTIVE	**AUTHOR**	**REVIEWER**
Ensure that users can select, record, and view their mood in the system.	Karen Yam Vei Xin	

Test Case ID (TC006_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to "Mood Tracking" section		Page is loaded successfully			Karen Yam Vei Xin	
2	Select a mood from the list	Mood options (e.g., Happy, Sad)	Mood is selected successfully			Karen Yam Vei Xin	

3	Save the mood entry	Click "Save" button	Mood entry is saved			Karen Yam Vei Xin	

Test Case ID **(TC006_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to "Mood Tracking" section		Page is loaded successfully			Karen Yam Vei Xin	
2	Select a mood and attempt to save it during a simulated server outage.		The system displays an error message indicating a submission failure and prompts the user to retry later.			Karen Yam Vei Xin	

8.10 TC004: Test <Performance and tracker alert > Subsystem: <Receive notification for child app usage (UC004)>

This test contains the following test cases:

(a) TC004_01: Test <Receive notification for child app usage (SD004)>

8.10.1 **TC004_01: Test <Receive notification for child app usage (SD004)>**

This test contains the following scenarios:

(a) TC004_01_01: Test <Receive notification for child app usage (SD004))>

(b) TC004_01_02: Test <alternate scenario of mental health assessment (SD004)>

(c) TC004_01_03: Test <Exception scenario of Receive notification for child app usage (SD004)>

DESCRIPTION	VERSION	TEST DATE
Test the functionality to give output of scoreboard, notification setting and alarm.	1.0	12-Jan-2024
TEST OBJECTIVE	**AUTHOR**	**REVIEWER**
Verify that users can view scoreboard, set notification and alam .	Ainurrulnajwa	

Test Case ID **(TC004_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Verify that the app sends a notification when the child logs into the app.	Child logs into the app.	Notification appears on the parent's app with the message "Your child is logged into the application."			Ainurrul najwa	
2	Key in 4 digit pin . Log into the app	Key in 4 digit pin	Parent received notification on application access: your child has log into the application			Ainurrul najwa	
3	3. Child Choose Mood	Child choose their mood every time they log into the application	Parent receive notification : your child mood is updated			Ainurrul najwa	
4	Child play therapy games	Child app usage: Play game for 30 minutes.	parent receive notification for result from therapy games.			Ainurrul najwa	

Test Case ID **(TC004_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	If the parent disable notification , the system logs the child's login activity instead.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The parent app receives a notification stating: "Your child used the app for 30 minutes."			Ainurrul najwa	
2.	Ensure alternate flow works when notifications are disabled	-	System logs activity instead of sending a notification.			Ainurrul najwa	

Test Case ID **(TC004_01_03)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Simulate a failure to send notifications..	user retrieves the application	Notification retry occurs, and activity is logged successfully.			Ainurrul najwa	
2	Check if the system retries automatically and logs the activity.	Simulated notification failure scenario.	Notification retry occurs, and activity is logged successfully.			Ainurrul najwa	

8.10.1 **TC008_01: Test <Send alarm for daily goals (UC008)>**

This test contains the following scenarios:

(a) TC008_01_01: Test <Receive notification for Send alarm for daily goals (SD008))>

(b) TC008_01_02: Test <alternate scenario of Send alarm for daily goals (SD008)>

(c) TC008_01_03: Test <Exception scenario of Send alarm for daily goals (SD008)>

Test Case ID **(TC008_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments

1	1. Log in to the parent app.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered on the parent app at 7 PM with a notification: "Reminder: Complete your daily goals!".			Ainurrul najwa	
2	2. Set a daily goal with a specific time for the alarm (e.g., "Remind to complete daily goals at 7 PM").	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered on the parent app at 7 PM with a notification: "Reminder: Complete your daily goals!".			Ainurrul najwa	
3	3. Wait for the scheduled alarm time.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered on the parent app at 7 PM with a notification: "Reminder: Complete your daily goals!".			Ainurrul najwa	

Test Case ID **(TC008_02_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments

1	1 Log in to the parent app.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	
2	2. Set multiple daily goals with alarms for different times (e.g., 7 PM and 9 PM).	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	
3	3. Wait for the scheduled alarm time.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	

Test Case ID **(TC008_03_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	1 Log in to the parent app.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	
2	2. Set a daily goal with a specific time for the alarm.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	
3	3. Disable app notifications before the alarm time.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	
4	4. Wait for the scheduled alarm time.	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	

8.10.1 **TC009_01: Test <View Game Scoreboard (UC009)>**

This test contains the following scenarios:

(a) TC009_01_01: Test <View Game scoreboard (SD009))>

(b) TC009_01_02: Test <alternate scenario of View Game Scoreboard (SD009)>

(c) TC009_01_03: Test <Exception scenario of View Scoreboard (SD009)>

Test Case ID **(TC009_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	1. Log in to the child app.	User credentials	The game scoreboard is displayed, showing the list of games played and their respective scores in descending order of scores.			Ainurrul najwa	
2	2. Navigate to the "Game Scoreboard" section.e daily goals at 7 PM").	User credentials	The game scoreboard is displayed, showing the list of games played and their respective scores in descending order of scores.			Ainurrul najwa	

3	3. View the scoreboard for all games played.	User credentials	The game scoreboard is displayed, showing the list of games played and their respective scores in descending order of scores.			Ainurrul najwws	

Test Case ID **(TC009_02_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	1. Log in to the child app.	Child app usage:Congratulations kids! Parent app notification settings: Your kid play puzzle game , with statistic of their mental health is good	The game scoreboard is displayed with filtered results, showing only scores for "Puzzle" games played in the last 7 days.			Ainurrul najwa	
2	2. Navigate to the "Game Scoreboard" section.	Child app usage:Congratulations kids! Parent app notification settings: Your kid play puzzle game , with statistic of their mental health is good	The game scoreboard is displayed with filtered results, showing only scores for "Puzzle" games played in the last 7 days..			Ainurrul najwa	

3	3. Wait for the scheduled alarm time.	Child app usage:Congratulations kids! Parent app notification settings: Your kid play puzzle game , with statistic of their mental health is good	The game scoreboard is displayed with filtered results, showing only scores for "Puzzle" games played in the last 7 days.			Ainurrul najwws	

Test Case ID **(TC009_03_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	1. Log in to the child app. .	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	

2	2. Navigate to the "Game Scoreboard" section..	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	
3	3. Attempt to view the scoreboard when no games have been played	Child app usage: Play game for 30 minutes. Parent app notification settings: Enabled.	The alarm is triggered at each scheduled time with the appropriate message for each goal: "Reminder: Complete your daily goals!".			Ainurrul najwa	

8.11 TC011: Test <Therapy and Game Interaction> Subsystem: <Play Therapy Game (UC003)>

This test contains the following test cases:

(a) TC011_01: Test <Child plays therapy game (SD011)>

8.11.1 **TC001_01: Test <Child Plays Therapy Game (SD011)>**

This test contains the following scenarios:

(a) TC001_01_01: Test <Normal scenario of Child plays therapy game (SD011)>

(b) TC001_01_03: Test <Exception scenario of Child plays therapy game (SD011)>

DESCRIPTION	**VERSION**	**TEST DATE**
Test the scenario for child playing therapy game	**1.0**	**12-Jan-2024**
TEST OBJECTIVE	**AUTHOR**	**REVIEWER**
To verify that the child can complete the game without any difficulties	**Ravinesh A/L Maran**	

Test Case ID **(TC011_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the game choosing interface		All the game is displayed for children to choose			Ravinesh A/L Maran	
2	Child selects a game that they find interested	Example: Chat Quest	The game that has been chosen displayed correctly			Ravinesh A/L Maran	
3	Child plays the game and completes the game		Child completes the game			Ravinesh A/L Maran	
4	Display scoreboard of the game to children		The scoreboard are displayed correctly			Ravinesh A/L Maran	

5	Send the report of child score to parent		Sends a correct and effective report to parent's account			Ravinesh A/L Maran	

Test Case ID **(TC011_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the game choosing interface		All the game is displayed for children to choose			Ravinesh A/L Maran	
2	Child selects a game that they find interested	Example: Chat Quest	The game that has been chosen displayed correctly			Ravinesh A/L Maran	
3	Child plays the game (fails to finish the game)		System saves the progress			Ravinesh A/L Maran	

4	The game will continue from the part where they leave it.		Child completes the game			Ravinesh A/L Maran	
5	Scoreboard is displayed.		The scoreboard are displayed correctly			Ravinesh A/L Maran	
6	Send the report of child score to parent		Sends a correct and effective report to parent’s account			Ravinesh A/L Maran	

8.12 TC012: Test <Therapy and Game Interaction> Subsystem: <Access Online Therapy Video (UC010)>

This test contains the following test cases:

(a) TC012_01: Test <Parent accessing online therapy videos (SD012)>

8.12.1 **TC001_01: Test <Access Online Therapy Video (SD012)>**

This test contains the following scenarios:

(a) TC012_01_01: Test <Normal scenario of Parent access online therapy video (SD001)>

(b) TC012_01_02: Test <Exception scenario of Parent access online therapy video (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test the scenario for parents accessing online therapy video	1.0	12-Jan-2024
TEST OBJECTIVE	**AUTHOR**	**REVIEWER**
To verify that parents can access the online therapy videos without difficulties.	Ravinesh A/L Maran	

Test Case ID **(TC012_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the online therapy video interface		List of the therapy videos are displayed to choose			Ravinesh A/L Maran	

2	User(parent) chooses the video	Example: ‘How to help a child with anxiety	The video that has been chosen streamed correctly			Ravinesh A/L Maran	
3	Parent watch the video		Display completed message			Ravinesh A/L Maran	

Test Case ID **(TC012_01_02)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the online therapy video interface		List of the therapy videos are displayed to choose			Ravinesh A/L Maran	
2	Parent chooses the video	Example: ‘How to help a child with anxiety	The video that has been chosen streamed correctly			Ravinesh A/L Maran	

3	Parent watch the video (connection error occurred)		Parent completes watching the video.			Ravinesh A/L Maran	
4	Parent resumes the video from the part where they lost the connection		Display completed message			Ravinesh A/L Maran	